

Polynomial vacuum representations and bridge mass for honeycomb walks

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Abstract

We prove that, under the all-length critical measure, a honeycomb bridge crossing a strip of h layers, with its initial port fixed and its terminal port free, has mean length $h^{4/3+o(1)}$. We also obtain central visit probability $R^{-2/3+o(1)}$ and mean length $R^{4/3+o(1)}$ for macroscopic free-boundary chords in a regular hexagon.

Contents

1	Critical bridge length and a polynomial polygon observable	2
1.1	Critical crossing mass and related laws	3
1.2	A polygon observable and its construction	4
1.3	Route through the proof	5
2	Planar and cylindrical setup	5
3	Row matrices and polynomial vacua	8
3.1	Polynomial vacuum	12
4	The balanced puncture observable	17
4.1	Interpolation of the contraction	19
5	The boundary exponent	20
5.1	Row susceptibility at infinity	20
5.2	Positive kernel estimate	22
5.3	Comparing the strip tails	26
6	Asymptotics of the balanced observable	27
6.1	Color formula and small-cycle gas	27
6.2	Homogeneous Pfaffian normalization	30
6.3	Radial periodization of the small-cycle integral	31
6.4	Upper-edge resummation	34
6.5	The residual kernels	37
6.6	A kernel positivity decomposition	41
6.7	Estimates and summation of the defect gas	44
6.8	Bulk calibration and puncture exponent	49

7	Localization of surrounding polygons	50
7.1	Weighted incompatibility from the zero-free disk	50
7.2	The six-rotation radius bound	51
7.3	Removing oversized polygons	55
8	Vertex visits and finite chord means	56
8.1	Exponential lateral tails in a strip	59
8.2	Macroscopic chords in a hexagon	59
8.3	A fixed-source strip version	60
9	A marked tripod comparison for bridge mass	61
9.1	Deletion and first contact	62
9.2	The marked tripod identity	62
9.3	Comparison of intersecting path pairs	65
9.4	Translation hits on a deep arch	65
9.5	The bridge first moment	67

1 Critical bridge length and a polynomial polygon observable

A walk can cross a wide strip while making many returns to a much smaller region. Its total crossing weight therefore does not determine its mean length. We study this distinction for critical self-avoiding walks on the honeycomb lattice. Our initial boundary point is fixed and the terminal point is summed over the opposite wall. The main result is a mean-length exponent of $4/3$ under this all-length measure.

We use the honeycomb lattice dual to the equilateral triangular tiling of side one. Its edges have length $1/\sqrt{3}$. A *port* is the midpoint of an edge cut by a tiling line. A path between ports visits distinct honeycomb vertices, and its length $|\gamma|$ is their number. Its critical weight is $x_*^{|\gamma|}$, where

$$x_* = (2 \cos(\pi/8))^{-1} = (2 + \sqrt{2})^{-1/2}.$$

The terminal half-edges have no additional weight. A strip of h layers has physical height $h\sqrt{3}/2$. With a bottom source at horizontal coordinate zero, its attainable top ports lie in $\mathbb{Z} + h/2$. Write $\mathcal{B}_h(e)$ for the paths confined to that strip from the source to top port e , and put

$$b_h(e) = \sum_{\gamma \in \mathcal{B}_h(e)} x_*^{|\gamma|}, \quad b_h = \sum_e b_h(e), \quad \ell_h = \sum_e \sum_{\gamma \in \mathcal{B}_h(e)} |\gamma| x_*^{|\gamma|}.$$

Thus ℓ_h is a first-length insertion, rather than a partition function conditioned on one length. The bridge law assigns each such path probability $x_*^{|\gamma|}/b_h$.

Theorem 1.1 (Strip mass and bridge length). *As the integer layer height h tends to infinity,*

$$b_h = h^{-1/4+o(1)}, \quad \ell_h = h^{13/12+o(1)}, \quad \mathbb{E}_h |\gamma| = h^{4/3+o(1)}.$$

All three sums are finite. These exponents are unchanged if the paths are also confined to horizontal distance $h \log^2 h$ from their initial port.

Throughout, $f(h) = h^{a+o(1)}$ means that $f(h) > 0$ for all sufficiently large h and $\log f(h)/\log h \rightarrow a$. In particular, the theorem gives subpower precision; it does not assert bounds by fixed multiples

of the displayed powers. Its last conclusion uses a logarithmically widening strip, and is distinct from confinement to a fixed multiple of h .

The same proof determines a finite free-boundary chord law. In a regular lattice hexagon of integer side R , sum critical weights over paths joining ordered boundary ports, retaining only paths of diameter at least $R/10$. Reversal gives a second ordered path and leaves the normalized law unchanged. Denote the resulting probability by $\mathbb{P}_R$.

Theorem 1.2 (Hexagon visits). *The mass normalizing $\mathbb{P}_R$ is $R^{3/4+o(1)}$. Uniformly at vertices in the concentric hexagon of side $R/3$,*

$$\mathbb{P}_R(v \in \gamma) = R^{-2/3+o(1)}.$$

The corresponding upper bound holds at every vertex, and $\mathbb{E}_R|\gamma| = R^{4/3+o(1)}$.

There is an intermediate strip law in which the distinction between arches and bridges matters. An *arch* returns to the wall where it started. From one fixed bottom port, allow both arches and bridges and condition the path to reach at least ϑh layers above the bottom, where $0 < \vartheta < 1/2$ is fixed.

Proposition 1.3 (Mixed exits). *The conditioned event has unnormalized mass $h^{-1/4+o(1)}$ and first-length mass $h^{13/12+o(1)}$. Hence its normalized mean length is $h^{4/3+o(1)}$. Restricting horizontal distance to $h \log^2 h$ preserves these exponents. The total first-length mass of all exits, without the height condition, is at most $h^{13/12+o(1)}$.*

The mixed mean follows directly from a finite vertex-visit identity. The bridge mean requires an additional argument that transfers mass from returning arches to crossing paths. It is this argument, rather than normalization alone, that completes Theorem 1.1. Both theorems concern means in positive critical measures; fixing a length or prescribing both ports asks a different question.

1.1 Critical crossing mass and related laws

Nienhuis's analysis of the dilute $O(n)$ model led to the honeycomb critical point and the critical-exponent predictions associated with planar self-avoiding walk [19]. Duminil-Copin and Smirnov proved the connective constant $\sqrt{2 + \sqrt{2}}$ by a parafermionic observable and its boundary cancellation [6]. We use that local cancellation, prove the particular finite flux identities needed here, and keep the visited-vertex convention explicit.

The sharper crossing-mass power $h^{-1/4}$ is recorded as a prediction in [6, Section 4]. Beaton, Bousquet-Mélou, de Gier, Duminil-Copin and Guttmann proved that the critical bridge mass tends to zero [2]. Glazman and Manolescu [13] obtained a quantitative subsequence bound, as well as invariance of boundary two-point functions in columnwise rhombic half-planes with Yang–Baxter weights. Krachun and Panagiotis proved a polynomial upper bound on the critical bridge mass [17]. These crossing-mass estimates do not by themselves control a first-length insertion. In particular, the all-length means considered here are distinct from displacement laws for uniform walks of a fixed length.

The companion paper [20, Theorem 1.1] proves the same all-length bridge-mean exponent and retains it inside a fixed corridor of sufficiently large width Ch . It also obtains bridge mass comparable to $h^{-1/4}$. Our proof of the strip and hexagon results is given independently below; we do not use the companion's mean theorem.

1.2 A polygon observable and its construction

The first-length insertion will be obtained by summing the mass of paths through a marked vertex. The corresponding algebraic observable counts surrounding polygons instead of paths. Identify horizontal translates by $2n$, forming an infinite honeycomb cylinder, and mark two face centers on one horizontal cut, separated by n sites. Let $F_n(\eta)$ sum over systems of pairwise vertex-disjoint unoriented simple polygons, each separating the two marks. A polygon P has weight $\eta x_*^{|P|}$, and the empty system has weight one. Each polygon crosses the intervening marked interval, so this is a polynomial in η ; its coefficients are convergent weighted sums over polygon shapes on the infinite cylinder.

Theorem 1.4 (Normalized separator pairing). *For every compact set $K \subset \{s \in \mathbb{C} : |\Im s| < \pi\}$ there is $C_K < \infty$ such that, for all $n \geq 1$ and $s \in K$,*

$$|F_n(2 \cosh s)| \leq C_K n^{1/6 + 2\Re(s^2)/(3\pi^2)}.$$

There is a disk about $\eta = 0$, independent of n , in which F_n has no zeros. For each fixed k , the order- k Taylor coefficient of $\log F_n$ at zero is $O_k(1 + \log n)$. At the positive fugacity two,

$$F_n(2) = n^{1/6 + o(1)}.$$

To construct F_n , record at each of $N = 2n$ edges on a cylinder cut whether the edge is empty or carries either orientation of a path. The oriented pieces carry local turn phases, and adding a row acts linearly on the resulting three-state tensors. We construct a common eigenvector of eigenvalue one whose entries are polynomials in parameters attached to those edges. Dividing by its empty component gives the normalized *vacuum*: at the physical honeycomb parameters its entries are the amplitudes of the half-cylinder path systems. Pairing two such states with a twist between the marked faces cancels nonseparating loops and assigns each separating loop weight $\eta = 2 \cosh s$.

The normalization $F_n(0) = 1$ is essential. Both the finite color formula for the numerator and the Pfaffian denominator have exponential growth in n^2 . The polynomial power survives only after those factors cancel in the same normalization. Uniformity for complex s also has a positive-measure consequence: the zero-free disk controls polygon incompatibilities when the cylinder is opened into the plane.

This construction uses polynomial exchange equations and reductions at special values of neighboring parameters. Di Francesco and Zinn-Justin [5] used these ideas for a multiparameter sum rule in the dense $O(1)$ model. Garbali and Nienhuis [11, 12] developed q KZ and fusion recurrences and determinant expressions for the dilute $O(1)$ model with open boundaries; they also discussed the polynomial construction at loop weight zero. Here the periodic twist- i state, its empty-component normalization and its physical separator pairing are constructed explicitly. The companion's disk states represent the same homogeneous separator sum; Section 4 compares the common scalar polynomial and physical observable. The inhomogeneous vectors and normalized complex estimates here are proved separately.

The local transfer weights belong to the dilute-loop and three-state vertex-model setting of Zhou and Batchelor [24]. Related finite recurrences and corner-transfer calculations for restricted-height dilute models were developed by Warnaar, Pearce, Seaton and Nienhuis [23]; Batchelor and Yung [1] obtained integrable boundary weights for the ordinary and special transitions of honeycomb polymers. Bleher and Fokin's six-vertex analysis with domain-wall boundaries likewise connects determinant partition functions to orthogonal-polynomial asymptotics [3]. These are methodological comparisons, not inputs for the vacuum or complex-twist estimate above.

1.3 Route through the proof

The proof first obtains two different mass estimates. After the flux and convergence bounds in Section 2, Sections 3 and 4 construct the polynomial vacuum and identify its pairing with F_n . Differentiating a row at zero loop fugacity then gives the boundary mass $b_h = h^{-1/4+o(1)}$ in Section 5. This calculation is independent of the bulk pairing asymptotics.

Section 6 proves Theorem 1.4. A finite color sum becomes contour integrals, with a separate Pfaffian lower bound having bounded logarithmic error. Periodizing the integrals in an auxiliary spectral direction permits a circle-determinant expansion. After that expansion, a positive Fourier decomposition controls the remaining interacting variables. Their sum stays bounded after its explicit pressure and twist factors are removed. This spectral period is distinct from the physical cylinder circumference $2n$.

Section 7 uses the zero-free disk and six lattice rotations to localize polygons around one face: their planar partition function at radius R has exponent $1/12$. The finite exploration identity in Section 8 converts this nesting mass into vertex-visit mass. Summing over h layers and using horizontal translation gives mixed-exit first-length mass $h^{1+1/12+o(1)}$. In the hexagon, dividing visit mass by the free-boundary normalizer $R^{3/4+o(1)}$ gives the visit probability $R^{-2/3+o(1)}$ and hence the stated mean.

For the strip, the mixed mass could still be carried by returning arches. Section 9 compares intersecting bridge–arch pairs with bridge–bridge pairs through a marked tripod. A translation argument supplies sufficiently many bridge hits on each deep arch, so bridges retain the same first-length exponent. Only then do we divide by b_h , obtaining

$$\mathbb{E}_h |\gamma| = \ell_h / b_h = h^{13/12+1/4+o(1)} = h^{4/3+o(1)}.$$

2 Planar and cylindrical setup

We begin with the positive path bounds that justify all later limits of finite cylinders. The same flux identity also verifies the critical normalization of the vertex weight.

Use the honeycomb lattice dual to the equilateral triangle tiling of side 1. Paths can begin/end at mid-edges ("ports"). A triangle visited has weight

$$x_* = (2 \cos \lambda)^{-1}, \quad \lambda = \pi/8.$$

Otherwise its weight is 1. Edges to/from terminal midpoints are traversed halfway; the paths counted are simple (in particular they cannot reverse immediately along an edge, nor repeat a lattice vertex (triangle)). Absolute SAW weights in what follows are products, without dividing out any partition function unless specified. A strip of height h (in layers, physical height $h\sqrt{3}/2$) has bottom and top on parallel tiling lines. Write A_h, b_h for the total weights from a given bottom port to bottom and to top respectively, inside the strip; the trivial path is excluded. All translated and reflected versions have the same totals.

On cuts formed by tiling lines we stop at the port just before leaving the allowed triangles. Centers of triangles are never on these cuts. The half-edge from a center up to (but not including) a port is inside that triangle, and the port is on its side but not at a tiling vertex. Thus a tiling line cutting a walk meets it through a perpendicular edge, not at a turn, and different nonparallel cuts never share a port.

The finite-domain identity below controls absolute path weights because convexity keeps all its phases in a fixed open half-plane.

Lemma 2.1 (Critical flux). *For an outer boundary port of a finite convex domain bounded by tiling lines, start a path inwards and let it exit anywhere. Then*

$$\sum_{\gamma} x_*^{|\gamma|} e^{i\sigma W(\gamma)} = 1, \quad \sigma = 3/8$$

where the sum is over the nontrivial walks to exit, $|\gamma|$ counts vertices, and W is total turning angle along honeycomb edges. Every such walk has $|W| \leq \pi$, so its real phased weight is at least $\cos(\sigma\pi)x_*^{|\gamma|}$.

Proof. At each interior lattice vertex take the sum of the phased weights of paths to its adjacent ports signed + when pointing away from that vertex and - when pointing towards it there (the initial empty path at the source points towards the interior). Sum also over vertices; internal terms cancel. Locally, paths pointing in towards an unused vertex each have two extensions out, exactly balancing them ($2x_* \cos(\sigma\pi/3) = 1$).

The remaining negative terms occur when the vertex was used earlier; the tail describes one of the two ways round a polygon coming back towards that vertex on its unused edge. Pair them by reversing the traversal after the stem, thus changing the terminal port but keeping the same number of vertices. The stem lies outside the polygon, since it connects to the source without crossing it. Relative to the end of the stem just before the vertex, the two extra turning angles are $\pm(2\pi - 2\pi/3)$: on a positively oriented traversal the turn at the vertex would be $+\pi/3$ (edge of the stem outside), and is replaced by $-\pi/3$. Their phases therefore cancel. This uses only trivalence and the polygon turning-angle theorem.

Paths to the outer boundary are simple arcs in a convex polygon, so $|W| \leq \pi$: close them by the counterclockwise boundary arc and subtract from 2π its intervening corner turns and the two right angles. In particular all real parts compare to absolute weights by the constant $\cos(\sigma\pi) > 0$. $\square$

Arches to the right from a horizontal bottom side have $W = -\pi$, those to the left $W = \pi$, and opposite-side exits have $W = 0$. Passing from finite convex domains to an infinite strip gives the following normalization and preliminary bounds.

Corollary 2.2 (Strip flux and crude bridge bounds). *The strip masses A_h, b_h are finite and satisfy*

$$b_h + \cos(\sigma\pi)A_h = 1. \quad (2.1)$$

Moreover,

$$A_{h+1} - A_h \leq x_*^{-1}b_{h+1}^2, \quad \frac{c}{h+1} \leq b_h \leq 1.$$

Proof. Cut the strip off at great lateral distances by tiling lines whose outward normals are rotated by $\pm\pi/3$ from vertical upward. Lemma 2.1 first gives finiteness of A_h, b_h . To obtain equality in (2.1), continue each side exit from its port into the part cut away, strictly increasing both height and outward projection, by a zigzag of vertical up and upward-outward edges to the top in the untruncated strip. The boundary edge crossing the cut is itself upward-outward. This costs a bounded factor at fixed h , without repetition or unbounded multiplicity, since the first hit of the cut recovers the original exit. The weight of bridges so obtained tends to zero by finiteness, so the side-exit mass vanishes.

An arch visiting the last layer uses a vertex immediately below a top port: it enters the layer by the unique downward edge of a lower triangle there, and next goes to an upper triangle. Split there into two bridges sharing that vertex and ending at that port, and drop their avoidance. The first bridge starts at the fixed source; for each endpoint of the first, the reverse of the second can go to

any bottom port, again with total weight bounded by b_{h+1} . The shared vertex accounts for the factor x_*^{-1} , proving the difference bound. By (2.1),

$$\frac{1}{b_{h+1}} - \frac{1}{b_h} \leq x_*^{-1} \cos(\sigma\pi) \frac{b_{h+1}}{b_h} \leq x_*^{-1} \cos(\sigma\pi).$$

Summation gives the lower bound; the upper bound follows directly from the strip identity. $\square$

Criticality of the weight. The unfolding argument follows the bridge-decomposition method of Hammersley and Welsh [16], as used on the honeycomb lattice in [6, Section 3]. We include the geometric normalization needed for our ports. A vertex-to-vertex weak bridge of positive span (intermediate heights between those of its endpoints inclusively) can be extended at start and end to bracketing layer cuts using only strictly more extreme vertices, at most one extra vertex per end. Hence by the strip upper bound the weighted sum for each span and fixed start is uniformly bounded.

Count vertex SAWs of length L from a fixed start by splitting at a lowest vertex and considering the two weak half-space walks from there. On each, cut at the successive last maximum, last minimum of the remainder, and so on. This gives weak bridges of alternating sign with decreasing positive spans (the first two can be equal), in units of one third of a layer, with sum at most $2L$ per half; there are no zero-height steps. There are $O(\sqrt{L})$ pieces and $\exp(O(\sqrt{L} \log(L+1)))$ span lists. One can enumerate the two walks relative to the splitting vertex (two translation types suffice) and then the translate back to the given start is fixed. Thus multiply the uniform bounds allowing also a bounded weight factor per joint. SAW counts have exponential rate at most $1/x_*$.

A strictly smaller rate would make the vertex SAW susceptibility (weighted sum over all lengths at x_*) finite, hence by trimming port half-edges would make $\sum b_h$ finite, contradicting the bridge lower bound. The rate exists, e.g. by submultiplicativity and lattice transitivity. Thus the weight x_* is critical.

A flat cylinder identifies horizontal translates by an integer $N > 0$. The following fixed-width estimate will justify the row-transfer limits. Its constants may depend on N .

Lemma 2.3 (Fixed-width cylinder convergence). *Fix N . Let $b_h^{(N)}$ and $A_h^{(N)}$ denote the cylindrical versions of the strip masses from a fixed bottom port, with terminal port summed on the top and bottom cut respectively. Then $\sum_h b_h^{(N)} < \infty$, the limit $A_\infty^{(N)}$ is finite, and there are $C_N, c_N > 0$ such that*

$$b_h^{(N)} + A_\infty^{(N)} - A_h^{(N)} \leq C_N e^{-c_N h}.$$

The same bounds apply when a terminal port is prescribed instead of summed.

Proof. A cylindrical bridge lifts to a plane strip bridge. Cut strip bridges at each interior horizontal layer line crossed just once. The weights of positive-height indecomposable pieces summed over all heights in the plane, from a fixed bottom start, total at most 1 by concatenation and $b_h \leq 1$. More explicitly, all sequences concatenate freely by translations. If the total restricted to some finite height bound H exceeded 1, their weight after k concatenations would grow exponentially in k , but would be bounded by $\sum_{h \leq kH} b_h \leq kH$.

The total mass of pieces whose projections are simple on the cylinder is strictly less than 1: already in a single layer one can take a bottom-to-top bridge by following the internal zigzag arbitrarily far horizontally, and some such pieces repeat a vertex after projection. Summing the resulting geometric series bounds the total cylindrical bridge mass over all heights, after dropping avoidance between distinct pieces. Same-boundary arc weights are bounded by lifting and the uniform bound on planar arches.

It remains to strengthen summability to exponential decay. The sequences $b_h^{(N)}$ and $A_{h+1}^{(N)} - A_h^{(N)}$ have rational generating functions. Reveal layer by layer, retaining the occupied ports and the connectivity of dangling ends and prescribed terminals; keep only diagrams of paths, rejecting a closed cycle as soon as it forms. The update has finitely many states and constant coefficients, including periodic identification in each layer. A complete arc below the current line can be retained in a completed state. This counts single arcs with fixed or free terminal along the final cut, and also confined arcs starting and ending on the first cut. Translation by a lattice period in the layer direction identifies successive updates.

These sequences are nonnegative and summable. Rationality and absolute summability exclude poles on or inside the unit circle, so both sequences decay exponentially. Summing the second sequence over heights at least h gives the asserted same-boundary tail. Prescribing a terminal only reduces these positive sums. $\square$

3 Row matrices and polynomial vacua

The algebra has two outputs. Commuting transfer rows and their simple physical eigenline will supply the boundary susceptibility in Section 5. A polynomial representative of that eigenline will supply the polygon pairing. We first establish the local relations that both constructions use.

The rhombus weights are the dilute-loop weights of [24, Eq. (1.2)], divided by the empty weight. Related dilute Yang–Baxter operators admit the two-colour braid–monoid formulation of Grimm and Pearce [15, Sections 4.2, 4.5–4.6]. We specify the gauge here and verify the local and transfer identities needed below.

The checked tile and its rank reductions

Write $q = e^{i\lambda}$, $t = q^2$, $T = t^3$, $V = \mathbb{C}^{\{0,+1,-1\}}$. We denote standard tensors by lists of spins in kets. The charge of a tensor is the sum of its spins. Transpose throughout this construction involves no conjugation. Put $n_0 = \sqrt{2} - 1$ and define a vector C and a map $I : V \rightarrow V^{\otimes 2}$, whose columns are I_h :

$$C = |00\rangle + \sum_{h=\pm 1} q^{5h} |h, -h\rangle, \quad I_0 = |00\rangle + x_* \sum_{h=\pm 1} q^{4h} |h, -h\rangle, \quad I_h = x_*(q^h |h, 0\rangle + q^{-h} |0, h\rangle).$$

Direct multiplication gives

$$I^t I = n_0 1_V, \quad C^t C = -n_0, \quad C^t I = 0.$$

Thus $P_1 = II^t/n_0$, $P_0 = -CC^t/n_0$, $P_2 = 1 - P_1 - P_0$ are mutually orthogonal idempotents for this bilinear form. The *checked* matrix is

$$F(z) = \sum_{j=0}^2 r_j(z) P_j, \quad (r_0, r_1, r_2) = (a, b, ab), \quad a = \frac{t^2 - z}{t^2 z - 1}, \quad b = \frac{T - z}{Tz - 1}.$$

It acts between ordered tensor slots of parameters u, w with $z = u/w$, and outputs in order w, u . It preserves charge, is symmetric, and satisfies

$$F(1) = 1, \quad F(z)F(1/z) = 1, \quad F(T) = CC^t = -n_0 P_0, \quad F(t^2) = II^t = n_0 P_1.$$

The last two specializations reduce two spin slots to a cup, or to one spin slot embedded by I .

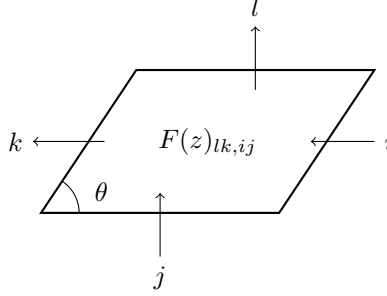


Figure 1: Port and tensor order for the checked rhombus matrix. The horizontal stream enters at the right and leaves at the left; the vertical stream enters below and leaves above. The arrows specify the positive spin directions, not a particular allowed path pairing. The local phase convention is imposed on each pairing separately.

Here is its geometric realization. On a rhombus of angle θ (bottom edge to side at the left, counterclockwise), set $v = 3\theta/8$, $z = e^{2iv}$, $D = \sin(2\lambda + v) \sin(3\lambda + v)$. Allowed unoriented pieces have these weights:

- empty 1; a single arc joining right to top or left to bottom has weight $u_1 = \sin(2\lambda) \sin(3\lambda - v)/D$;
- a single arc round either other corner has weight $u_2 = \sin(2\lambda) \sin v/D$; opposite-edge connection $o = \sin(3\lambda - v) \sin v/D$;
- a pair of disjoint arcs at the first two corners has weight $w_1 = \sin(3\lambda - v) \sin(2\lambda - v)/D$; at the other two, $w_2 = \sin(v - \lambda) \sin v/D$.

Orient all arcs, giving them the phase $e^{iW/4}$ for their turns (normal incidence, turns in $(-\pi, \pi)$). Horizontal-stream labels are i, k at right, left respectively (+ pointing leftwards across), vertical-stream j, l at bottom, top (+ towards top); vacant means 0. Multiply by the further (gauge) factor $e^{5iv(i-k)/3} = e^{5iv(l-j)/3}$. The resulting sum is $F(z)_{lk,ij}$. Indeed the blocks on $(|h0\rangle, |0h\rangle)$ have diagonals $u_1 e^{ihv}, u_1 e^{-ihv}$ and off-diagonal o ; on $|00\rangle, |+, -\rangle, |-, +\rangle$ the matrix is symmetric, first diagonal 1, the couplings to $|h, -h\rangle$ are $u_2 q^{2h} e^{ihv}$, the latter diagonals $(w_1 + t^{2h} w_2) e^{2ihv}$, and their mutual coupling w_2 ; on charge ± 2 it is w_1 . These entries also follow by substituting r_j above $(a, b = \sin(k\lambda - v)/\sin(k\lambda + v), k = 2, 3)$, giving $F = w_1 1 + x_*^{-2} o I I^t - x_*^{-1} \sin(2\lambda - v) \sin v C C^t / D$. At $v = \lambda$ the rhombus splits into two triangles and $u_1 = x_*, u_2 = o = w_1 = x_*^2, w_2 = 0$, exactly giving the disjoint honeycomb paths. Reflection gives the corresponding realization at $v = 2\lambda$. From the same entries the matrix at $z = 0$ has $k \geq i$, at $z = \infty$ has $k \leq i$. Its diagonal in stream order ($k = i, l = j$) at 0 has value 1 if either spin vanishes, t^5 if they are equal nonzero, t^3 if opposite nonzero; at infinity the inverse values.

For any of the nondegenerate real rhombi used geometrically the allowed pairings can indeed be drawn disjointly with these turns: perturb straight chords between ports by giving them short inward-normal segments at their endpoints (the chord points strictly into the first supporting line's half-plane and out towards the last line, so the two direction changes each have magnitude $< \pi/2$). In the two-triangle case the actual honeycomb connections have the same turns.

Lemma 3.1 (Local slide and interchange relations). *As rational identities, the maps from V and $V^{\otimes 2}$, respectively, to $V^{\otimes 3}$ satisfy*

$$F_{12}(z)F_{23}(Tz)(C \otimes 1) = 1 \otimes C, \quad F_{12}(z)F_{23}(t^2 z)(I \otimes 1) = (1 \otimes I)F(tz).$$

The checked matrices also satisfy the Yang–Baxter equation

$$F_{12}(y)F_{23}(zy)F_{12}(z) = F_{23}(z)F_{12}(zy)F_{23}(y).$$

Proof. We first prove the slides by projecting onto the three channels. Write $m = (1, 1+1/\sqrt{2}, 1/\sqrt{2})$, $c = (1/n_0, \sqrt{2}, 2+\sqrt{2})$. The needed small contractions are

$$(1 \otimes C^t)(C \otimes 1) = 1, \quad (1 \otimes C^t)(I \otimes 1) = (C^t \otimes 1)(1 \otimes I) = I^t, \quad (1 \otimes I)C = (I \otimes 1)C,$$

$$(1 \otimes I^t)(I \otimes 1)/n_0 = \sum m_j P_j, \quad 1 \otimes C - C \otimes 1 = (2 + \sqrt{2})(1 \otimes I - I \otimes 1)I.$$

All are direct products of the tensors above; e.g. one can use $C_{ab} = \delta_{a,-b}q^{5a}$ and $I_{ab,h} = \delta_{h,a+b}J(a,b)$, with $J(0,0) = 1, J(h,-h) = x_*q^{4h}, J(h,0) = x_*q^h, J(0,h) = x_*q^{-h}$ for $h \neq 0$ (invalid labels zero); the last identity reads

$$\delta_{ah}C_{bc} - C_{ab}\delta_{ch} = x_*^{-2}\delta_{a+b+c,h}[J(b,c)J(a,b+c) - J(a,b)J(a+b,c)].$$

Similarly $(1 \otimes I^t)(I \otimes 1) = x_*^2(1 - CC^t) + II^t$ reduces entrywise at $a+k = h+c$ to

$$J(a, h-a)J(h-a, c) = x_*^2(\delta_{ah} - \delta_{h,-c}q^{5(a+h)}) + J(a, k)J(h, c),$$

using $q^8 = -1, q+q^{-1} = x_*^{-1}$. Consequently $P_{j,12}(1 \otimes C) = c_j P_{j,12}(1 \otimes I)I$ (first project onto 0 and 1, then subtract). Project the first slide onto $P_{j,12}$; the scalar required, relative to its right side, is

$$r_j(z)[- \delta_{j0}n_0r_2(Tz) - (r_0 - r_2)(Tz)/n_0 + (r_1 - r_2)(Tz)/(n_0c_j)] = 1.$$

For the second slide sandwich between $P_{j,12}$ and P_k on the source. Relative to $P_{j,12}(1 \otimes I)P_k$ the multipliers on the two sides are

$$r_j(z)[\delta_{j1}r_2(t^2z)/m_k - \delta_{k1}c_j(r_0 - r_2)(t^2z) + m_k(r_1 - r_2)(t^2z)] = r_k(tz)$$

whenever $j = 1$ or $k = 1$ or $j = k = 2$. The scalar equalities are substitutions of a, b above: with $p = t^2$ use the common denominator $(pu - 1)(Tu - 1)$ for $(r_2, r_0 - r_2, r_1 - r_2)(u)$, with numerators $(p - u)(T - u), (p - u)(T + 1)(u - 1), (T - u)(p + 1)(u - 1)$. At $u = Tz, pz$ the denominators are $T(p - z)(T - z), (1 + z)(1 + tz)$ respectively; $a(tz) = (p - tz)/(Tz - 1), b(tz) = -(T - tz)/(z + 1)$. In the remaining cases the tensor factor vanishes by the contractions. This proves the slides. Inverse and transpose slides follow as well, using unitarity as rational identities.

Explicitly, denoting the three numerators just given by D_2, D'_0, D'_1 , expansion using $p^2 = -1, t = (1 + p)/\sqrt{2}, T = (p - 1)/\sqrt{2}$ reads (indices ordered 0, 1, 2)

$$[-\delta_{j0}n_0D_2 - D'_0/n_0 + D'_1/(n_0c_j)](Tz) = T \begin{pmatrix} (T - z)(pz - 1) \\ (p - z)(Tz - 1) \\ (pz - 1)(Tz - 1) \end{pmatrix}_j,$$

$$[\delta_{j1}D_2/m_k - \delta_{k1}c_jD'_0 + m_kD'_1](pz) = \begin{pmatrix} * & -t(1 + tz)(pz - 1) & * \\ -p(1 + z)(t - z) & p(p - z)(Tz - 1) & T(p - z)(t - z) \\ * & p(pz - 1)(Tz - 1) & T(t - z)(pz - 1) \end{pmatrix}_{jk}$$

where $*$ entries are not needed.

For Yang–Baxter, clear the denominators and interpolate in z . For this use of degeneration values in dilute-loop Yang–Baxter relations, compare [18, Section 4]. Both sides have degree at most four, and for generic y the five points

$$1, \quad T, \quad t^2, \quad T/y, \quad t^2/y$$

are distinct. Equality at $z = 1$ is immediate. At $z = T, t^2$, the z factor is EE^t , with $E = C, I$, respectively; sliding E on the left and E^t on the right gives equality. Here the reversed-order slide before transposing is the inverse slide at $1/(zy)$. The cases $zy = T, t^2$ follow from the preceding cases at $(zy, 1/y)$ by unitarity. This proves the rational identity. $\square$

Twisted rows and the physical eigenline

For $X = (x_1, \dots, x_N)$ and $\kappa \neq 0$, bring an auxiliary slot of parameter z past the sites in their listed order and trace its output against its input, weighting the identified spin h by κ^h . This defines the row transfer $\mathcal{T}_\kappa(z; X)$ on $V^{\otimes N}$. Explicitly, for input spins $j_1, \dots, j_N$ and output spins $l_1, \dots, l_N$,

$$[\mathcal{T}_\kappa(z; X)]_{l,j} = \sum_{i_0, \dots, i_{N-1} \in \{0, \pm 1\}} \kappa^{i_0} \prod_{a=1}^N F(z/x_a)_{l_a i_a, i_{a-1} j_a}, \quad i_N = i_0.$$

Thus site order follows the horizontal auxiliary stream, from right to left in the tile convention of Figure 1.

Lemma 3.2 (Row relations). *At fixed twist, the row transfers commute in their row arguments. Exchanging adjacent sites by F intertwines the corresponding rows. At generic sites, and hence meromorphically,*

$$\mathcal{T}_\kappa(x_j) \mathcal{T}_\kappa(Tx_j) = 1, \quad \mathcal{T}_\kappa(x_j) \mathcal{T}_\kappa(t^2 x_j) = \mathcal{T}_\kappa(tx_j).$$

For adjacent sites (x, Tx) , insertion of C intertwines the row with the row omitting those sites. For $(x, t^2 x)$, insertion of I intertwines it with the row having the single site tx . On charge zero, $\mathcal{T}_\kappa(0) = (1 + \kappa + \kappa^{-1})1$.

Proof. For commutation, insert a generic invertible crossing of two auxiliary lines, move it through the sites by Yang–Baxter, and remove it in the trace; charge conservation preserves the product twist. The same identity intertwines a site exchange.

For the two row fusion identities, take the indicated auxiliary pair in that order. The second auxiliary crosses each site first. The slide preserves its embedded C or I subspace throughout, and at the site x_j the rank reduction followed by a checked identity sends the whole auxiliary space into that subspace. The twisted trace is therefore the fused trace: the product twist on C is 1, and on the I image it is the single twist under the identification. The inverse slides give the corresponding statements for a pair of sites. Finally, at row argument zero, triangularity forces the auxiliary spin to remain constant around the trace. The stated diagonal values of $F(0)$, together with charge zero, give $1 + \kappa + \kappa^{-1}$. $\square$

Lemma 3.3 (Physical eigenline). *At $X = (1, \dots, 1)$, $z = t$, and $\kappa = \pm i$, powers of the row transfer converge exponentially to a rank-one projection whose empty-to-empty entry is 1. In particular, eigenvalue 1 is simple and all other eigenvalues have modulus less than 1. The same conclusions hold at the reflected honeycomb point $z = t^2$.*

Near either set of physical sites and twist, the rows have a common analytic eigenline of charge zero. Its eigenvalue is rational in the row argument and analytic in the other data near regular arguments. At $\kappa = \pm i$ this row eigenvalue is identically 1.

Proof. The empty basis tensor has all spins zero. At the homogeneous point, closed loops cancel after orientation summation: contractible loops have turns $\pm 2\pi$, with trace twist net 1, while simple essential loops have winding ± 1 , trace twist $\kappa^{\pm 1}$, and total turn zero. The latter total turn follows from the exterior-angle sum on the annular region between the curve and a distant parallel geodesic, which is zero by triangulation and Euler’s formula. The winding assertion also follows by identifying the cylinder with a punctured plane. Gauge factors telescope, so only prescribed boundary-to-boundary arcs remain.

At least one crossing arc gives an exponentially small contribution by the finite-width cylinder bounds in Lemma 2.3. With no crossing arc, the systems along the two cuts factor in the limit into

half-cylinder amplitudes. Their error is exponentially small, since a system preventing factorization must reach halfway across the cylinder height. There are only finitely many boundary pairings and orientations, and dropping mutual avoidance bounds each of them by products of arc weights. This gives the rank-one limit. Its empty-to-empty entry is 1 because every nonempty closed system cancels already at finite height. Reflection gives the same argument at the other honeycomb realization.

Finite-dimensional perturbation of $\mathcal{T}_\kappa(t)$ now gives a simple analytic eigenline near the physical data. Commutation makes it a common eigenline, and charge conservation keeps it in charge zero. For fixed sites and twist its row eigenvalue is rational in z .

To identify this eigenvalue at $\kappa = \pm i$, first take real nearby spectral angles $z = e^{2iu}$, $x_j = e^{2i\alpha_j}$. Lay the rows using unit rhombus side vectors with directions $8u/3$ and bottom vectors with directions $8\alpha_j/3$. The slightly zigzag spatial boundary repeats by translation and projects monotonically onto the direction perpendicular to the row side vector, so the rows do not overlap. Loop turns and twists still cancel. The gauge factors telescope because $v = u - \alpha_j$ and $i - k = l - j_{\text{spin}}$ at every crossing: the u term sums along auxiliary strands and the site-angle term along site strands. Hence every empty-to-empty power remains 1.

Analyticity extends this equality to complex nearby data. With z also near t , all other eigenvalues remain strictly inside the unit disk and the empty entry of the near-1 projection is nonzero. The constant power entry therefore forces the distinguished eigenvalue and that projection entry to equal 1. Rational continuation in z finishes the proof. $\square$

3.1 Polynomial vacuum

Polynomial exchange and fusion constructions occur in the dense and dilute $O(1)$ models [5, 11, 12]. We construct the present normalized vacuum directly by polynomial interpolation and determine its empty component. Its Pfaffian formula supplies the denominator of the polygon observable.

Proposition 3.4 (Polynomial vacuum). *Use twist $\zeta = t^2 = i$. Let $\rho X = (x_N, x_1, \dots, x_{N-1})$, and Π_κ the corresponding rotation of spin slots with factor κ^{h_N} from the last (now first) spin. We construct a charge-zero vector polynomial $\Psi_N(X)$ with*

$$F_{i,i+1}(x_i/x_{i+1})\Psi(X) = \Psi(s_i X), \quad \Pi_\zeta \Psi(X) = \Psi(\rho X)$$

(s_i adjacent transposition), homogeneous of degree $\binom{N}{2}$, with degree at most $N - 1$ in each variable. For a spin component $\Psi_{N,h}$, the refined bounds are

$$h_j = +1 \implies \deg_{x_j} \Psi_{N,h} \leq N - 2, \quad h_j = -1 \implies x_j \mid \Psi_{N,h}.$$

It has the adjacent-pair reduction

$$\Psi_N(\dots, a, Ta, \dots) = s_C(a; Y) \iota_C \Psi_{N-2}(Y), \quad s_C(a; Y) = (1 + T)a \prod_{x \in Y} (x - t^5 a)(x - t^6 a),$$

where Y is the remaining list and ι_C inserts a C tensor in the indicated positions; and at last parameter zero its projection onto last spin zero is $(\prod_{j < N} x_j) \Psi_{N-1}$.

Proof. We induct on N . The construction has five steps: prescribe cup values and interpolate; prove the spectator degree bounds; propagate all cup reductions; prove the last exchange; and prove cyclicity. The refined bounds are the reason the last two discrepancies vanish.

The initial vectors are $\Psi_0 = 1$ and $\Psi_{1,0} = 1$, with the other size-one components zero. At sizes two and three, the empty components are

$$S_2(a, b) = a + b, \quad S_3(a, b, c) = \sum_{i \neq j} x_i^2 x_j + 2\sqrt{2} abc, \quad (x_1, x_2, x_3) = (a, b, c).$$

At size three, components with first pair $(h, -h)$ and last spin zero are

$$f_h(a, b; c) = 2 \sin \lambda t^h L_h(a, b) P(a, b; c),$$

$$L_+(a, b) = b, \quad L_-(a, b) = a, \quad P(a, b; c) = (c + ta)(c + b/t).$$

At size two omit P . Cyclicity specifies the remaining components. We verify these initial cases at the end of the proof.

Cup values and interpolation. Assume the proposition at lower sizes and let $N \geq 4$. Set $Y_i = (x_1, \dots, \hat{x}_i, \dots, x_{N-1})$. To prescribe the value at $x_N = Tx_i$, insert C in the parameter list $(x_1, \dots, x_i, Tx_i, x_{i+1}, \dots, x_{N-1})$, then carry its second slot to the end. With rightmost factor acting first, this is

$$V_i = s_C(x_i; Y_i) F_{N-1, N}(Tx_i/x_{N-1}) \cdots F_{i+1, i+2}(Tx_i/x_{i+1}) \iota_{C, i} \Psi_{N-2}(Y_i).$$

The product is empty when $i = N - 1$. At the crossing with x_d , the two denominator factors are proportional to $(x_d - t^5 x_i)(x_d - t^6 x_i)$, already present in s_C . Thus each V_i is polynomial.

These prescriptions are covariant under exchanges among the first $N - 1$ sites, carrying the marked value with its site. An exchange away from the mark uses lower-size exchange and Yang–Baxter. At the mark, the local identity is the cup slide

$$F_{12}(a/b) F_{23}(Ta/b)(C \otimes 1) = 1 \otimes C :$$

it moves the cup based at a past its neighbor b , without changing $s_C(a; Y)$. The inverse slide handles the opposite order; Yang–Baxter moves a distant exchange into this local configuration.

For last spin plus, interpolate the $N - 1$ values with degree $N - 2$. For last spin minus, interpolate the value divided by x_N with that degree. For last spin zero, add the prescribed zero node and use degree $N - 1$. More explicitly, put

$$\ell_i = \prod_{\substack{d < N \\ d \neq i}} \frac{x_N - Tx_d}{T(x_i - x_d)}.$$

The multiplier of V_i is ℓ_i for last spin plus and $x_N \ell_i / (Tx_i)$ otherwise. In last spin zero add $\prod_{d < N} (x_d - x_N/T) \Psi_{N-1}$.

The apparent poles at $x_i = x_d$ cancel: coincident nodes have identical prescriptions by exchange covariance. Indeed exchanging equal parameters through intervening slots is the identity by unitarity, initially at generic other parameters. The only other possible poles, the simple poles at $x_i = 0$, cancel because $x_i \mid V_i$. The result is polynomial, with total degree $\binom{N}{2}$ by homogeneity. Interpolation also proves all exchanges among the first $N - 1$ sites.

Spectator bounds. Fix $j < N$. For V_i with $i \neq j$, the lower-size bound and the quadratic factor in s_C give degree at most $N - 1$ in x_j . If slot j is crossed, then as $x_j \rightarrow \infty$ the crossing argument tends to zero. Its leading output spin at that site is no larger than the input spin, so an output plus spin retains the lower-size degree loss. At $x_j = 0$ the opposite triangularity preserves the required minus-spin zero. The Lagrange multipliers are bounded at both ends. The extra zero-node term has the same bounds directly from size $N - 1$.

The term V_j has larger raw degree because x_j is the cup parameter. Its orders and the two possible multipliers are

	$x_j \rightarrow \infty$	$x_j \rightarrow 0$
V_j	$O(x_j^{2N-3})$	$O(x_j)$
ℓ_j	$O(x_j^{-(N-2)})$	$O(1)$
$x_N \ell_j / (Tx_j)$	$O(x_j^{-(N-1)})$	$O(x_j^{-1})$

These give all unrefined bounds. There are only two extra cases to check. At infinity, if both site j and the final site have spin plus, the carried cup spin starts minus and cannot increase under the leading $F(\infty)$ transport. The leading coefficient is therefore zero, supplying one more degree of decay. At zero, if site j has spin minus and the final spin is not plus, the carried spin starts plus and cannot decrease under $F(0)$. Its leading term again vanishes, compensating for the possible simple pole in the multiplier. This proves both refined bounds at every spectator.

Compatibility of cup reductions. To prove the adjacent reduction at $x_{j+1} = Tx_j$ among the first $N - 1$ sites, check it at the last-variable interpolation nodes. Disjoint cup specializations commute by the slides and lower-size exchange and reduction. Their scalar factors commute as well: for two cup parameters a, d , their cross factor is

$$\prod_{y \in \{d, Td\}} (y - t^5 a)(y - t^6 a) = T^2 \prod_{r \in \{t^2, T, t^5, t^6\}} (d - ra).$$

The four roots occur in reciprocal pairs and have product one, so this expression is unchanged on exchanging a, d .

At the overlapping node $x_N = Tx_{j+1}$, both sides vanish: one has the x_N spectator factor, and the other the x_j factor in V_{j+1} , whose slot is not crossed. At $x_N = Tx_j = x_{j+1}$, the first crossing in V_j is a checked identity. It leaves the required cup and the lower-size exchange transport of the remaining spectator. At the zero node in spin zero, the scalar factors agree because $t^{5+6} = T$. Interpolation proves the adjacent reduction. Together with the exchanges already proved, this gives every specialization $x_k = Tx_i$, $i < k$, with the stated transport for nonadjacent slots.

The last exchange. Clear denominators in the remaining exchange discrepancy:

$$\mathcal{D} = (t^2 x_{N-1} - x_N)(Tx_{N-1} - x_N)[F_{N-1, N}(x_{N-1}/x_N)\Psi(X) - \Psi(s_{N-1}X)].$$

It vanishes on every cup hyperplane $x_k = Tx_i$, $i < k$, except possibly the pair $(N - 1, N)$. For an unaffected pair this is lower-size exchange after cup reduction. When the second member is exchanged, the cup transports acquire precisely that exchange or its inverse. Consequently $\mathcal{D} = QE$, where

$$Q = \prod_{\substack{i < k \\ (i, k) \neq (N-1, N)}} (x_k - Tx_i).$$

Each spectator x_j , $j \leq N - 2$, occurs in exactly $N - 1$ factors of Q , exhausting the degree bound on $\mathcal{D}$. Thus E is independent of every spectator. The plus-spin degree loss and minus-spin zero force all those spectator spins to be zero. For adjacent spectators, however, $Q(s_j X)/Q(X) = b(x_j/x_{j+1})$. Their exchange equation therefore forces $FE = bE$, so their pair lies in channel 1. A nonzero pure $|00\rangle$ does not belong to $\text{im } I$, and there are at least two spectators. Hence $E = 0$, proving full exchange.

Cyclicity. Lower-size cyclicity and the cup reductions show that

$$\Pi_\zeta^{-1}\Psi(\rho X) - \Psi(X) = \prod_{i < k \leq N-1} (x_k - Tx_i) E(X).$$

The specialized pair never straddles the rotation seam. The quotient has degree at most one in each of its first $N - 1$ variables, so write

$$E(X) = \sum_{\epsilon \in \{0,1\}^{N-1}} x_1^{\epsilon_1} \cdots x_{N-1}^{\epsilon_{N-1}} E_\epsilon(x_N).$$

By the refined bounds, bit zero permits spins $0, +$ and bit one permits spins $0, -$. Exchange of adjacent variables among these first $N - 1$ acts on E by

$$E(s_j X) = (aP_2 + P_1 + (a/b)P_0)_{j,j+1} E(X), \quad a = a(x_j/x_{j+1}), \quad b = b(x_j/x_{j+1}).$$

Claim 3.5. *A tensor polynomial multiaffine in at least three consecutive variables, with the bit-dependent spin support and exchange relation just stated, is zero.*

Proof of the claim. For an adjacent pair, the two uncanceled denominator factors of a/b , namely $t^2x_j - x_{j+1}$ and $Tx_{j+1} - x_j$, must divide P_0E . Their product has degree two in each variable, whereas P_0E is multiaffine. Hence $P_0E = 0$. The channel-1 part is symmetric. The channel-2 part is divisible by $x_{j+1} - t^2x_j$, with quotient independent of those two variables. Consequently, with all other bits fixed, the adjacent coefficients obey

$$E_{10} = (P_1 - t^2P_2)E_{01}.$$

This operator is invertible on channels 1 and 2.

It suffices, therefore, to kill coefficients with sorted bits. Equal adjacent bits require channel 1. With the permitted spins, their pair must be proportional to I_+ for 00 or I_- for 11. Three equal bits are impossible: a first pair proportional to I_h and either third spin $0, h$ would produce a forbidden second pair 00 or hh . For 0011 the four slots must factor as $I_+ \otimes I_-$. Their middle cup contraction is zero because $P_0E = 0$, but fixing the end spins $+, -$ makes that contraction nonzero unless the coefficient vanishes. For 001, fix the first spin plus and then zero. The cup contraction on the last two excludes third spin zero and then minus, killing the coefficient through the first-pair combination. The argument for 011 is the same with the ends reversed. These exhaust sorted binary strings of length at least three: either some bit occurs three times, both occur at least twice, or the string is 001 or 011. This proves the claim. $\square$

Apply the claim to the first $N - 1$ variables. Since $N \geq 4$, the cyclic discrepancy vanishes.

Verification of the initial cases. The size-two and size-three formulas above meet the cup, zero, and degree conditions directly. To check the first exchange by the explicit blocks of F , write S for the corresponding empty polynomial and substitute $e^{2iv} = a/b$ in the local weights. The scalar identities used are

$$\begin{aligned} P(b, a; c) &= w_1 P(a, b; c) + u_2 S / (2 \sin \lambda \sqrt{ab}), \\ P(c, b; a) &= u_1 \sqrt{a/b} P(c, a; b) + o P(b, c; a), \\ P(a, c; b) &= o P(c, a; b) + u_1 \sqrt{b/a} P(b, c; a). \end{aligned}$$

(with square-root branch corresponding to v); at size 2 only the first, with the two P 's replaced by 1. For the first identities compare coefficients, using

$$1 - w_1 = u_2 \cos v / \sin \lambda,$$

$$\cos(2\lambda - v) - w_1 \cos(2\lambda + v) = (u_2 / (2 \sin \lambda))(\cos(2v) + \sqrt{2}).$$

For the other two use $u_1 e^{iv} + o/t = u_1 e^{-iv} + ot = \sin(3\lambda - v) / \sin(3\lambda + v) = (b + ta) / (a + tb)$. The w_2 terms on the occupied pair cancel since $t^2 = i$; likewise the two contributions to the change of the empty component. Remaining exchanges follow by rotation not wrapping the pair. This verifies the bases, and finishes the construction. $\square$

The physical normalization and its empty component

The polynomial vector now determines the physical common eigenline. We identify its empty component, which will be the denominator of the separator pairing.

First,

$$\mathcal{T}_\zeta(z; X) \Psi(X) = \Psi(X).$$

At $z = x_N$ the last crossing is a checked identity. The row is Π_ζ followed by transport of its first parameter to the end, so the exchange and cyclic equations prove the equality there. Exchange gives it at all x_j , row fusion/inversion gives it at all Tx_j , and triangularity gives it at zero. After clearing row denominators, the degree is at most $2N$. The $2N + 1$ generic points $0, x_1, \dots, x_N, Tx_1, \dots, Tx_N$ therefore suffice.

Similarly $\Psi(X^{-1})^t$, with X^{-1} the entrywise reciprocal list, is a left eigenvector of eigenvalue 1 for twist ζ^{-1} . At $z = x_N$, reverse transport through the symmetric crossings has exactly the exchange arguments that move the last reciprocal parameter to the first; the final rotation cancels because $\Pi_{\zeta^{-1}}^t \Pi_\zeta = 1$. The other interpolation points follow as above by inverse and transpose exchanges.

Let S_N be the empty component, which is not identically zero by cup reduction, and put $\Omega = \Psi / S_N$. For generic sites near homogeneity, Lemma 3.3 identifies Ω with the lower half-cylinder state obtained from empty input far below, normalized to have empty component 1. Its reciprocal transpose is the analogously normalized left state at twist ζ^{-1} .

The symmetry of S_N requires this geometric normalization: it does not follow just by taking the empty component of exchange, since that component mixes with occupied pairs. Take real nearby angles with $\alpha_i > \alpha_{i+1}$ in the earlier embedding. From left to right the upper boundary has site steps of directions $8\alpha_{i+1}/3, 8\alpha_i/3$. Adding the rhombus above those steps exchanges their order, entering from right and bottom with parameters x_i, x_{i+1} , without crossing the twist seam. With empty final output, all gauge factors telescope and all loops cancel even at finite height. The added crossing therefore takes one normalized half-cylinder state to the other with no scalar factor. Comparing with polynomial exchange gives $S_N(s_i X) = S_N(X)$. This extends polynomially from the open set of real angles.

Proposition 3.6 (Empty component). *Put $D(a, b) = a^2 + \sqrt{2}ab + b^2$ and $K(a, b) = (a^2 - b^2) / D(a, b)$. The symmetric empty component is*

$$S_N(X) = \prod_{i < j} \frac{D(x_i, x_j)}{x_i - x_j} \text{Pf}[K(x_i, x_j)]$$

where for odd size an extra last Pfaffian index has $K_{i, N+1} = 1$, $i \leq N$, corresponding to parameter zero. It satisfies

$$S_N(X^{-1}) = \frac{S_N(X)}{\prod_i x_i^{N-1}}$$

and the second fusion identity

$$S_N(a, t^2 a, Y) = \sqrt{2} ta \prod_{y \in Y} (y - t^5 a) S_{N-1}(ta, Y).$$

Proof. The displayed Pfaffian expression is a symmetric polynomial of the required degrees. Its cleared numerator is alternating, so the Vandermonde denominators cancel; the remaining possible poles cancel against the factors D . The bounded kernel at infinity gives the one-site degree bound. At a first pair (a, Ta) , only its own Pfaffian pairing survives and gives $s_C S_{N-2}$. A last zero parameter gives the zero-node prescription; if a dummy zero is present, only their mutual pairing remains. Symmetry and interpolation identify the expression with S_N . Inversion of all parameters gives the reciprocity formula. The zero-node identity and reciprocity also give the leading-coefficient identity

$$[x_N^{N-1}] S_N(X) = S_{N-1}(x_1, \dots, x_{N-1}).$$

For the second fusion identity, compare its two sides in a spectator variable. Cup reduction at another spectator is compatible on both sides because

$$(a - t^5 d)(a - t^6 d)(t^2 a - t^5 d)(t^2 a - t^6 d) = (d - t^5 a)(Td - t^5 a)(ta - t^5 d)(ta - t^6 d)$$

for a spectator pair d, Td . Each of the other $N - 3$ spectators therefore gives two roots, at ratios T and T^{-1} , by symmetry. The zero-node identity gives one more root. The leading-coefficient identity makes the top degrees agree, lowering the degree of the difference to at most $N - 2$. Thus its $2(N - 3) + 1$ roots suffice for $N \geq 4$; sizes two and three follow from the initial formulas. $\square$

Corollary 3.7 (Normalized reductions). *At generic parameters, the normalized states reduce by insertion of C at adjacent sites (a, Ta) , and by insertion of I at $(a, t^2 a)$, replacing those sites by the single site ta . When one parameter tends to zero or infinity, the spin-zero component at that slot tends to the reduced Ω , with scalar factor one.*

Proof. The cup assertion follows from the identical scalar reductions of Ψ and S_N . At $(a, t^2 a)$, the pole in the exchange equation forces channel 1. On that subspace the row intertwines the reduced row by Lemma 3.2. Its eigenvalue-1 space is a unique line for generic reduced sites: it is one-dimensional at homogeneous sites, and rational matrix minors preserve that property generically. The reduced site ta can be generic. Empty-component normalization fixes the insertion scalar to be one.

At zero, use rotation and the zero-node prescription. At infinity, the limit exists generically by the degree bounds and has no plus spin at the extreme slot. After projection to spin zero, triangularity at crossing argument zero makes the transfer the reduced row. Uniqueness again identifies the limit, and the nonzero generic leading coefficient of S_N fixes its normalization. $\square$

4 The balanced puncture observable

A contraction of the two vacua will count polygons separating two specified cylinder faces. We first establish that positive interpretation and then the interpolation data that characterize the contraction.

Divide the ordered sites into a first block A of size n and remaining block B of size $m = N - n$. Define

$$X_{n,m}(X; \kappa) = \Omega(X^{-1})^t D_{\kappa/\zeta}^A \Omega(X)$$

where the diagonal multiplies by $(\kappa/\zeta)^{\sum_{j \in A} h_j}$, for $\kappa \neq 0$.

Proposition 4.1 (Physical separator pairing). *For two homogeneous blocks of equal size n ,*

$$X_{n,n}(1, \dots, 1; \kappa) = F_n(\kappa + \kappa^{-1}).$$

Here the left side is the bilinear vacuum contraction just defined and the right side is the positive polygon polynomial from Section 1.

Proof. Work first at finite height with homogeneous sites. If $|0\rangle$ is the empty tensor, the lower and upper boundary states are

$$|u_H\rangle = \mathcal{T}_\zeta(t)^H |0\rangle, \quad \langle v_H| = \langle 0| \mathcal{T}_{\zeta^{-1}}(t)^H.$$

Every empty-to-empty entry is 1 by loop cancellation. Lemma 3.3 and the normalized left and right eigenvectors therefore give

$$|u_H\rangle \longrightarrow \Omega(1, \dots, 1), \quad \langle v_H| \longrightarrow \Omega(1, \dots, 1)^t.$$

The upper state is obtained by rows between the gluing cut and an empty upper boundary; its twist is ζ^{-1} . This identifies the reciprocal left eigenvector geometrically by the rank-one limit. At nonsingular physical data we always use these analytic states, including when their rational formulas are evaluated by a limit.

The finite contraction $\langle v_H| D_{\kappa/\zeta}^A |u_H\rangle$ glues matching occupied ports to make disjoint polygons. Internal gauge factors telescope under this gluing. It has three directed seams: from the lower boundary to f , with weight ζ ; from f to the upper boundary, with weight ζ^{-1} ; and from the block-division face f' to f along block A, with weight κ/ζ . See Figure 2. Positive crossing is toward the left of a directed seam. For an oriented simple polygon, let $L(a)$ indicate whether a lies on its left. The signed crossing number of a seam $a \rightarrow b$ is $L(a) - L(b)$, so its three seam factors multiply to

$$\zeta^{L(-\infty)-L(f)} \zeta^{-L(f)+L(+\infty)} (\kappa/\zeta)^{L(f')-L(f)}.$$

Here the two end indicators may be taken at the empty finite boundaries. For a contractible polygon oriented with interior left, the turn factor is $i = \zeta$ and both end indicators are zero. For an essential polygon oriented with upper end left, the turn factor is 1 and the end indicators are zero and one. In both cases the total factor is

$$\zeta^{1-L(f)-L(f')} \kappa^{L(f')-L(f)}.$$

The reverse orientation contributes its inverse. Equal indicators therefore give $\zeta + \zeta^{-1} = 0$, whereas unequal indicators give $\eta = \kappa + \kappa^{-1}$. Exactly the separating polygons survive, with their critical vertex weights.

At each finite height the contraction is consequently a polynomial in η with nonnegative coefficients. Its degree is at most n , since every separating polygon uses a distinct port of block A. The coefficients increase as the height grows. At any fixed positive η , convergence of the two boundary states bounds the contractions, and hence bounds every coefficient. Their finite limits are exactly the infinite-cylinder polygon sums. Passing to the limit coefficientwise proves the claimed identity for every $\kappa \neq 0$.

□

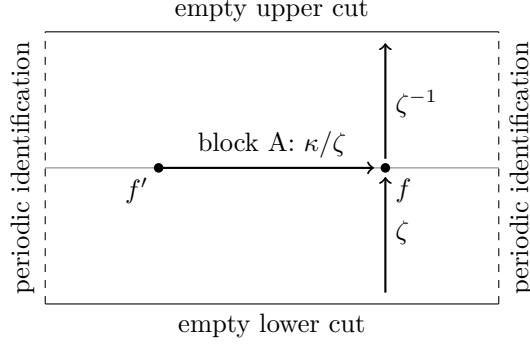


Figure 2: The finite cylinder, cut open along the two identified dashed sides. The middle line is the gluing cut; the arrows are directed twist seams, not polygon segments. Block A follows the site order from f toward f' , while its seam is directed from f' to f . The empty upper and lower cuts recede to the cylinder ends in the limit.

Remark 4.2 (Comparison of the scalar and physical pairings). For $N = 2n$, reversing both the numerator and Vandermonde conventions leaves the empty polynomial unchanged:

$$S_{2n}(X) = \prod_{i < j} \frac{D(x_i, x_j)}{x_j - x_i} \text{Pf} \left[\frac{x_j^2 - x_i^2}{D(x_i, x_j)} \right].$$

The prefactor changes by $(-1)^{n(2n-1)}$ and the Pfaffian by $(-1)^n$, whose product is one. This is exactly the scalar convention of [20, Section 5.2].

Multiplying our lattice coordinates by $\sqrt{3}$ gives honeycomb edge length one. It preserves polygon vertices, critical weights, and which polygons separate the marks, and therefore preserves F_n on the rescaled cylinder. For $n \geq 3$, the disk construction in [20, Section 5.3] identifies its physical pairing with this same polygon sum. These comparisons concern the scalar polynomial and the physical observable; the inhomogeneous vectors and their denominator estimates remain separate constructions.

4.1 Interpolation of the contraction

The following characterization will identify the finite color formula in Section 6.1. It applies at generic parameters; it does not assert that S_N is nonzero at every complex specialization. In the displays below we separate the site lists of the two blocks by a semicolon.

Proposition 4.3 (Interpolation of the pairing). *Fix $\kappa \neq 0$ and put $\eta = \kappa + \kappa^{-1}$. The family $X_{n,m}$ is uniquely determined among rational functions by the following properties. It is symmetric separately in its two blocks, and $S_N^2 X_{n,m}$ is polynomial of degree at most $2(N-1)$ in each site, where $N = n + m$. Within block A its reductions are*

$$\begin{aligned} X_{n,m}(a, Ta, A'; B) &= X_{n-2,m}(A'; B), \\ X_{n,m}(a, t^2 a, A'; B) &= X_{n-1,m}(ta, A'; B), \\ \lim_{a \rightarrow 0 \text{ or } \infty} X_{n,m}(a, A'; B) &= X_{n-1,m}(A'; B), \end{aligned}$$

The pair reductions require $n \geq 2$, and the extreme reduction requires $n \geq 1$; the analogous rules hold in block B with m in place of n . The initial value is $X_{0,0} = 1$, and at every balanced size $n = m > 0$,

$$X_{n,n}(a_1, \dots, a_n; Ta_n, \dots, Ta_1) = (1 + \eta)^n.$$

Proof. Separate block symmetry follows from exchange and its inverse, transpose symmetry of F , and charge conservation. Reciprocity of S_N shows that $S_N^2 X_{n,m}$ is the contraction of two polynomial vectors, each of degree at most $N - 1$ per site; this gives the degree bound.

For the extreme reductions, the complementary vanishing of nonzero spins on the two sides leaves only the spin-zero limit in Corollary 3.7. At an adjacent cup the right vector inserts C . In the reciprocal vector, the projection onto the same channel inserts $C/r_0(T)$: exchanging its reversed pair into cup order acts by $F(T) = r_0(T)P_0$. Contraction cancels the scalar because $C^t C = r_0(T) = -n_0$, and the block twist preserves the pair. At the other fusion, use $F(t^2) = n_0 P_1$ and $I^t I = n_0 1$ in the same way. The block twist on the image of I is the single-site twist. These are rational identities on the generic domain.

At the balanced specialization, successive nested cups in the right vector force spins h on A with opposite mates in B and amplitudes $\prod q^{5h}$. Rotate B to the beginning in the reciprocal vector. Its cups now face the other way, so in the original order its amplitudes are $\prod \zeta^h q^{-5h}$. Together with the block insertion, each pair contributes $\sum_{h=0,\pm 1} \kappa^h = 1 + \eta$. This gives the last specialization.

For uniqueness, suppose two families agree at smaller sizes, and let $A(X)$ be their difference at (n, m) , multiplied by S_N^2 . The zero and infinity reductions give respectively $x_i \mid A$ and $\deg_{x_i} A \leq 2N - 3$ at each site. The cup and second fusion, in both orders, give four factors for each intra-block pair. Thus

$$Q(X) = \left(\prod_{i=1}^N x_i \right) \prod_{P \in \{A, B\}} \prod_{\substack{i < j \\ i, j \in P}} \prod_{d \in \{2, 3, 5, 6\}} (x_j - t^d x_i) \text{ divides } A(X).$$

In a site of A this divisor has degree $4(n - 1) + 1$, leaving

$$\deg_{x_i}(A/Q) \leq 2N - 3 - [4(n - 1) + 1] = 2(m - n).$$

In a site of B the remaining degree is at most $2(n - m)$. An unbalanced difference is therefore zero. In a balanced case the quotient is a scalar; the nested-cup specialization fixes it to zero, since it is generically off the zero sets of both Q and S_N . Induction from $X_{0,0} = 1$ completes uniqueness. $\square$

5 The boundary exponent

We now determine the strip mass needed to normalize chordal walks. The row derivative reduces this problem to a positive polynomial-kernel estimate and a comparison of geometric tails.

Proposition 5.1 (Boundary exponent). *For critical bridges across h layers from a fixed bottom port to all top ports, $b_h = h^{-1/4+o(1)}$. The half-plane arch mass to the right is $1/(2 \cos(3\pi/8))$; its height and diameter tails both have exponent $-1/4$.*

The proof has three stages. A row derivative gives an exact excluded-arch mass. A positive polynomial-kernel estimate gives its power, and a two-direction flux comparison changes the cylinder exclusion into a strip-height tail. This argument uses the row relations and physical eigenline from Lemmas 3.2 and 3.3, independently of the polynomial vacuum and separator pairing.

5.1 Row susceptibility at infinity

Use even widths $N = 2m$. Let G_N be the half-plane mass, from a fixed bottom port, of arches with positive endpoint displacement $0 < d < N$ whose projections to the cylinder of period N are simple. The strip flux bound gives $G_N \leq 1/(2 \cos(3\lambda))$. We will express this deficit through one Taylor coefficient of a row eigenvalue.

For the common eigenline near twist $\kappa = i$, differentiate its row eigenvalue in $\eta = \kappa + \kappa^{-1}$ at zero. Call the answer in homogeneous data $H(\theta)$, with $z = e^{2iv}$, $\theta = 2v - 3\lambda$; we also write $H(z)$ when using the transfer argument. It tends to 1 as $z \rightarrow 0, \infty$ by triangularity. It is even: at real v near λ , empty-to-empty powers agree with those at $3\lambda - v$ by reflection of the unoriented loop sum (u_1, u_2 and w_1, w_2 interchange). The simple eigenvalue strictly dominates nearby at both parameters, so they agree there on that eigenvalue and then by continuation. Its denominator may be taken to be $(\cos \lambda + \cos \theta)^N$.

Before making sites homogeneous, transfer fusion gives for the derivative as a function of transfer argument z both relations

$$H(z) + H(Tz) = 0, \quad H(z) + H(t^2 z) - H(tz) = 0$$

at every $z = x_j$ (base eigenvalue 1). Thus the relations have zero jets of orders below N at the common site 1, by analyticity in sites and coalescence (all the denominators here regular locally; e.g. divide by $\prod(z - x_j)$ for nearby distinct sites and bound inside a small circle). In θ notation the site and its shifts t, t^2, T correspond to $-3\lambda, -\lambda, \lambda, 3\lambda$. By evenness the jets are

$$H^{(2j)}(3\lambda) = 0, \quad H^{(2j+1)}(3\lambda) = 2H^{(2j+1)}(\lambda) \quad (0 \leq j < m).$$

Represent the cosine-rational function as

$$H(\theta) = 1 + \int_{\mathbb{R}} e^{\theta p} \frac{\sinh(\lambda p)U(p^2) + p \cosh(\lambda p)V(p^2)}{\sinh(\pi p)} dp$$

with polynomials of degrees below m . Indeed, near the real angles used here,

$$\int_{\mathbb{R}} \frac{e^{\theta p} \sinh(\alpha p)}{\sinh(\pi p)} dp = \frac{\sin \alpha}{\cos \theta + \cos \alpha}.$$

Its successive even θ -derivatives and their α -derivatives give precisely a basis by increasing pole order. The elementary transform follows by taking differences of $\text{pv} \int e^{up} / \sinh(\pi p) dp = \tan(u/2)$. For real $|u| < \pi$, take the integral on $\mathbb{R} + i\epsilon$, $0 < \epsilon < 1$, shift that line by i , and get $-2ie^{iu}/(1 + e^{iu})$ by the pole at i . As $\epsilon \downarrow 0$, the line integral gives the principal value minus i , from $1/(\pi p)$ at zero.

Put $x = \lambda p$, $s = p^2$, and let μ be the measure on $s \geq 0$ pushed forward from

$$\frac{\cosh(3x) \sinh x}{\sinh(8x)} dp.$$

Define the test and trial spaces

$$\begin{aligned} \mathcal{F}_m &= \text{span}\{s^j, s^j p \tanh x : 0 \leq j < m\}, \\ \mathcal{G}_m &= \text{span}\{s^j, s^j p \coth x : 0 \leq j < m\}. \end{aligned}$$

The even functions of p here are regarded as functions of s , with their continuous values at zero. The jets say precisely that $R(s) = U(s) + p \coth x V(s) \in \mathcal{G}_m$ satisfies

$$\int_{[0, \infty)} [-R(s)] f(s) d\mu(s) = f(0) \quad (f \in \mathcal{F}_m). \quad (5.1)$$

This uses the identities

$$\cosh 3x = (2 \cosh 2x - 1) \cosh x, \quad \sinh 3x - 2 \sinh x = (2 \cosh 2x - 1) \sinh x.$$

At $z = 0$ the first Taylor coefficient is

$$[z]H = 2 \sin \lambda e^{-3i\lambda} R(-1), \quad (5.2)$$

by shifting the p contour upwards just past i at fixed $\Re \theta = 0$, $\Im \theta \rightarrow +\infty$: the pole there contributes $2 \sin \lambda R(-1) e^{i\theta}$ and the new integral is $o(|z|)$ by exponential decay.

There is also a path interpretation. Take a homogeneous finite stack of honeycomb rows with one further row of parameter v above, empty boundaries, and divide by the empty-to-empty value without the further row. This tends to the row eigenvalue uniformly near twist i , by strict spectral dominance and commutation. Its derivative at $\eta = 0$ counts a single essential loop using the further row. This follows by the same arc/turn embedding for angles near λ , hence identically. With empty top the loop is either purely horizontal in that row (weight o^N) or has horizontal intervals connecting bottom ports there, using equal positive numbers of u_1, u_2 and otherwise only factors o along intervals. Now

$$o = \frac{(1-z)(1-t^{-3}z)}{(1-t^2z)(1-Tz)}, \quad u_1 u_2 = o \sin^2(2\lambda) / [\sin(2\lambda + v) \sin(3\lambda + v)] \sim -2e^{5i\lambda} z.$$

Thus the only other contribution to $[z]$ is a single lower arch between distinct ports plus one horizontal connection in the further row. Such an arch lifts with endpoint displacement strictly below N in absolute value (a larger displacement interlaces its translate), and of the two horizontal connections exactly one closes it with essential winding. Conversely every arch counted by G_N contributes, with this choice of starting end unique for each unoriented cylinder arc. The reflected version has the same weight. Passing to the infinite lower stack therefore gives $[z]H = N([z]o - 2e^{5i\lambda} G_N)$. This passage on derivative and coefficient is also valid by spectral convergence, uniformly in the twist perturbation and in regular further-row arguments near $z = 0$. Combining $[z]o = 2e^{5i\lambda} / (2 \cos(3\lambda))$ with (5.2) gives the promised identity:

$$\frac{1}{2 \cos(3\lambda)} - G_N = \frac{\sin \lambda}{N} [-R(-1)]. \quad (5.3)$$

5.2 Positive kernel estimate

By (5.3), it remains to prove

$$-R(-1) = N^{3/4+o(1)}. \quad (5.4)$$

The two spaces in (5.1) are different, so that equation does not itself give a positive polynomial kernel. We first express its solution as a positive average of ordinary kernels over two pole sets. A comparison valid for every pole set gives the upper bound. For the lower bound, low poles occur with probability tending to one; on that event a smaller polynomial space admits the reverse comparison.

The partial fractions of $p \tanh x$ and $p \coth x$ are respectively $(2/\lambda)s \sum_a 1/(s+a)$ and $1/\lambda + (2/\lambda)s \sum_b 1/(s+b)$, with $a = (8l)^2$ at positive half-integers $l = 1/2, 3/2, \dots$, $b = (8l)^2$ at positive integers. Equivalently the all-pole products $\prod(1+s/a), \prod(1+s/b)$ equal $\cosh x, \sinh x/x$. For example integrate $f(w)/(w^2 - x^2)$ for $f = \tanh, \coth$ on expanding squares spaced away from poles (where f stays bounded); residues at $\pm x$ sum to $f(x)/x$, while the poles of f each have residue 1 for f . The products follow by integrating logarithmic derivatives from zero.

For subsets A, B of size m of the respective pole sets, write $\pi_A(s) = \prod_{a \in A} (1+s/a)$, similarly π_B , and $dw = d\mu/(\pi_A \pi_B)$. Let K_d^w denote the ordinary polynomial evaluation kernel through degree $d-1$: if P_j are orthonormal for w , then $K_d^w(u, v) = \sum_{j < d} P_j(u) P_j(v)$.

Lemma 5.2 (Positive pole representation). *The weights*

$$v_A v_B Z_N(w), \quad v_A = \Delta(A)^2 \prod_{a \in A} a^{1-N}, \quad Z_N(w) = \frac{1}{N!} \int \Delta(s_1, \dots, s_N)^2 \prod_i dw(s_i)$$

with Δ Vandermonde have a finite, positive sum over A, B , and hence define a joint probability law after normalization. For this law,

$$-R(-1) = \mathbb{E}_{A,B} \frac{K_N^w(0, -1)}{\pi_A(0)\pi_B(-1)} \quad (5.5)$$

and $\pi_A(0) = 1$, $\sin \lambda / \lambda \leq \pi_B(-1) \leq 1$. In particular all terms in the expectation are positive.

Proof. A determinant evaluating the test basis at N nodes factors as a nonzero signed constant times

$$\Delta(s_1, \dots, s_N) \sum_A \frac{v_A}{\prod_i \pi_A(s_i)};$$

the trial determinant has the corresponding expansion over B . To obtain these formulas, expand the last m rows by the partial fractions, discard polynomial-row terms from division, and antisymmetrize the pole choices. One Vandermonde in the poles comes from powers of poles, the other from the Cauchy–Vandermonde determinant with rows $1, s, \dots, s^{m-1}, 1/(s+a)$: clear denominators and evaluate at $-a$. Truncate pole sums first. After removing the common sign and Vandermonde, the scalar sums increase positively for nodes above -16 ; the determinant integrals also converge directly by exponential tails.

For column-valued lists f, g of the test and trial bases, set $M = \int f g^t d\mu$. Andréief’s identity [22, Eq. (1.3)] expresses $\det M$ as the integral of the product of the two evaluation determinants, divided by $N!$. Substituting their pole expansions gives a common nonzero constant times $\sum_{A,B} v_A v_B Z_N(w)$. This proves finiteness, positivity of the normalizing sum, and invertibility of M .

Equation (5.1) now gives $-R(-1) = g(-1)^t M^{-1} f(0)$. Its numerator with denominator $\det M$ is

$$\frac{1}{(N-1)!} \int \det f(s_1, \dots, s_{N-1}, 0) \det g(s_1, \dots, s_{N-1}, -1) \prod_i d\mu(s_i),$$

by expanding the final columns and then minors. Inserting the pole expansions again, the same cofactor identity for monomials gives $Z_N(w) K_N^w(0, -1)$, with the remaining factors $\pi_A(0)\pi_B(-1)$ in the denominator. The common constants cancel in the ratio, proving (5.5). These integrals have nonnegative sign after removal of the common constant because the two external nodes lie below the integration nodes. Finally, $\pi_A(0) = 1$, and completing the product over all b -poles gives

$$1 \geq \pi_B(-1) \geq \prod_b (1 - 1/b) = \frac{\sin \lambda}{\lambda} > 0.$$

□

Reference weights and uniform comparisons. For weights on $s \geq 0$, zeros of the degree- d orthogonal polynomial are eigenvalues of compression of multiplication by s to degrees below d : the eigen-equation states orthogonality of $(s - \rho)P$. An increasing likelihood ratio shifts each zero up by min-max, since every Rayleigh quotient increases by monotone tilting. We also compare diagonal kernels by their extremal property,

$$K_d^w(u, u) = \sup_{\substack{\deg Q < d \\ Q \neq 0}} \frac{|Q(u)|^2}{\int |Q(s)|^2 dw(s)}.$$

Let ν_α be the pushforward to $s = p^2$ of $dy/\cosh(\pi y)$, where $y = \alpha p/\pi$ and $\alpha > 0$. Orthonormal polynomials of degree j in s have values, up to sign, given by coefficient $2j$ in

$$(1 - v^2)^{-1/2}((1 + v)/(1 - v))^{iy}.$$

Indeed integrating a product of two generating functions gives $1/(1 + uv)$ by $\int e^{iay} dy/\cosh(\pi y) = 1/\cosh(a/2)$ (shift contour by i); use just even indices. At $s = 0$ the magnitudes are of order $(j + 1)^{-1/2}$; at $s = -1$ and $r = \alpha/\pi \in (1/2, 1)$, of order $(j + 1)^{r-1/2}$. Indeed the coefficients needed there are those of $(1 - v)^{-1/2-r}(1 + v)^{r-1/2}$. The first factor has increasing coefficients of order $(k + 1)^{r-1/2}$ with ratio tending to 1 (binomial series); the second has summable coefficients with nonzero sum, giving the claim by convolution.

The all-poles lower bound for w comes from density $w_\infty(p) = x(2 \cosh 2x - 1)/\sinh(8x)$ before pushing forward. Its ratio to the reference density at $\alpha = 6\lambda$ is increasing on $p \geq 0$: up to a constant this ratio is $x(2 - 1/\cosh 2x)/(\tanh 2x + \tanh 6x)$. Removing poles only increases the upward tilt. Conversely w relative to reference $\alpha = 2\lambda$ has decreasing ratio since μ times $\cosh 2x$ has density proportional to $(2 \cosh 2x - 1)/\cosh 4x$, decreasing. It follows that

$$K_N^w(0, -1) \leq C\sqrt{\log(2N)} N^{3/4}.$$

(Indeed w dominates a constant times the 6λ reference uniformly, so the two diagonal kernel values by the supremum characterization are at most constant times the reference sums of squared coefficients; now use Cauchy–Schwarz.) Moreover, for every orthogonal polynomial P_j of degree j for any of the conditional weights w ,

$$1 \leq P_j(-1)/P_j(0) \leq C(j + 1)^{3/4}$$

by positivity and comparison of zeros with $\alpha = 6\lambda$.

The upper kernel bound holds for every pole set. The lower bound will use pole sets that contain all low poles; the next lemma shows that these account for asymptotically all of the probability in (5.5).

Lemma 5.3 (Low-pole saturation). *Fix $0 < \delta < 1/2$ and put $J = N^{1-\delta}$. Under the law of Lemma 5.2, let $\mathcal{E}_J$ be the event that A and B each contain every pole in their respective sets with index $l \leq J$. Then*

$$\mathbb{P}(\mathcal{E}_J) = 1 - o(1).$$

For either subset, the probability of missing such a pole is $o(1)$ uniformly conditional on the other subset.

Proof. Fix the other subset. If an index $j \leq J$ is missing, replace the smallest occupied $k \geq 2J$ in the chosen subset by j . Such a k exists for all large N , since the subset has $N/2$ elements and only $O(J) = o(N)$ indices lie below $2J$. For fixed k the replacement is injective. The ratio of old to new weight is

$$(k/j)^2 \prod_{l \neq k} \left(\frac{k^2 - l^2}{j^2 - l^2} \right)^2 \mathbb{E} \prod_{i=1}^N \frac{s_i + (8j)^2}{s_i + (8k)^2}$$

(the product is over other indices of that subset, and the expectation is under the new polynomial ensemble proportional to $\Delta^2 \prod_i dw_{\text{new}}(s_i)$). The first factors cost at most $\exp(CJ)(k/J)^{CJ}$. Larger indices than k cost nothing; by minimality of k , smaller occupied indices lie below $2J$, where

$$\sum_l [\log^+(J/l) + \log^+(J/|j - l|)] \leq CJ$$

by unit spacing, excluding j .

The expectation is the ratio of degree- N monic orthogonal polynomial values at $-(8j)^2, -(8k)^2$ for weight $w_{\text{new}}/(s + (8k)^2)$. Indeed, averaging $\prod_{i=1}^N (u - s_i)$ against a normalized Δ^2 -ensemble gives its monic orthogonal polynomial. Its integral against u^r , $r < N$, for the one-point base weight vanishes: it contains $\Delta(s_1, \dots, s_N, u)$, while antisymmetrizing the other factor $\Delta(s_1, \dots, s_N)u^r$ gives zero.

The extra denominator $s + (8k)^2$ preserves the decreasing likelihood ratio relative to $\nu_{2\lambda}$. The zeros are therefore no larger than those for this reference weight. Since $(s + (8j)^2)/(s + (8k)^2)$ increases in s , zero comparison bounds the polynomial ratio above by the reference ratio. Define

$$c_M(R) = [v^M](1 - v)^{-1/2-R}(1 + v)^{-1/2+R}, \quad M = 2N.$$

We have bounded the old-to-new weight ratio, uniformly in both subsets apart from j, k , by

$$\exp(CJ)(k/J)^{CJ} \frac{c_M(2j)}{c_M(2k)}. \quad (5.6)$$

We next show that (5.6) sums to $o(1/J)$ over $k \geq 2J$, uniformly in $j \leq J$. Coefficients increase positively with R , since the generating function contains $\exp(2R \operatorname{atanh} v)$. Replacing $2k$ by $2k - 1/2$ gives the lower bound

$$c_M(2k) \geq \binom{M + 2k - 1}{2k - 1}.$$

For $2J \leq k \leq N^{1-\delta/2}$, bounding the numerator on radius $1 - k/N$ gives $\exp(O(k))(CN/k)^{2j+1}$, while the denominator is at least $(N/k)^{2k-1}$. The logarithms of the replacement cost are $O(k)$, whereas

$$(2k - 1 - 2j - 1) \log(N/k) \gtrsim \delta k \log N,$$

which absorbs that cost and makes the sum negligible.

For $N^{1-\delta/2} < k < N$, the denominator already grows as $\exp(c_\delta N^{1-\delta/2} \log N)$, while costs and numerator, now estimated on radius $1 - J/N$, are at most $\exp(CJ \log N)$. Finally, for $k \geq N$, the binomial gives $\exp(cN)(k/N)^M$, sufficient also on summing to infinity. Injectivity for fixed k bounds the conditional probability of missing each j by $o(1/J)$. The union bound over $O(J)$ low indices, and then over the two subsets, proves the lemma without any independence assumption. $\square$

From saturation to the lower kernel bound. On $\mathcal{E}_J$, the weight ratio to w_∞ is bounded by $\exp(Cp^2/J)$, since only poles with indices above J can be missing. Fix $\alpha' < 6\lambda$ close to 6λ . On $|p| \leq \epsilon J$, choosing $\epsilon > 0$ small makes w bounded above by a constant times $\nu_{\alpha'}$. This pointwise comparison need not hold on the whole line. Put $d = \lfloor N^{1-2\delta} \rfloor$. What the kernel estimate requires, and what does hold uniformly for all saturated pole sets, is the bound

$$\int |Q(s)|^2 dw(s) \leq C \int |Q(s)|^2 d\nu_{\alpha'}(s) \quad (\deg Q < d). \quad (5.7)$$

To include the tails, expand Q in the reference orthonormal polynomials. Their diagonal kernel satisfies

$$|Q(p^2)|^2 \leq \|Q\|_{L^2(\nu_{\alpha'})}^2 \exp(C_r d + C_r |p|),$$

using the generating function on a small fixed radius r . The density of w before pushing forward is always at most $Ce^{-4\lambda|p|}$. Choose r small enough to leave exponential decay in $|p|$; the integral over $|p| > \epsilon J$ is then at most

$$C \exp(C_r d - c\epsilon J) \|Q\|_{L^2(\nu_{\alpha'})}^2.$$

Since $d = o(J)$, this proves (5.7) together with the comparison on the central interval.

The zeros of every P_j lie in $(0, \infty)$, so $P_j(0)P_j(-1) > 0$. The bound $P_j(-1)/P_j(0) \leq C(j+1)^{3/4}$ and the extremal kernel comparison from (5.7) therefore give, on $\mathcal{E}_J$,

$$K_N^w(0, -1) \geq C^{-1}d^{-3/4}K_d^w(-1, -1) \geq cd^{2\alpha'/\pi-3/4}.$$

Here the second inequality uses the reference sum of squared values at -1 . The event has probability $1 - o(1)$ by Lemma 5.3, and the denominator in (5.5) is bounded above and below. Together with the uniform upper bound, letting $\delta \downarrow 0$ and $\alpha' \uparrow 6\lambda$ proves (5.4).

5.3 Comparing the strip tails

The total half-plane arch weight to the right is now exactly $1/(2\cos(3\lambda))$: this was the upper bound by flux, and G_N tends to it. Equations (5.3) and (5.4) give

$$d_N := \frac{1}{2\cos(3\lambda)} - G_N = N^{-1/4+o(1)}$$

for even N . This is the mass excluded by displacement or failure of simple cylinder projection. We first transfer its exponent to a spatial maximum, and then use a second tilted direction to force the same exponent for height.

Two tilted flux identities. Use these positive-endpoint arches above a horizontal bottom from coordinate 0, and write their endpoint as $D > 0$, height maximum as Y , and

$$M_+ = \max(Ax + By), \quad M_- = \max(Ax - By), \quad A = \sqrt{3}/2, \quad B = 1/2$$

in planar coordinates. Write P for their unnormalized weighted law. At large positive levels r on the respective tiling grids,

$$P(M_{\pm} - AD \geq r) = q_{\pm}P(M_{\pm} \geq r), \quad q_{\pm} = \frac{\sin(\sigma(\pi - \phi_{\pm}))}{\sin(\sigma(\pi + \phi_{\pm}))}, \quad (\phi_+, \phi_-) = (\pi/3, 2\pi/3).$$

Indeed, in a large convex exhaustion add the cut $Ax \pm By \leq r$ to the domain. Subtracting the two identities of Lemma 2.1 puts the phased sum over old exits crossing that cut on the ray $e^{-i\sigma\phi_{\pm}}$ by convexity. Tiling cuts of differing directions never share ports. Old non-bottom exits vanish in weight in the exhaustion by the now saturated arch mass: bottom real parts increase to 1 in total, and all real parts control absolute weights. Bottom exits crossing the cut have phases $e^{-i\sigma\pi}$ to the right and $e^{i\sigma\pi}$ to the left, with the latter events corresponding by path reversal and translation to the shifted maxima above. Taking the imaginary part after rotating the ray to angle zero gives the identity. Note $0 < q_- < q_+ < 1$.

From cylinder exclusion to spatial extent. Put $U(r) = P(M_+ \geq r)$. First, $P(D \geq r) \leq r^{-1/4+o(1)}$ by d_N . Conversely, an arch excluded at period N has horizontal range at least of order N : either its displacement is at least N , or two vertices that differ by a nonzero multiple of N prevent simple projection. Its maximum horizontal coordinate or the negative of its minimum is therefore at least of order N . Reflection with reversal, replacing x by $D - x$, turns the far-left case into the far-right case and preserves the positive-endpoint arch law. Thus $U(r) \geq r^{-1/4-o(1)}$. Both comparisons use even periods on the appropriate side by constant factors.

Splitting on large endpoint and using the plus ray identity gives, for any fixed $\rho < 1$ close to 1,

$$U(r) \leq P(D \geq c_\rho r) + q_+ U(\rho r)$$

for large r , rounding lines on grids of bounded spacing. Choose ρ sufficiently close to 1 that $q_+ \rho^{-1/4} < 1$; iteration gives $U(r) = r^{-1/4+o(1)}$. In particular the weight of arches reaching distance r is at most $r^{-1/4+o(1)}$, by covering both sides using also the reflected maxima.

From spatial extent to height. To get a lower height tail, use $M_- \leq M_+ \leq M_- + 2BY$ and both ray identities. Again for any $\rho < 1$ close to 1,

$$U(r) \leq q_+^{-1} P(Y \geq c'_\rho r) + (q_-/q_+) U(\rho r).$$

Indeed start at a plus-grid level just below r , use its shifted-max identity, and except when Y is large compare this event to the analogous shifted minus event at a grid level just above ρr . Choose $(q_-/q_+) \rho^{-1/4} < 1$ and iterate for k steps with $\rho^k r$ of order $r^{1-\epsilon}$. The remainder $(q_-/q_+)^k U(\rho^k r)$ is negligible compared to $U(r)$ by the latter's exponent. The sum of the height-tail terms is at most a constant times $P(Y \geq cr^{1-\epsilon})$, by monotonicity and the geometric coefficients. Hence

$$P(Y \geq cr^{1-\epsilon}) \geq r^{-1/4-o(1)}.$$

As ϵ is arbitrary, this gives the matching height exponent, and therefore the diameter exponent as well. Finally (2.1) gives $b_h = 1 - \cos(3\lambda)A_h$. The missing arch mass is the height tail, up to the constant and rounding of heights, proving Proposition 5.1.

6 Asymptotics of the balanced observable

The boundary exponent alone does not determine marked bulk mass. We estimate the balanced polygon observable by comparing its color-sum numerator with the squared homogeneous empty component. Their leading exponential factors cancel; control of the remaining powers must be uniform in the complex twist.

6.1 Color formula and small-cycle gas

For the balanced homogeneous case (circumference $2n$) the target is the following estimate, with C uniform on each compact subset of $|\Im s| < \pi$:

$$|X_{n,n}(\kappa = e^s)| \leq C n^{1/6+2\Re(s^2)/(3\pi^2)}, \quad |\Im s| < \pi,$$

The normalization is $X_{n,n} = 1$ at $s = i\pi/2$. Together with positivity the estimate will give the matching exponent at $s = 0$ and analytic control near zero polygon fugacity. The calculation uses (i) a finite color interpolation formula converted to residues; (ii) a one-sided homogeneous normalization estimate for the Pfaffian; (iii) analytic radial periodization of the residues; upper-circle resummation creates hole variables, after which only a small defect gas around an explicit pressure remains. We bound that gas after a Gaussian count summation and recover the nonperiodized value by Cauchy at zero radial nome. The radial period L is an auxiliary **spectral** period, not the circumference of the physical SAW cylinder. Matrix and scalar abbreviations will be local to the computations.

Here is a rational product sum for

$$\mathcal{P} = S_N^2 X_{n,m} / \prod_{i < j} x_i x_j$$

(with the denominator meaning $\prod_{i < j} (x_i x_j)$). Assign labels $a_i \in \{1, 0, -1\}$, total sum zero. The type of site $P = A, B$ has sign $s_P = +, -$ respectively. Take single-site activities $\lambda_0^P = 1$, $\lambda_+^P \lambda_-^P = \sqrt{2} - 2$ and $\lambda_+^A / \lambda_+^B = T\kappa$. Pair factors in order $P \rightarrow Q$ with argument $r = x_{\text{target}} / x_{\text{source}}$ and labels a, b are

$$l_{ab}^{PQ}(r) = u_{ab}^{PQ} r^{-1-ab} \prod_{j \bmod 8} (r - t^j)^{d_{ab}^{PQ}(j)}.$$

The leading multipliers are $t^{-s_P b}$ for $a = 0$; for $a \neq 0$ they are $t^{s_Q a(1-2b^2)}$. Writing e_j for unit arrays modulo 8 and $*$ for index negation, the tables in order $+, 0, -$ are

$$d^{AA} = \begin{pmatrix} H & J & L \\ J^* & H - e_1 - e_{-1} & J \\ L^* & J^* & H \end{pmatrix},$$

$$H = e_2 + e_3 + e_5 + e_6, \quad J = e_3 + e_5 + e_6 - e_0, \quad L = e_5 + e_6 - e_0 - e_1,$$

$$d^{AB} = 2 \begin{pmatrix} e_5 + e_6 & e_5 & 0 \\ e_5 & e_4 & e_3 \\ 0 & e_3 & e_2 + e_3 \end{pmatrix}.$$

Obtain BB and BA by reversing **both** labels in AA and AB respectively. Then $\mathcal{P}$ is the sum over these neutral assignments of the products of all site and pair factors.

For checking, normalize by the source sign, putting $g = s_P a$, $h' = s_P b$, $\epsilon = s_P s_Q$. Thus the divisor array is $D_{gh'}^\epsilon = d_{gh'}^{AA}$ or $d_{gh'}^{AB}$ for $\epsilon = +, -$, of total sum $2 + 2gh'$. Write $t^{U^\epsilon(g, h')}$ for the leading multiplier ($U = -h'$ at $g = 0$, else $\epsilon g(1 - 2h'^2)$). Put $\tau_k e_j = e_{j+k}$. The tables have $D_{\epsilon h', \epsilon g}^\epsilon = (D_{gh'}^\epsilon)^*$ and $U^\epsilon(g, h') - U^\epsilon(\epsilon h', \epsilon g) = -\sum_j j D_{gh'}^\epsilon(j)$ modulo 8.

We check the characterization in Proposition 4.3. Factors are reciprocal on exchanging ends, so order does not matter. Simple poles at equal same-block sites cancel by exchanging unequal labels. At ratio t the two polar assignments $(s, -s), (0, 0)$ in type P ($s = s_P$) cancel including activities. At ratio T only $(s, -s)$ remains; its local multiplier including activities is $2 - \sqrt{2}$. At ratio t^2 the possible pairs $(s, 0), (0, -s), (s, -s)$ coalesce to their sum with local multiplier twice the single activity. These are the two S^2 reductions with the monomial denominator. To see the spectator factors explicitly put $B_{ab,c}(r; k) = l_{ac}^{PQ}(r) l_{bc}^{PQ}(r/t^k)$. Substitution in the two tables gives

$$\begin{aligned} B_{s,-s,c}(r; 1) &= t^{3sc} B_{00,c}(r; 1), \\ B_{s,-s,c}(r; 3) &= T^{sc} \frac{(r - t^5)^2 (r - t^6)^2}{T r^2}, \\ B_{ab,c}(r; 2) &= \frac{(r - t^5)^2}{tr} l_{a+b,c}^{PQ}(r/t) \begin{cases} 1 & a + b = \pm 1, \\ t^{3sc} & (a, b) = (s, -s). \end{cases} \end{aligned}$$

on the allowed pairs. Indeed for a fixed target label c , abbreviating rows $D_g = D_{g,sc}^\epsilon$, one reads $D_+ + \tau_1 D_- = (1 + \tau_1) D_0$, $D_+ + \tau_3 D_- = 2(e_5 + e_6)$, and $D_g + \tau_2 D_{h'} = 2e_5 + \tau_1 D_{g+h'}$ for $(g, h') = (+, 0), (0, -), (+, -)$; compare the powers and U 's at infinity. The phases disappear by neutrality. Locally $l_{00}^{PP} / l_{s,-s}^{PP}$ tends to $2 - \sqrt{2}$ at t , $l_{s,-s}^{PP}(T) = -1$, and the three factors at t^2 for the indicated surviving pairs are $2, 2, -(2 + \sqrt{2})$. At extreme parameters a site of label a grows with degree $N - 1 - a^2$ in either divergent direction by neutrality; only its zero label contributes at top order and the leading phases cancel. This gives polynomiality after restoring $\prod_{i < j} (x_i x_j)$, the degrees and both extreme decimations.

Finally at the balanced specialization cross factors at T require the sum of matched labels nonnegative, hence zero at every match. The inter-match factors do not depend on labels: with A label a matched by $-a$, and similarly $b, -b$, they are

$$l_{ab}^{AA}(r)^2 l_{a,-b}^{AB}(Tr) l_{b,-a}^{AB}(T/r) = r^{-4} \prod_{d \in \{2,3,5,6\}} (r - t^d)^2,$$

as required by the cross factors in the cup reduction of S^2 with the monomial denominator (here $2D_{ab}^+ + (\tau_3 + \tau_{-3})D_{a,-b}^- = 2H$). Local multipliers including activities are $(2 - \sqrt{2})\kappa^a$. This verifies the last interpolation condition and proves the sum formula.

Now specialize to two homogeneous blocks of size n . Factor out the default term (all labels s_P), which equals

$$B_n = (2D_0)^{n^2} D_0^{-2n} \kappa^n, \quad D_0 = 2 + \sqrt{2}.$$

The remaining gas has objects in each type of length one (denoted E) or two (D); the total virtual length M is the same in both types. An E object changes a site label from its default s_P to zero; a D object changes it to $-s_P$. Object length counts virtual integration variables, rather than visited vertices of a walk. Integrate each top variable z around a small positive circle surrounding 1 with measure $dz/(2\pi iz)$; the virtual variables of an object are z alone or $(z, z/t)$ respectively. Each virtual variable carries h^n evaluated there, and distinct objects carry all pairwise interactions between their virtual variables, as follows:

$$h(r) = \frac{(r+t)^2}{(r-1)(r-t^2)}, \quad K_s(r) = \frac{(r-1)^2(1+r^2)}{tr(r-t)(r-1/t)}, \quad K_x(r) = \frac{tr(1+r)^2}{(1+\sqrt{2}r+r^2)^2}$$

(pairwise argument = ratio, same and different colors respectively). Sum with inverse factorials of the numbers of indistinguishable objects and activities

$$A_P = \frac{1}{a_0 \lambda_{s_P}^P}, \quad a_0 = \frac{t^2}{1-t^2}, \quad B_P = -A_P^2 \operatorname{Res}_{r=t} (K_s(r)/r)$$

for E,D respectively. This gives $\mathcal{P}/B_n$.

For the residue check first separate base sites slightly, preserving equal block sizes and dividing by their corresponding default term, using instead of h^n products at virtual/site ratios of $p_s(r) = t^2 r / [(r-1)(r-t^2)]$ on same-color sites and $p_x(r) = t^6 (r-t^5)^2 / r$ on the others. Objects now select distinct sites of their color (coincident tops have same-color zero of order 2,1,2 for EE,ED,DD in the combined kernels). Length i means new label $s_P(1-i)$. By the pair tables, against a still-default site the change ratio equals the string of p 's times $t^{3s_P s_Q 1_{i=2}}$. Between two changed sites the correction (on top of their two single-site changes) is the string of K 's times $t^{-3s_P s_Q (i 1_{j=2} + j 1_{i=2})}$; these identities follow just by multiplying linear factors. The constants thus give net t^3 per D: for a length-2 site in type P the assigned exponent is $3s_P \sum s_Q(1-j)$ over other sites of lengths j (taking $j=0$ if unchanged), which equals 3 by neutrality. The extra self-site residue factors are a_0 , or $a_0 p_s(1/t)$ for a D, consistent with $B_P = t^3 \lambda_{s_P}^P / (\lambda_{s_P}^P a_0 p_s(1/t))$. No other poles intervene on these small contours. This proves the integral formula (one can truncate to $M \leq 2n$).

For instance use the ratio (top at type P)/(top at Q), with $\epsilon = s_P s_Q$, $g = 1-j$, $h' = \epsilon(1-i)$ normalized now at source Q . The correction divisor read from the tables is

$$D_{gh'}^\epsilon + D_{1,\epsilon}^\epsilon - D_{1,h'}^\epsilon - D_{g,\epsilon}^\epsilon = \sum_{u=0}^{i-1} \sum_{v=0}^{j-1} \tau_{u-v} K_\epsilon^\#, \quad (K_+^\#, K_-^\#) = (2e_0 + e_2 + e_6 - e_1 - e_7, 2e_4 - 2e_3 - 2e_5),$$

and the analogous difference of four U 's modulo 8 is $\epsilon[-ij + ij(j-i)/2 - 3(i 1_{j=2} + j 1_{i=2})]$, giving the stated phase including the shifts in the kernel string.

6.2 Homogeneous Pfaffian normalization

Proposition 6.1 (Pfaffian denominator). *The homogeneous empty component satisfies ($D_0 = 2 + \sqrt{2}$, $\delta = 3/8$)*

$$S_{2n}(1, \dots, 1) \geq c D_0^{-n} (4D_0^2/9)^{n^2} n^{5/48}.$$

Proof. The lower bound includes bounded error in its logarithm, because a larger normalization error would affect the final power. The kernel at variables e^{2ia}, e^{2ib} equals $i \sin(2(a-b))/(\cos(2(a-b)) + 1/\sqrt{2})$, represented by $i \operatorname{pv} \int e^{(a-b)p} W(p) dp/p$ as in the elementary transforms above, with

$$W = \frac{p \cosh(\lambda p)}{2 \sinh(4\lambda p)} = W_0 e^{f(p)}, \quad W_0 = \frac{p}{4 \sinh(\delta \pi p)}, \quad f(p) = \log(1 + 1/(2 \cosh(\pi p/4))).$$

Taking the homogeneous Taylor/Vandermonde limit in the Pfaffian gives

$$S_{2n}(1) = (D_0/2)^{\binom{2n}{2}} \frac{\det[\int p^{2j+2k} W(p) dp]_{j,k=0}^{n-1}}{\prod_{j=0}^{2n-1} j!}.$$

By multilinearity the Pfaffian over Taylor coefficients uses indices $0, 1, \dots, 2n-1$ (leading alternating order); paired evens and odds give this determinant, signs cancelling against the prefactor ($i^{\binom{2n}{2}}(-i)^n = 1$). For W_0 instead, using $y = \delta p$, $C = \int W_0 = 1/(8\delta^2)$, even coefficients of

$$(1-v)^{-1-iy}(1+v)^{-1+iy}$$

are real orthogonal polynomials with squared norms $C(m+1)$ at index $m = 0, 2, \dots$. Just integrate two generating functions: the normalized push $2y dy / \sinh(\pi y)$ has Fourier transform $1/\cosh^2(a/2)$, giving $(1+vu)^{-2}$. Thus the monic squared norms are $Cm!(m+1)!/(2\delta)^{2m}$, and the formula with W_0 gives precisely $D_0^{-n} (4D_0^2/9)^{n^2}$.

Log of the moment determinant ratio is bounded below by the trace of multiplication by f in the reference even-polynomial space, by scalar Jensen in an orthonormal eigenbasis of the positive Gram matrix. We give details to order $O(1)$. For real $|y| \leq c_1 \log m$ at large even m the unnormalized generating coefficient equals

$$2\Re \left[2^{-1+iy} m^{iy} H(y) (1 + O((1+y^2)/m)) \right] + O(m^{-1/2}), \quad H(y) = \lim_{k \rightarrow \infty} k^{-iy} \prod_{j=1}^k (1 + iy/j).$$

Choose $c_1 > 0$ sufficiently small. Indeed subtract from the generating function the two singular terms $2^{-1+iy}(1-v)^{-1-iy}, 2^{-1-iy}(1+v)^{-1+iy}$. The derivative of the remainder in the disk is bounded by $C(1+y^2)e^{C|y|}/|1-v^2|$ by a first-order estimate of the regular factor at each end. On radius $1 - 1/m$ the angular integral of $1/|1-v^2|$ is $O(\log m)$, proving the additive error via coefficient extraction of the derivative. The multiplicative error follows by the convergent logarithmic products (tails after subtracting first-order terms are $O(y^2/m)$). Also $|H(y)|^2 = \sinh(\pi y)/(\pi y)$.

Drop the range $|y| > c_1 \log m$ by positivity in the trace lower bound. The remaining integral per even index is at least

$$\frac{1}{\pi(m+1)} \int_{\mathbb{R}} f(y/\delta) \left[1 + \Re((2m)^{2iy} H(y)/\overline{H(y)}) \right] dy$$

up to summable errors: integrate the square of the coefficient against $2yf(y/\delta) dy/[(m+1) \sinh(\pi y)]$; $|H|^2$ cancels its weight, and f decays exponentially (also to restore the omitted range on the right). The oscillatory integrals are $O((\log m)^{-2})$ before the prefactor, by parts twice; the first two

logarithmic derivatives of the phase factor $H/\bar{H}$ have polynomial growth by its product. Finally $\int f(p)dp = 5\pi/9$, by expanding $\log(1 - e^{-3u}) - \log(1 - e^{-u}) - \log(1 + e^{-2u})$, $u = \pi|p|/4$ (using $\sum k^{-2} = \pi^2/6$, e.g. by Parseval on the linear function). The even-index sum now gives the claimed lower bound. $\square$

The denominator is now fixed. We return to the numerator integral and deform it without changing its physical value at zero radial nome.

6.3 Radial periodization of the small-cycle integral

Write $\alpha = \pi/4$, $\kappa = e^s$, and use coordinates $x = \log(z/t)$, periods $L, 2\pi i$, radial nome $q = e^{-L/2}$, and $P = q^2$. Define

$$S(x) = 2 \sinh(x/2) \prod_{a \geq 1} (1 - P^a e^x)(1 - P^a e^{-x}).$$

Use kernels $\prod_j S(x - i\alpha j)^{d_j}$ specified by these divisors (actual integer shifts, so 4, -4 denote two factors):

$$h : d_{\pm 1} = -1, d_{\pm 4} = 1; \quad K_s : d_0 = 2, d_{\pm 1} = -1, d_{\pm 2} = 1; \quad K_x : d_{\pm 3} = -2, d_{\pm 4} = 1.$$

They are for the external field in x and for pair differences respectively. At $P = 0$ these agree with the small-cycle kernels with multipliers t^{-1}, t stripped from the pair kernels; these phases would contribute $t^{M+\#D}$ in total. Set $C_L^2 = \lim_{x \rightarrow 0} K_s(x)/x^2$ (square root positive at $P = 0$), $k_L = \text{Res}_{x=i\alpha} K_s(x)$. Use E activity $a_L = -iC_L e^{-s/2}$ in each type and D activity $-k_L a_L^2$ (incorporating the phases at $P = 0$; indeed $C_\infty^2 = D_0$ and $tA_A A_B = -D_0/\kappa$ there). Multiply the integrand by

$$\Theta(C)^2, \quad C = \exp\left(\sum_\nu q_\nu x_\nu\right), \quad \Theta(C) = \sum_{b \in \mathbb{Z}} e^{-Lb^2/2} C^b$$

where ν ranges over virtual variables including both constituents of D, and q_ν is their color sign. D constituents are now $x, x - i\alpha$. Integrands are elliptic in each object's top coordinate, by neutrality and the shift formula $S(x + L) = -e^{x+L/2} S(x)$ (the multiplier of the pair kernels moving one object of charge p_i is $C^{-2p_i} \exp(-p_i^2 L)$, canceled by $\Theta(C)^2$).

Denote by $\mathcal{N}_n(q, s)$ the resulting small-cycle gas around tops $x = -i\alpha$, summed over $0 \leq M \leq 2n$ with its factorial measures and deformed activities, and multiplied by B_n . This includes the factor κ^n in B_n . The numerator $\mathcal{N}_n$ is analytic for small q ; at zero nome the homogeneous monomial denominator in $\mathcal{P}$ is one, so

$$\begin{aligned} \mathcal{N}_n(0, s) &= \mathcal{P}(1, \dots, 1; \kappa = e^s) \\ &= S_{2n}(1, \dots, 1)^2 X_{n,n}(\kappa = e^s). \end{aligned}$$

Even if the small-cycle sum is extended, any fixed $M > 2n$ contributes zero: resolve $h(x)^n$ into a product with slightly displaced fields and use distinct top poles as before, then pass to the limit.

Here a single sufficiently small contour size and disk of analyticity in q work independently of n . For top differences near zero the same-color string products have orders 2, 1, 2 for EE, ED, DD, and the opposite-color ones have no poles; other real-period images stay away. So after splitting the external poles each color selects distinct top poles (at most n , length at most 2 each). Products defining S, Θ converge normally where needed, and C_L, k_L are analytic deformations of their values at zero radial nome.

We describe term transformations first at fixed M , large real L . Write a radial cycle of shift θ as $x = r - i\alpha\theta$, $-L/2 < r < L/2$, positive towards the right. Expand each small cycle as lower minus upper (shifts $1 \pm \epsilon_0$, tiny ϵ_0). Move upper E's up to shift 0. The pole crossed against each still-lower E of the same color is simple, producing a two-string with coefficient $-k_L a_L^2$ (minus sign of the moving cycle). These cancel all upper D contours, which have opposite coefficient. More explicitly one can move the E contours successively; merged pairs have top near shift 0 and can move to shift $1 - \epsilon_0$. They have no further poles against remaining E or D's in this step (same-color nearby poles of string products cancel against zeros), hence the binomial expansion over disjoint mergings gives the stated cancellation with the factorials. Cycles can be infinitesimally separated during moves. Move both lower contours together to shifts $2 + \ell$ for remaining E (now called L) and $2 + d$ at D tops, with $\ell = 1/2, d = -1/8$. No pole is crossed (opposite-color pair distances from the upper contour remain below 3; lower top relative shifts stay smaller than 1). The other E's are now at shift 0 (U) with reversed sign.

In this deformation the small-cycle expansion holds by periodicity since all top contours in the initial small band have no pair poles. For each choice of r' disjoint mergings in one color, the two E factorials after summing matchings leave factorials for the two residual E counts and an extra $1/r'!$. The residue equation is $x_{\text{moving}} = x_{\text{lower}} + i\alpha$, so after moving the created contours next to the upper D contours, a fixed total d' there has relative coefficient $\sum_{r'=0}^{d'} (-1)^{d'-r'} / [r'!(d'-r')!]$ times the D activities. Hence only $d' = 0$ remains.

For instance an E approaching a D with their virtual shift differences at $(1, 2)$, $(-1, 0)$ or $(0, 1)$ sees no same-color pole in the product; two D's likewise have no pole for top differences $0, \pm 1$. Newly created tops can go from ϵ_0 to $1 - \epsilon_0$ with bottoms strictly beyond 1. At the last move take linear interpolation of both lower top shifts: no U-lower constituent distance reaches 3 or recrosses 1, and the L,D top relative shift has size < 1 .

Here are explicit zero and nonzero mode conventions. Put $\varepsilon = 2\pi\alpha/L$, $\phi = 2\pi r/L$, $y = e^{i\phi}$. For a pair of constituents with kernel divisor d_j , shift difference θ (target minus source), $p = q_{\text{source}} q_{\text{target}} = \sum d_j/2$, use

$$W(v) = \sum_j d_j \frac{\exp[v((j + \theta) \bmod 8 - 4)]}{2 \sinh 4v} = \frac{p}{4v} + W^{[0]} + W^{[1]}v + O(v^2), \quad k = \sum_j d_j(1/2 + [(j + \theta)/8])$$

taking half averages of the two branches at a wall. For whole objects sum the arrays over constituents, obtaining W_{ij} when source and target are lower objects (also use copies at identical data for $i = j$), a two-color column J_i^+ from U to lower i (reversed = J_i^-), a symmetric matrix A between U variables, and $H\mathbf{1}$ (for U) or h_i^+ (for lower, reversed = h_i^-) from the external field once ($p = 0$). In particular $H = (1 - \cosh 3v)/\sinh 4v$, A_{ss}, A_{sx} (diagonal and off-diagonal) equal $(\cosh 4v - \cosh 3v + \cosh 2v)/\sinh 4v$, $(1 - 2 \cosh v)/\sinh 4v$, both eigenvalues of A positive.

These conventions follow from expanding the Gaussian-modified kernel (multiply by $\exp[-p(x_j - x_i)^2/L]$). It has winding phase $(-y_j/y_i)^k$, mean $\log pL/6 + p\alpha^2\theta^2/L - \varepsilon W^{[1]}$ apart from phase, and log coefficient $-W(m\varepsilon)/m$ at $(y_j/y_i)^m$, $m > 0$, with reversed shifts at negative frequency. Winding powers here can be grouped to integer powers for whole objects (off the walls this follows from even total divisor order per kernel; at the final shifts the same-contour EE walls have even order, and for DD the $+1, -1$ differences of constituents enter together along with the two zero differences). One direct derivation uses, for each shifted sine at $r - i\alpha u$,

$$S = \text{sgn}(r) e^{(|r| - i\alpha u \text{sgn}(r))/2} \prod_{l \in \mathbb{Z}} (1 - e^{-|r - lL|} e^{i\alpha u \text{sgn}(r - lL)}) \quad (|r| \leq L/2).$$

Signs cancel; the log jump at 0 after summing and Gaussian compensation is $-2\pi i k$, giving the indicated phase. The mean of the product log per shifted factor is $-2\pi^2 B_2(\{u/8\})/L$, $B_2(w) =$

$w^2 - w + 1/6$. At nonzero frequency expand the logs; integrating on the two half-lines gives sums of $1/(a^2 + \omega^2)$ against $\cos(a\alpha u)$ and $(\omega/a) \sin(a\alpha u)$, $\omega = 2\pi m/L$. Combining with the exponential and winding terms yields the formula for W (equivalently use the Fourier series of $e^{\omega(x-\pi)}/(2 \sinh \pi\omega)$ on $0 < x < 2\pi$). At zeros take log-moduli singularities by radial limits.

In detail, for $\beta = \alpha u \bmod 2\pi$, per unit divisor the positive log coefficient after winding removal is $-1/L$ times

$$\frac{1}{\omega^2} + \frac{\beta - \pi}{\omega} + 2 \sum_{a \geq 1} \frac{\cos(a\beta) + (\omega/a) \sin(a\beta)}{a^2 + \omega^2} = \frac{2\pi}{\omega} \frac{e^{\omega(\beta-\pi)}}{2 \sinh(\pi\omega)}$$

(average endpoints), using $i(\phi - \pi \operatorname{sgn}(r))$ as the two-half log of $-y$ and symmetry of the shifts to distribute the Gaussian terms.

Poisson-transform the two theta factors (dual indices b_1, b_2 , sum b). Let $p_i = \sum_{\nu \in i} q_\nu$, $z_0 = \sum_{\nu} q_\nu \theta_\nu$. Let u be the column of U counts (length M minus lower virtual counts in each type); let e be the counts of lower objects (D counted once). Apart from activities, measures, powers e^{-sM} , and lower-object self corrections specified below, the scalar mean is $2\pi/L$ times

$$\exp \left\{ -\frac{\varepsilon}{2} \left[(z_0 + 4b)^2/4 + 4(b_1 - b_2)^2 + [v] \{ u^t A u + 2u^t (nH\mathbf{1} + \sum J_i^+) + \sum_{ij} W_{ij} + 2n \sum h_i^+ \} \right] \right\},$$

where $[v]$ takes the coefficient of v^1 at zero. Indeed the raw-pair Gaussian compensation sums to $-(\sum q_\nu x_\nu)^2 + \#D \alpha^2/L$ by neutrality, the square canceling the modular exponential. Pair mean offsets sum to $\alpha^2(-z_0^2 - \#D)/L$ (the internal D pairs are omitted). The $[v]$ terms use the full ordered square of all objects and require diagonal corrections below. Phases outside the upper expectation are

$$\prod_i y_i^{-bp_i} (-y_i)^{u_{c(i)} - n|p_i|} \prod_{i < j} (-y_j/y_i)^{k_{ij}}$$

where $c(i)$ denotes color. Upper means and smooth self factors convert upper cycles exactly to independent circular unitary measures in each color (Vandermonde modulus squared with normalized Haar circles and factorials). The inserted winding power on each upper y is $-e_c - bq_c$. Additionally insert, in the exponent for each frequency $m > 0$, $v = m\varepsilon$, with $T_\pm$ the columns of sums of upper $y^{\pm m}$, the quadratic term

$$m^{-1} [T_-^t (1 - A) T_+ - T_-^t (nH\mathbf{1} + \sum_i J_i^+ y_i^m) - (nH\mathbf{1} + \sum_i J_i^- y_i^{-m})^t T_+].$$

The remaining nonzero-mode log is

$$-m^{-1} \left[\sum_{ij} W_{ij} y_j^m y_i^{-m} - \sum_i W_{ii}(+\infty) (1 - e^{-v}) + n \sum_i (h_i^+ y_i^m + h_i^- y_i^{-m}) \right]$$

per mode.

Indeed from U to a lower object the winding k is 1 in the same color and 0 in the other, and for its external field is $-|p_i|$. Fourier sums with coincident cycles are understood away from coincidences or with the pure log singularities factored. Including self terms requires the factor $\exp(\Gamma_i)$ per lower object (note $W_{ii}^{[0]} = 0$ by reversal symmetry), where

$$\Gamma_i = -p_i^2 L/12 + \varepsilon W_{ii}^{[1]}/2 + \sum_{m>0} [W_{ii}(m\varepsilon) - W_{ii}(+\infty)(1 - e^{-m\varepsilon})]/m = O(1)$$

uniformly at large $\Re L$, bounded $\Im L$ (subtract $p_i^2/(4m\varepsilon)$ in the series to cancel the linear growth). At U instead subtract 1 from the same-color coefficient at every frequency for the Haar separation:

the resulting self log correction is exactly $-\log(LC_L/(2\pi))$ by the limiting same-color kernel divided by $|1 - e^{i\phi}|^2$. Thus it cancels the signed measure conversion with a_L , modulo the length fugacity already counted. Lower objects still have factorial measures and activities with e^{-sM} removed. These fixed- M identities continue to complex L (radial cycles along the period) near large positive real parts with bounded imaginary parts.

In this bookkeeping $\sum_{i<j} p_i p_j L/6 = -\sum_i p_i^2 L/12$ over distinct whole objects (including U). Explicitly for real period the U same-color compensated kernel at shift zero divided by $|1 - e^{i\phi}|^2$ tends at coincidence to $(LC_L/(2\pi))^2$. Its logged limit by the mean and Fourier coefficients above is

$$L/6 - \varepsilon A_{ss}^{[1]} - 2 \sum_{m>0} (A_{ss}(m\varepsilon) - 1)/m.$$

Negative half of this is the self correction for writing the smooth residual U interactions with both indices independently summed. The kernel at this alignment is positive on the real cycle away from its zeros, so there is no extra phase in the conversion.

6.4 Upper-edge resummation

The upper contours still contain an arbitrarily large number of points. The following circle determinant expansion sums them first and replaces them by holes on two separated contours. The dependence of its sign on the winding and circle size must be retained: it is what later permits the Gaussian sum over the original count M .

The zero-winding case is closely related to the Toeplitz–Fredholm identity of Borodin and Okounkov [4, Theorem 1]. The calculation below retains the winding and its size-dependent sign; its analytic interchange with the Hubbard fields is justified separately.

Here is the winding version of the expansion. Take potentials $V_+(y), V_-(y)$ with strictly positive and negative powers respectively, analytic on a neighborhood of a closed annulus spanning the unit circle. For a circular unitary expectation of size u insert winding y^{-d} and $\exp(V_+ + V_-)$ on each variable. Expand in additional points called holes, with signs $\tau = +1$ (outer circle > 1 , O) or -1 (inner < 1 , I), counts r, l with $r - l = d$. Use normalized positive circle measures divided by factorials. The scalar prefactor is

$$(-1)^{ud} \exp \sum_{m>0} m[y^m] V_+[y^{-m}] V_-.$$

Hole factors are

$$\prod Y_h^{-u\tau_h} \exp \sum_h \tau_h (V_-(Y_h) - V_+(Y_h)) \prod_{h<h'} E(Y_h, Y_{h'})^{\tau_h \tau_{h'}}, \quad E(a, b) = (a - b)^2 / (ab)$$

times a sign depending on r, l, d only (not u); only this independence is needed. Circles can be taken wherever the series potentials converge in the indicated annulus.

We give proof details to specify both convergence and the dependence of phases. It suffices first to use polynomial positive and negative times, as formal series. For $d \geq 0$, expand the two exponentials against Vandermonde determinants by minors and integrate: one gets $\sum_\lambda s_{\lambda+(d^u)}(V_+) s_\lambda(V_-)$ with $\ell(\lambda) \leq u$. Here $s_\lambda(t) = \det[h_{\lambda_i - i + j}]$, $h(z) = \exp(\sum_{k>0} t_k z^k)$ for the respective coefficient list. Transposing partitions gives $(-1)^{ud} \sum_{\nu_1 \leq u} s_{(u^d, \nu)}(-V_+) s_\nu(-V_-)$. Indeed the partition determinant is a Toeplitz minor on columns $u - i$, rows $u - i + \lambda_i$; the complementary inverse minor on a sufficiently large finite index interval starting at 0 uses the complement $u + j - 1 - \lambda_j^t$ and h^{-1} , giving the transpose and sign.

Explicitly $\det[y_j^{u-i}] \prod_j h(y_j)$ expands by matrix multiplication into determinants with rows $y_j^{u-i+\mu_i}$ for partitions μ (padded to size u), with precisely these minor coefficients. The reciprocal-variable determinant expands likewise. Their product integrates with $\prod_j y_j^{-d}$ and the circle factorial measures by orthogonality of monomials, requiring positive-list indices to be the negative-list indices shifted up by d ; this gives the asserted first sum. In the transposed sum (u^d, ν) means prepend d parts of size u .

The charged-wedge calculation is the classical boson–fermion correspondence [9]; see the Clifford-operator and Schur-function treatment in [10, Theorems 8.1 and 8.3]. We give its coefficient calculation to fix the charge convention and the projection onto permitted occupied levels, before taking any analytic specialization.

Use wedges with decreasing occupied positions $c + \lambda_i - i$, charge c , vacancy of the partition denoted $|c\rangle$. Insert/remove level j by ψ_j, ψ_j^* (sign by exterior order). The currents $A_k = \sum_j \psi_{j-k} \psi_j^*$ ($k \neq 0$) satisfy

$$[A_k, \psi_j] = \psi_{j-k}, \quad [A_k, \psi_j^*] = -\psi_{j+k}^*, \quad [A_k, A_h] = k\delta_{h,-k}.$$

Indeed terms act finitely on each state, and the last bracket commutes with all insertions/removals so is scalar (commute first with occupation projectors); evaluate on the vacancy, using the k possible moves up by $k > 0$. The partition amplitudes of $\Gamma_-(t)|c\rangle$ with $\Gamma_\pm(t) = \exp(\sum_{k>0} t_k A_{\pm k})$ are the minors above by exterior multiplication with the positive shift generating function h (can truncate the sea deep below all possible movements at any fixed weighted degree). Thus after removing the transpose sign our sum is

$$\langle d | \Gamma_+(-V_+) \psi_{u+d-1} \cdots \psi_u \prod_{j \geq u+d} (1 - \psi_j \psi_j^*) \Gamma_-(-V_-) | 0 \rangle.$$

Expand the projectors (finitely effective in each weighted degree) and use generating currents $\psi(w) = \sum \psi_k w^k$, $\psi^*(v) = \sum \psi_k^* v^{-k}$, with creations outside, annihilations inside. Commuting the Γ 's past the currents gives precisely the exponential factors stated. The vacancy expectation with $r = d + l$ creations and l annihilations (ordered separately, signs from reordering depending only on counts) is up to sign the determinant with columns w_i^a ($0 \leq a < d$), $v_j/(w_i - v_j)$, by summing over distinct removed negative levels and creating these plus $0, \dots, d-1$. The mode extractions, symmetrized using antisymmetry and summed by minors over the projection levels, give a second determinant with columns w_i^{-a} , $(v_j/w_i)^d/(1 - v_j/w_i)$, apart from $\prod_h Y_h^{-u\tau_h}$ and inverse factorials. Indeed the fixed creations use levels $u + a$, and the l projector pairs distinct levels $u + d + n'$, $n' \geq 0$. After the $Y_h^{-u\tau_h}$ extraction, antisymmetrization in the w 's gives columns w_i^{-a} , $w_i^{-d-n'}$, in the v 's the minor on powers $d + n'$. Sum over subsets of the n' 's (listed in order) by determinant multiplication using $|v_j| < |w_i|$; the antisymmetrizations divide by $r!l!$, with ordering signs depending only on counts. Subtract polynomial columns; the product of the two Cauchy–Vandermonde determinants gives exactly the pair factors up to sign independent of u . All mode sums at fixed counts converge on the separated radii. The series thus obtained converges absolutely for deterministic analytic potentials (Hadamard bounds $r^r C^r$ for the two determinants at fixed d , absorbed by the factorials), giving the analytic identity. Negative d follows by inverting circles and interchanging the signs.

To elaborate, in the second determinant use the polynomial columns and the finite geometric sum so its non-polynomial columns become $1/(1 - v_j/w_i)$; thus it is a Cauchy–Vandermonde in the reciprocal variables. The product with the first determinant has Vandermonde squares divided by $\prod (w_i - v_j)^2$, and the remaining powers per w_i, v_j are respectively $l - r + 1, r - l + 1$, exactly as in the E factors. Projector subset signs, antisymmetrization signs and Cauchy–Vandermonde signs here do not depend on u . For finitely many times the absolute bound above is uniform on their compact sets; Taylor coefficients agree formally (only finitely many projector subsets can act at

fixed weighted degree before commuting). Then approximate analytic potentials by truncations on separated circles.

Now use Hubbard linearization of the upper nonzero modes in the preceding integral. Write $B^\pm = nH\mathbf{1} + \sum_i J_i^\pm y_i^{\pm m}$ at positive frequency m . In the color basis jointly use coefficient columns

$$V_{\pm m} = -B^\mp/m + \sqrt{1-A}\xi_{\pm,m}/\sqrt{m},$$

where the columns $\xi_{+,m} = \overline{\xi_{-,m}}$ are independent standard circular complex Gaussians across modes (matrix square root in the fixed real orthogonal eigenbasis). Their average reproduces the quadratic term. Apply the winding expansion in each color, with its upper size and its winding $d_c = e_c + bq_c$ (distinct from lower contour shift notation). Set $Y_h = e^{\varepsilon h}y_h$, where the shift h is initially tiny positive on O, negative on I. Let $c^\pm$ collect $\sum \tau Y_h^{\pm m}$ by color. Averaging the exponential factors from the expansion gives exactly

$$\det A(v)^{-1} \exp\left(m^{-1}\{(B^- - c^-)^t A(v)^{-1}(B^+ + c^+) + (c^-)^t c^+\}\right)$$

per mode. Hole pair factors $E^{\tau\tau'}$ apply within each color. Their winding/positive-modulus separation uses mean $\log \varepsilon|h-h'|\tau\tau'$, phase power $-\text{sgn}(h-h')\tau\tau'$ on $y_{h'}/y_h$ (up to fixed signs). Thus the resulting nonzero Fourier hole-hole kernel, including the displayed Gaussian terms, is $\tau\tau'e^{(h'-h)v}(A^{-1} + \text{sgn}(h-h')1)$, with self subtractions for the original circle pair singularities. Hole-lower kernel in forward direction is $\tau e^{-hv}A^{-1}J_i^+$, in reverse $-\tau(J_i^-)^t A^{-1}e^{hv}$. The two factors $(-1)^{u d_c}$ combine with the preceding lower winding powers to leave no count sign depending on M since $u = M\mathbf{1} - (\text{lower virtual counts})$.

For the Gaussian/series interchange at fixed L real and fixed discrete data before introducing holes, the random potentials are analytic on a small annulus (exponentially decaying high modes). Moreover the series is integrable absolutely. Choose shifts $\pm h_0$ sufficiently tiny inside this annulus. Integrating the modulus against the Gaussians uses the positive stiffness $A(v)$ after the Szegő prefactor. Apart from one-body terms (bounded per hole at fixed data), the absolute real nonzero-mode Hessian after including Cauchy factors and Gaussian gains, conjugated by charge signs τ , is, per color eigenvalue a of A ,

$$e^{-|h-h'|v} - (a^{-1} - 1)_+ \sinh(hv) \sinh(h'v) - (1 - a^{-1})_+ \cosh(hv) \cosh(h'v).$$

Indeed the linear real Gaussian sources from holes have coefficients proportional to $2 \sinh(hv)\Re(\xi y^m)$ for a real-coupled mode, $2 \cosh(hv)\Im(\xi y^m)$ for an imaginary-coupled one. The matrix on the two shift classes is positive definite when $\tanh(h_0v) < a < \coth(h_0v)$. This holds for tiny fixed $h_0 > 0$ by the explicit A (eigenvalues asymptotic to $(9/8)v, 1/(2v)$ at zero and $1 + O(e^{-v})$ at infinity). At fixed ε , subtract $(1 - e^{-cv})1$ for very small $c > 0$, leaving positive smooth Fourier forms whose missing diagonal terms cost at most constants per hole. The subtracted kernels have pairwise gain at most $O(c\varepsilon)$ per pair (sum the logarithms), which is absorbed by the favorable Cauchy mean $-2\varepsilon h_0 \sum_c r_c l_c$ up to constants since count differences are fixed. The factorial measures now ensure absolute convergence.

Here the small subtraction is the identity on both color and shift indices. Its distinct-point terms only concern identical shifts in one color (hence equal charge signs), with pair gain bounded by $2 \log \sup_{|z|=1} |(1-z)/(1-e^{-c\varepsilon}z)| \leq c\varepsilon$. It preserves positivity if c is small: at high modes the Hessian tends exponentially to identity, and on the rest $v \geq \varepsilon$ use strict positivity. Completing the residual (positive) ordered sums with their diagonals then only pays $\sum_{m>0} e^{-cm\varepsilon}/m$ per hole: the Gaussian gains already have their diagonals, and the original missing Cauchy diagonal coefficient 1 was split into $1 - e^{-cv}$ (kept on distinct points only) and e^{-cv} . Also $\binom{r_c}{2} + \binom{l_c}{2} \leq r_c l_c + O(1)$ at fixed

count difference for the two same-shift classes in each color (the subtraction constant can be chosen below $2h_0$). Gaussian integrations can also be truncated in modes first (for each fixed number of points the high coefficients are exponentially small, giving integrable domination of that tail).

The absolute bounds work uniformly over the lower radial positions here at fixed preceding counts and indices. With Gaussian cutoffs the high-mode gain terms in the real Hessian simply drop out, preserving the bound, and the Gaussian cross terms with the deterministic sources are bounded by a constant per hole by high-mode decay on the tiny annulus. For each fixed number of holes the Gaussian formula passes to the infinite product as well (summable coefficient tails; one can take absolute moments of order slightly above 1 using strict Gaussian stiffness).

6.5 The residual kernels

The circle determinant has eliminated the U variables. We first carry out the remaining transformations for a fixed nonnegative integer M and a large real period L . Here M is the common total number of virtual integration variables per color in the original small-cycle gas, including variables assigned to lower objects. It is unrelated to the visited-vertex length of a walk. If g_c is the number of lower virtual variables of color c , the eliminated upper ensemble had size $u_c = M - g_c$.

We now give the remaining interactions as a matrix on a fixed list of species. The exact one-object measures and factorials are retained, together with the implicit constant phase signs. The one-object prefactors obey the stated uniform bounds and are independent of M at fixed defect data; the implicit phases have modulus one and depend only on the discrete configuration. The position-dependent phases remain explicit below. Section 6.7 will justify absolute summability of the assembled expression, complex-period continuation, and the subsequent sum over M .

The hole sign $\tau = 1, -1$ refers to the outer or inner contour of an eliminated upper ensemble. Both signs occur in each gas color, whose sign is q . In the moves below we continue the Fourier coefficients and their winding powers as functions of the offsets, re-expanding a pure singular factor when its Fourier branch changes. The upper circles have already been integrated out.

After integrating the Hubbard fields move hole contours O to $h = 3/8$; then move I to $h = -1/4$. In the second move the pole with a D of the same color creates an object P = (D, I) at equality of their radial variables, with $d = h$ (here $2 + d$ is the lower top shift of D). Put free D at $d = 1/8$ and P at $d = h = 0$. Their unshifted radial variables within the period are denoted x_i in what follows (so $y_i = \exp(2\pi i x_i / L) = e^{i\phi_i}$, and e.g. a hole has $Y = e^{\varepsilon h} y_i$).

Use $\mathbf{1} = (1, 1)^t$, $Q = (1, -1)^t$, and unit columns by color. Let $t_i = \tau_i$ times the corresponding unit column for holes, otherwise zero. For a lower gas object write p_i for its signed length, $z_i = \sum q_\nu \theta_\nu$ for its virtual angular moment, λ_i for its color unit column; all three are zero on holes. Use addition of data for P. The following table collects the resulting five species in either color of sign q . Here a_i means respectively $\ell, d, h, h, d = h$, the jump charge is $j_i = \mathbf{1}^t(\lambda_i - t_i)$, and the external-drive coefficient is $\xi_i = |p_i| - j_i$.

species	constituents	a_i	p_i/q	j_i	ξ_i
L	one lower variable	1/2	1	1	0
D	two lower variables	1/8	2	1	1
O	outer hole	3/8	0	-1	1
I	inner hole	-1/4	0	1	-1
P	bound D and I	0	2	2	0

For fixed species, counts and positions, we will isolate all M -dependence in a Gaussian count factor. We first define its center and the residual interactions, state the resulting factorization, and then verify the contour and scalar bookkeeping.

Continue $J_D^\pm$ by retaining the initial branches (coefficient dependence $e^{\pm dv}$). Define columns

$$\begin{aligned} F_i^+ &= J_i^+ + t_i e^{h_i v} = T_i/v + f_i + r_i v + \cdots, \\ F_i^- &= J_i^- - t_i e^{-h_i v} = -F_i^+(-v), \quad T_i = Qp_i/4, \quad f_i = Qz_i/4 - \lambda_i + t_i. \end{aligned}$$

(terms only on applicable constituents). The dressed oriented Fourier kernel $G(v)$ is

$$G_{ij} = W_{ij}^g - (F_i^-)^t A^{-1} F_j^+ + t_i^t t_j \operatorname{sgn}(h_i - h_j) e^{(h_j - h_i)v} + C_{ij}.$$

Here W^g is just the preceding bare W between lower gas parts, with actual lower-lower shift branches. The crossing correction for each I,D pair with relative offsets h, d satisfying $d > h$, writing $R = 1$ on equal colors and $R = -2$ otherwise, is

$$C_{ID} = R e^{(d-h)v}, \quad C_{DI} = -R e^{(h-d)v}.$$

Use half at equality, zero for $d < h$. Expressions on constituent pairs are added. Use the winding k_{ij} from the direct interactions: gas winding already defined, plus $-t_i^t t_j \operatorname{sgn}(h_i - h_j)$, plus $-R$ from I to D in the crossing correction (antisymmetric, half at equality).

Write g for the column of lower virtual counts by color, $u = M\mathbf{1} - g$. Thus $M \geq \max g_c$ and the winding constraints for holes read

$$\sum (t_i - \lambda_i) = bQ.$$

Let P_+, P_- be projections onto $\mathbf{1}, Q$, $a = 9/8$, and put

$$\nu_i = -2T_i - P_+ r_i/a, \quad M_0 \mathbf{1} = n\mathbf{1} + \sum_i \nu_i + g.$$

Indeed $Q^t \nu_i = -p_i$, so the last equality is consistent. $M_0 = n + O(\#\text{objects})$ is real and independent of positions and periods. The table's charges $j = (1, 1, -1, 1, 2)$ have sum zero over the configuration, by the winding constraint. At small v ,

$$G(v) = \frac{jj^t}{2av} + K_0 + O(v), \quad (K_0)_{ij} = \frac{p_i z_j - z_i p_j}{4} - k_{ij} + \nu_i^t f_j - f_i^t \nu_j, \quad G(-v) = -G(v)^t.$$

This follows by multiplying the columns using $A = P_-/(2v) + A_1 v + O(v^3)$, $P_+ A_1 = aP_+$; the bare pole cancels in the minus channel.

Fixed-sector reduction. For a fixed nonnegative M at large real L , the termwise contour operations on the corresponding small-cycle sector of $\mathcal{N}_n(e^{-L/2}, s)$ give configurations of the five species above satisfying $\sum_i j_i = 0$, $\sum_i (t_i - \lambda_i) = bQ$, and $M \geq \max_c g_c$. Each term retains its exact one-object measures and factorials, the κ -free part of B_n , and a constant phase of modulus one depending only on the discrete configuration. The factor κ^n from B_n combines with the length fugacity e^{-sM} to give the displayed count twist $e^{-s(M-n)}$. The one-object prefactors are bounded by $C|dx_i|$, uniformly for large $\Re L$ and bounded $\Im L$. These prefactors and the implicit phases are independent of M at fixed defect data. Apart from them, the term is the product of the following four factors; all remaining M -dependence occurs in the count factor.

Pressure and determinant. The overall scalar factor is

$$\frac{2\pi}{L} \prod_{m>0} (\det A(m\varepsilon))^{-1} \exp \left[n^2 \varepsilon \left(\frac{1}{2} H_*(0) + \sum_{m>0} H_*(m\varepsilon) \right) \right], \quad H_*(v) = \frac{2 \cosh 3v - 2}{v(2 \cosh v - 1) \sinh 4v}.$$

Count Gaussian and twist. The count factor is

$$\exp[-2\varepsilon(b_1 - b_2)^2 - a\varepsilon(M - M_0)^2 - s(M - n) + i(M - M_0)\mathcal{S}], \quad \mathcal{S} = \sum_i j_i \phi_i.$$

External drive. Its logarithm is

$$-n \sum_i \xi_i \left[i\phi_i + a_i \varepsilon + \sum_{m>0} \frac{\rho(m\varepsilon)}{m} (e^{a_i m \varepsilon} y_i^m - e^{-a_i m \varepsilon} y_i^{-m}) \right], \quad \rho(v) = \frac{1}{2 \cosh v - 1}.$$

Interaction. Its logarithm, in addition to a linear phase, is

$$-\frac{\varepsilon}{2} \sum_{ij} [v] G_{ij} - \sum_{m>0} \frac{\sum_{ij} G_{ij}(m\varepsilon) y_j^m / y_i^m - \sum_i (1 - e^{-m\varepsilon})}{m}.$$

Its additional linear phase is precisely $-i \sum_{ij} \phi_j (K_0)_{ij}$.

The displayed count twist and linear phase retain their position dependence; only the implicit constant phases are suppressed. The verification below is termwise at fixed counts. Section 6.6 bounds the interaction matrix frequency by frequency. Section 6.7 then proves absolute summability, justifies the assembled fixed-sector identity, continues it to complex periods, and permits the sum over the original count.

Contour crossings and one-object factors. The growing term in the I-to-D $-(F_i^-)^t A^{-1} F_j^+$ is $-Re^{(d-h)v}$. Transferring the corresponding singular factor to the opposite Fourier direction adds C as above. Between two P objects at equal shifts both directions are transferred or averaged together (the powers are integer powers of the combined two-sided factor, up to a sign); on the internal I,D pair the singular factor is omitted by the residue.

Indeed at high positive v , the only growing or nondecaying singular part of J_D^+ when crossing is the column with entries Re^{dv} , whereas J_D^- decays at least as fast as $O(e^{-(1+d)v})$, and $J_L^\pm = O(e^{-v/2})$. This follows by summing the given divisors. Since $A^{-1} = 1 + O(e^{-v})$, and D always stays at offset strictly inside O, the troublesome product among these factors is precisely $-Re^{(d-h)v}$ from I to D, giving $(1 - e^{(d-h)\varepsilon} y_D / y_I)^{-R}$; other than the direct hole pair and gas factors all the remaining mode sums are analytic across the moves. Re-expansion gives the correction, mean and phase above (log mean changes by $-\varepsilon R(d-h)$). The pole is simple and taking its residue removes this factor and the normalized I circle measure. Bound pairs have no further poles in these moves: equal-color DD and II each have a double zero at coincidence, compensating the possible simple pole with an unbound object or the two poles between bound pairs. Opposite colors had a zero rather than a pole; remaining bare factors between lower contours have no poles for these offset differences. Thus only disjoint matchings are taken, with factorial denominators now those for the combined species and signs depending on the defect data. This reasoning can move contours successively with separated radii. The natural expansions of lower-lower and hole-hole factors after each move can always be taken directly. Constants and phases for the continued upper-lower modes track the retained branches: under an offset increment δ , a coefficient continued by $e^{\delta v}$ changes its mean $p\alpha^2\theta^2/L - \varepsilon[v]W$ by exactly $\varepsilon k\delta$. In paired-paired terms at equal offsets the two transferred factors have opposite orientations and can take the symmetric boundary values together up to signs.

More explicitly the leading terms of J_D^+ come from index -2 at the top (same color) or -3 at the bottom (opposite color), retained with wrapped value $8 + d$; the rest is $O(e^{-(1-d)v})$. One can cross by moving the I contours while all D have $d = -1/8$, and then move free D's and bound pairs ($h = d$) to final positions. Gas top relative offsets between two D gas parts then have size < 1 (the

possible same-color constituent poles at zero top difference already cancel); differences from L to the two constituents have size < 1 . Two P's on one contour combine their I-to-D pure factors to $((1 - y'/y)(1 - y/y'))^{-R}$, agreeing with the averaged expansions. The gas windings group to integers (at equal D shifts the $+1, -1$ constituent differences occur together), as do the crossing powers (the equal-offset halves occur in opposite directions together).

Self subtractions in the final series are $G_{ii}(+\infty)(1 - e^{-v})$ (always $G_{ii}(+\infty) = 1$); self and measure prefactors besides powers already isolated are bounded by $C|dx_i|$ per object, uniformly for large $\Re L$ and bounded $\Im L$. Indeed the lower self corrections were given earlier. An unbound hole has a missing 1 on the diagonal from the circle-pair log. Using $1 - e^{-v}$ instead leaves an activity $\exp(\sum e^{-m\varepsilon}/m)$ with its measure dx_i/L . For a bound pair, removing the pure pole adds 1 on its effective diagonal, so the two old subtractions ($1 - e^{-v}$ for D and 1 for I) now give exactly $1 - e^{-v}$ without extra activity. The residue has only the D radial measure. At infinity the joint kernel has same-color limit the identity in species, opposite-color limit 2 at P,P only, by the divisor rules.

Zero modes and count dependence. The direct constant of G at zero is $(p_i z_j - z_i p_j)/4 - k_{ij}$ by symmetry of divisor shifts. Dressing contributes $-2T_i^t f_j + 2f_i^t T_j$ and $(-r_i^t P_+ f_j + f_i^t P_+ r_j)/a$ there. For means during the contour moves, retaining a branch with offset increase δ multiplies its coefficient by $e^{\delta v}$; the winding-power increment of the constant log is then as above, using $W^{[0]} = p\theta/4 - k$ on a virtual pair. In total the z -square term precisely accounts for changes in all Gaussian offsets and the theta winding offsets by neutrality. Natural re-wrappings just use the actual direct factors (hole Cauchy, bare gas, or transferred pole).

Indeed the retained $[v]$ changes by $(p\theta/4 - k)\delta + p\delta^2/8$ per virtual pair; the sum of p times squared shift differences changes by minus the change of $(\sum z_i)^2$. The winding monomials from the old formulas change by the same required scalar when an old radial argument is moved ($y_{\text{old}} = e^{\varepsilon\delta} y_{\text{new}}$ for a single increment). On the internal bound pair $[v]C$ costs zero; in the nonzero modes deleting the pure ID pole subtracts $1/m$ per frequency after identification, giving the self subtraction specified above.

For external insertions the dressed columns (rows similarly) couple with $h_i^+ - H\mathbf{1}^t A^{-1} F_i^+ = \xi_i \rho e^{a_i v}$, reversed $-\xi_i \rho e^{-a_i v}$; the exterior-exterior kernel is $2H\rho = -vH_*$. Indeed $-A^{-1}H\mathbf{1} = \rho\mathbf{1}$. On gas divisors, cyclic neighbor sum minus the divisor itself for h equals minus the sum of the two pair divisors. In multiplying h_i^+ by $2\cosh v - 1$ the unwrapped neighbor exponents therefore leave only terms passing the seam. At the initial shift branches with $d < 0$ none do so for L or D bottom; at D top the -1 index passes down through zero, leaving $+e^{dv}$. This continues identically. For zero modes before dressing the bracket in the mean, now including the factors $Y^{-\tau u_c}$, is

$$2|f_{\text{tot}}|^2 + 4(b_1 - b_2)^2 + [v]\{u^t A u + 2u^t(nH\mathbf{1} + \sum F_i^+) + B(v)\}.$$

Here $f_{\text{tot}} = (\sum z_i/4 + b)Q$, the norm square is real, and $B(v)$ denotes the sum of the direct kernels $W^g, K^{\text{hole}} = t_i^t t_j \text{sgn}(h_i - h_j) e^{(h_j - h_i)v}$, C and direct external $n(h_i^+ + h_i^-)$. Hole means and crossings indeed match these coefficients of v . Replacing $B(v)$ by

$$B(v) - (nH\mathbf{1} + \sum F_i^-)^t A^{-1} (nH\mathbf{1} + \sum F_i^+)$$

incorporates all the other terms except $4(b_1 - b_2)^2 + 2a(M - M_0)^2$. This is completion of the square coefficientwise, using $P_- u = -2\sum T_i, H = -av + O(v^3)$, $\sum f_i = f_{\text{tot}}$. Finally the winding phases on y_j have powers $\sum_i k_{ij} + u^t(\lambda_j - t_j) - n|p_j| - bp_j$, with count signs independent of M as seen above. Splitting u by M_0 , the identity $\sum_i ((K_0)_{ij} + k_{ij}) = bp_j - \sum_i \nu_i^t(\lambda_j - t_j)$ gives exactly the stated phases.

Explicitly in the square completion the minus-channel pieces with $P_- u = -2T_{\text{tot}}$ leave $-2|f_{\text{tot}}|^2$ at order v , canceling the explicit norm square. The plus-channel part is $a|P_+ u + P_+ r_{\text{tot}}/a - n\mathbf{1}|^2$ since $P_+ f_{\text{tot}} = 0$.

To reiterate the count-sign point, the upper cycle measure was $-a_L dx/(2\pi i) = C_L e^{-s/2} dx/(2\pi)$ with exactly the positive-sign self conversion on real periods. The signs from $(-y_i)^{u_{c(i)}}$ over gas parts and $(-1)^{\sum u_c(e_c + bq_c)}$ from the winding determinant combine to a sign independent of M (e_c here counts those gas parts). Transferring pure factors and taking their normalized circle residues uses only signs fixed by defect data. These constant phase signs can simply be kept unchanged in continuation.

6.6 A kernel positivity decomposition

Our next task is to bound the residual particle sum uniformly in the spectral period. A positive Fourier form will control high local populations; a separate pole term will control the cumulative integer charge. The following decomposition supplies both controls.

In this calculation only the five species (per color) index the matrices; the symmetric part of $G(v)$ for $v > 0$ is meant. One can subtract the following pieces:

- $jj^t \text{sech}^2(Kv)/(2av)$, with K a sufficiently large constant, all colors;
- on the opposite-color blocks only, the entrywise repulsion

$$R(v) = \tanh(v/2)(R_1/\cosh v + R_2/\cosh(v/2)) + 2 \tanh v \, e_5 e_5^t,$$

where

$$8R_1 = \begin{pmatrix} 6 & 15 & 0 & 3 & 16 \\ 15 & 22 & 0 & 9 & 26 \\ 0 & 0 & 2 & 0 & 0 \\ 3 & 9 & 0 & 0 & 5 \\ 16 & 26 & 0 & 5 & 5 \end{pmatrix}, \quad 8R_2 = 2(e_2 e_5^t + e_5 e_2^t) + 5(e_4 e_5^t + e_5 e_4^t);$$

- on the diagonal, $\tanh(cv)$, for sufficiently small $c > 0$.

The result divided by v is an even, smooth, exponentially decreasing positive Fourier form; it even dominates $c_1 \text{sech}(c_2 v)$ times the identity, for some positive constants. The repulsions (second and third pieces divided by v) have entrywise nonnegative Fourier transforms. Indeed $\tanh(cv)/v = \int_0^c \text{sech}^2(uv) du$, and sech has positive Fourier transform (by shifting through one pole), so squares and products here give positive convolutions.

Here are verification details of the matrix inequality. Work in either color channel $h = \pm 1$, i.e. the eigenblock same plus h times opposite, and first subtract only $hR(v)$. Call the resulting symmetric matrix $S_h(v)$. With $x = e^{v/8}$, $w = x^8$, put $D = w^4 - w^{-4}$ and

$$A_n = w^{-4}(w^2 - w + 1)(w^3 - h)^2.$$

Thus the single upper eigenvalue is A_n/D . After removal of the offsets (factor $x^{8\beta}$, $\beta = a_j - a_i$), the oriented entry of G in this channel is

$$m_{ij}(w)/D - p_i(w^{-1})p_j(w)/(DA_n) + b_{ij}.$$

Here m_{ij} is the divisor exponential sum for bare gas parts, p_i that from an upper variable plus $\tau_i D$, and b_{ij} the hole-direct and crossing entries; explicit arrays are given below. At every $w^8 = 1$, $m_{ij}A_n = p_i(w^{-1})p_j(w)$, since wrapping and the branch choice do not change any powers. At

$w = 1, h = 1$ the difference vanishes to order at least two because $p_i(1) = 0$ and A_n has a double zero. Thus, putting $z = w^{-1}$,

$$\begin{aligned} N &= 44 \ (h = 1), \quad 40 \ (h = -1), \\ \Delta_+ &= (1 - z)(1 + z)^2(1 + z^2)(1 + z + z^2)^2(1 - z + z^2), \\ \Delta_- &= (1 + z)^2(1 + z^2)(1 - z + z^2)^3, \end{aligned}$$

the matrix $4x^N \Delta_h S_h$ is a Laurent polynomial of parity h under $x \mapsto 1/x$, with powers at most N (indeed $S_h \rightarrow 1$ at infinity). Denote its coefficients at x^k , $k \geq 0$ by P_k , taking only half the constant coefficient. In the Taylor expansion in $\log x$ the relevant matrices are

$$B_p = \sum_{k=0}^N (k/N)^p P_k$$

for even p in the plus and odd p in the minus channel. $B_0 = 128jj^t$ for the plus channel (here j is the five-component vector). All the other relevant matrices are positive definite.

We include small algebra details for checking this last sign test. Take $K_i = (2), (2, 3), \emptyset, \emptyset, (2, 3)$, respectively, and $\tau_i = 0, 0, 1, -1, -1$. Let $d_l = (2, -1, 1, -2h, 2h, -2h, 1, -1)_l$ for residues starting at 0. Put

$$L_b(m) = \begin{cases} z^{8-(m \bmod 8)} & m \not\equiv 0, \\ z^8, 1, (1 + z^8)/2 & m \equiv 0, \quad b > 0, b < 0, b = 0 \text{ respectively.} \end{cases}$$

Then

$$m'_{ij} = z^4 m_{ij}(w) = \sum_{r \in K_i, s \in K_j, l} d_l L_\beta(l + s - r), \quad u_j = z^4 p_j(w) = \tau_j(1 - z^8) + \sum_{s \in K_j, l} d_l L_{\sigma_j}(l + s),$$

where $\sigma_j > 0$ on L, negative on D or P. Finally

$$b_{ij} = -\tau_i \tau_j \operatorname{sgn} \beta + (1 - 2h)[\mathbf{1}_{\tau_i = -1, |K_j| = 2} \chi(\beta) - \mathbf{1}_{|K_i| = 2, \tau_j = -1} \chi(-\beta)]$$

with χ the positive step (half at zero).

The following table gives integer parts of the five successive leading minors; the last row in each

block instead gives integer parts of $100 \sum_{k < N} \sum_j (k/N)^p |P_{k,ij}|$, by rows i .

$p(+)$	1	2	3	4	5	$p(-)$	1	2	3	4	5
2	22	149	542	2294	10192	1	1	11	9	9	3
4	10	32	31	54	18	3	1	9	9	11	9
6	7	15	28	97	174	5	1	12	14	17	21
8	5	13	38	135	211	7	2	15	22	31	59
10	4	12	44	150	177	9	2	17	29	43	122
12	4	13	47	158	144	11	3	18	33	52	185
14	4	13	49	165	125	13	3	17	36	63	238
16	4	14	50	173	122	15	3	17	38	77	289
18	4	14	52	180	133	17	3	16	40	96	347
20	4	14	53	188	156	19	3	16	43	115	412
22	4	15	54	196	188	21	3	16	45	135	482
24	4	15	56	203	227	23	3	16	48	153	550
26	4	15	56	209	271	25	3	15	50	169	614
28	4	15	57	215	318	27	3	15	52	183	672
30	4	15	58	220	366	29	261	374	287	271	231
32	4	15	59	225	414						
34	4	15	59	228	461						
36	4	15	60	232	506						
38	4	15	60	235	549						
40	4	15	60	237	589						
42	178	295	150	177	382						

Since $P_N = 4I$, each last row proves strict diagonal dominance at its index. The factors $(k/N)^p$ for $k < N$ decrease with p , so diagonal dominance continues at every larger relevant index. The preceding rows prove positivity of the remaining B_p by their leading principal minors.

Here is one way of doing the additions/multiplications for the table. Form to order six inclusive

$$H_{ij} = \Delta_h(m'_{ij} + b_{ij}) - Q_h(z)z^8 u_i(1/z)u_j(z), \quad Q_+ = (1+z)^2(1+z^2) \sum_{d=0}^6 z^d, \quad Q_- = 1+z^2.$$

For the two channel blocks separately, set

$$D_{l,ij} = 2[z^{l/8+a_j-a_i}]H_{ij} - 2h[z^{l/8}]\Delta_h \left(2\frac{1-z}{1+z} \left(\frac{zR_1}{1+z^2} + \frac{\sqrt{z}R_2}{1+z} \right) + 2\frac{1-z^2}{1+z^2}e_5^t e_5^t \right)_{ij}.$$

A missing power means coefficient zero. Starting at $l = 0$ through $N - 1$, use $D_l + D_l^t$ (so the first is $4I$). Indeed expansion of the displayed oriented entry gives H_{ij} , since $(1 - z^8)^{-1}$ can be dropped here. To obtain column b in a minor row, take

$$\left| \frac{\det[\sum_{l=0}^{N-1} (D_{l,ij} + D_{l,ji})(N-l)^p]_{i,j \leq b}}{N^{pb}} \right|;$$

in the last row add $|D_{l,bj} + D_{l,jb}|(N-l)^p$ for $l > 0$ instead and multiply by $100/N^p$. These coefficient formulas use only polynomials of small degree.

For some further detail on the common denominators: at an eighth root w put $U = \sum_l d_l w^{l-4} = \sum_l d_l w^{4-l} = A_n$. Then $m_{ij} = U(\sum_{r \in K_i} w^{-r})(\sum_{s \in K_j} w^s)$, and $p_i(w) = U \sum_{r \in K_i} w^r$ there. Once

the D poles cancel this leaves the factor $(1 - z + z^2)(1 - hz^3)^2$, with at most a simple pole at $z = 1$ in the plus channel; the repulsion denominators use $(1 + z)^2(1 + z^2)$. Their combination thus divides Δ_h . The offsets only give integral powers of x . Symmetrization is odd under $v \mapsto -v$ before multiplication by $4x^N \Delta_h$; this gives the stated parity. The plus constant Taylor value is $2B_0$ and also follows immediately from $\Delta_+ \sim 72v$ and the pole. For moments at positive index p , only coefficients strictly above power zero in x matter, so $(N - 1)/8 + \max |a_i - a_j|$ controls the truncation in z .

To pass from these coefficient tests to every $v > 0$, expand the Laurent polynomial in $\log x = v/8 > 0$. Its parity and the half-constant convention give

$$4x^{44} \Delta_+ S_+ = 2 \sum_{r \geq 0} \frac{(44 \log x)^{2r}}{(2r)!} B_{2r},$$

$$4x^{40} \Delta_- S_- = 2 \sum_{r \geq 0} \frac{(40 \log x)^{2r+1}}{(2r+1)!} B_{2r+1},$$

where each line uses its own channel's B_p . All scalar weights are nonnegative, and $\Delta_h > 0$ for $x > 1$, proving that S_h is positive definite. Moreover B_1 gives positivity of S_-/v at zero, while B_2 gives the positive order- v coefficient of S_+ on the orthogonal complement of its pole; the denominator corrections to $B_0 = 128jj^t$ remain in the pole direction. Subtracting the indicated smoothed pole leaves a positive definite order- v coefficient for sufficiently large K ; away from zero this subtraction can be made arbitrarily small by further increasing K . At infinity the result is still $I + O(e^{-c_3 v})$, $c_3 > 0$, so subtracting $\tanh(cv)$ times identity for sufficiently small c is also allowed and leaves the asserted lower bound at all frequencies.

Indeed in the plus block the pole removed is $jj^t \operatorname{sech}^2(Kv)/(av)$, improving the regular linear coefficient in its pole direction arbitrarily as K increases. Once this works in a neighborhood of zero, increasing K only improves the pointwise form there. Similarly one can decrease c further on compact intervals after ensuring the inequalities near zero and infinity (at infinity first take $2c < c_3$).

6.7 Estimates and summation of the defect gas

The preceding matrix calculation is only a frequency-by-frequency statement. We must turn it into an absolutely summable particle expansion, continue it to complex periods, and then sum the original count M . We do these operations in that order; the final complex-twist estimate will be obtained only after all three steps.

Take $R = \Re L$ large, $|\Im L|$ bounded, s in a compact subset of the strip $|\Im s| < \pi$. Constants can depend on these strip bounds. Positions may run along a common contour $\gamma(r) = r + i\chi(r)$, $-R/2 < r < R/2$, extended by $\gamma(r + R) = \gamma(r) + L$, with χ odd, bounded on the segment, piecewise smooth with Lipschitz constant σ sufficiently small. Split the parametrizing period into boxes of widths between 1 and 2 with populations n_d ; put $m = \sum n_d$. We give details also justifying convergence of the expansions.

Population control. Isolate the smoothed Coulomb (pole) piece of the symmetric kernel as in the decomposition. Its coefficient of degree 1 at zero costs nothing in the mean term (sum of j vanishes). For the remaining symmetric kernel divided by v , say f , the interaction (zero mode included) uses the periodized line kernels

$$-\frac{1}{2} \sum_{k \in \mathbb{Z}} \hat{f}_{ij}((x_j - x_i + kL)/\alpha), \quad \hat{f}(t) = \int f(v) e^{itv} dv.$$

On the diagonal species, the repulsion with numerator $\tanh(cv)$ is taken here only between distinct objects; its self log with the $(1 - e^{-v})$ subtraction is bounded per object, by summing (the difference vanishes at zero and decays at infinity). No opposite-color self term occurs.

We record the analytic bounds underlying this use of periodization. The regular functions involved are even, analytic near zero and in a thin strip about the real axis, and exponentially decreasing at both real ends also on tilted rays with sufficiently small slopes. The same holds for the functions giving symmetric repulsions except that they need only decrease as $O(1/|v|)$. Indeed they have the displayed hyperbolic and exponential formulas, with poles only away from zero on the imaginary axis. In a small wedge $|\Im t| \leq \sigma |\Re t|$ their transforms are exponentially decreasing at infinity together with first derivatives; at zero the smooth rapidly decreasing pieces give bounded values and derivatives, the others need bounds $O(1 + |\log |t||)$ and $O(1 + 1/|t|)$. To see this for $\Re t > 0$ deform the line into $v = u + i(\delta + 2\sigma_0|u|)$, small fixed $\delta, \sigma_0 > 0$, $\sigma \leq \sigma_0$; similarly reflect for the other end. The integrals for real nonzero t allow the deformation also for decay of order 1 by oscillatory limits. Poisson summation in the form $\varepsilon \sum_{d \in \mathbb{Z}} f(d\varepsilon) e^{id\varepsilon x/\alpha} = \sum_k \hat{f}((x + kL)/\alpha)$ then gives the kernels above, first for real periods and positions (for a logarithmic singularity by Fourier series off the diagonal, or smoothing by a narrow Gaussian first), then by continuation. Here the degree-1 mean of the original regular numerator supplies the half-weighted value of $f(0)$.

On the real comparison line of period R , the smooth positive form costs at least $c_4 \sum n_d^2$. Here is the passage from frequency positivity to this population bound. Subtract its diagonal minorant $c_1 \operatorname{sech}(c_2 v)I$; the ordered point-measure quadratic form of the remainder, including its diagonal and zero mode, is nonnegative by the periodized Fourier series. The transform of the subtracted scalar kernel is nonnegative at every real distance and bounded below for two points in a box of width at most 2. If $n_{d,t}$ counts type t in box d , the retained energy is therefore at least

$$c \sum_d \sum_{t=1}^{10} n_{d,t}^2 \geq \frac{c}{10} \sum_d n_d^2.$$

This uses matrix positivity for the smooth remainder, not entrywise positivity of its transform. The separate repulsions between distinct objects are nonnegative in space, and the diagonal self correction costs only $O(m)$. Changing to complex γ costs only $C\sigma \sum n_d^2$: every lifted real difference changes by at most σ times its size, so the derivative bounds above give losses $C\sigma$ times exponentially decreasing pair kernels. Their sums are bounded by the box square sum (Cauchy–Schwarz). Coincident positions can be disregarded. Thus these parts of the real log are bounded by $-c_5 \sum n_d^2 + Cm$.

For the antisymmetric part $K = (G - G^t)/2$, even, smooth and exponentially decreasing, include the additional linear phase involving K_0 in the interaction. This gives on each ordered pair a primitive, vanishing at difference zero, of

$$-\frac{i}{2\alpha} \sum_k \widehat{K}((x + kL)/\alpha).$$

Indeed the linear phase inserts half the zero mode of the derivative of the sine sum. It is imaginary on the real comparison. On the complex curve its extra real part again costs only $C\sigma \sum n_d^2$, applying the same endpoint comparisons to the primitives of each image term (subtract at kL), whose derivatives have the exponential bound. Thus small enough slopes allow absorption.

Here the lower endpoint comparison at $k = 0$ has exactly zero change, and those at nonzero k cost $O(\sigma e^{-c'R})$ summed per pair, harmless also for the box bound. For the transform arguments with $1/|v|$ tails, the nonnegative real transforms used above are locally integrable functions (use the displayed positive mixtures of scaled sech^2 transforms, with convolutions as applicable), with the pointwise bounds off zero; Poisson's formula there also follows by periodic Fourier coefficients in L^1 .

For example on a real period these series off coincidences converge in the ordinary paired-frequency sense too, since at positive frequencies one only has a constant over frequency plus a rapidly decreasing remainder at infinity. On straight complex periods, with x/L fixed real in the Poisson formula, both sides continue (for the image transforms keep the positive or negative real-end branch according to the lifted real difference). For the antisymmetric primitive, real parts can be compared image by image using a line primitive real on the real axis, evaluated at a lifted difference and at kL , before multiplying by the imaginary prefactor. The joining paths can be chosen on the appropriate wedge side, since for differences inside one period the endpoint real parts do not straddle zero strictly.

Integer-charge variance and contour deformation. The population bound does not yet localize the objects along the period. For this we retain the pole term, whose charges are the integer jumps j_i . Let $w(r)$ on the real parametrizing period have initial value zero at $-R/2$, with jumps j_i at the positions r_i , periodically extended. Smooth this step function on the curve by convolution in contour length coordinate (complex differential dz) with

$$H(z) = \frac{1}{2\pi\alpha} \sum_k \widehat{\text{sech}(K \cdot)}((z + kL)/\alpha)$$

(here K in sech is the pole smoothing constant). Denote the result by w_* , and its average by $\bar{w}_*$; this average is that of the steps using complex differential, $-\mathcal{S}/(2\pi)$. The nonzero log of the smoothed pole is exactly

$$-c_0 \int_{\gamma} (w_* - \bar{w}_*)^2 dz, \quad c_0 = \pi^2/(2\pi\alpha a).$$

On straight contours these facts follow from Fourier series, differentiating the steps to the jump masses. On bent contours the identities continue: convolutions and integrals of steps use arcs with moving endpoints, the line kernels analytic with exponential bounds in neighborhoods of all the contour differences as contours and endpoints move together. Thus the resulting integrals continue holomorphically in endpoint data along the deformation (locally deform arcs by Cauchy's theorem). Equivalently the nonzero pole pair potential has second derivative given by a periodized smooth line kernel (numerator sech^2) minus its constant mode.

In fact the unsmoothed step has nonzero Fourier coefficient $\sum_i j_i e^{-ib\phi_i}/(2\pi ib)$ at integer $b \neq 0$, by integration by parts, and the smoothing multiplier is $\text{sech}(Kb\varepsilon)$ (unit at zero).

We deform the product torus of spatial contours jointly. Three checks make this passage precise.

Coincidences. Every lifted pair difference remains in the preceding wedge. Near a contact modulo L , separate the high-frequency constant of the *combined* pair coefficient from its exponentially decreasing remainder. The former exponentiates to the nonnegative integer power $G_{ij}(+\infty)$ of $(1 - y_j/y_i)(1 - y_i/y_j)$; the remainder converges normally in a joint neighborhood. Thus the full exponential, rather than each separated logarithm, extends across contact loci. The external kernel used below has no singularity along this homotopy.

Period seams. The original winding factors show periodicity: for each integer value of the original count M , the powers from winding and crossings are integers after constituents are combined. Continue the same straight-torus germ while following both representatives at a period seam. Their values remain equal. Small joint Cauchy deformations therefore create no vertical-end term. The common graph has a global Lipschitz bound on its imaginary part, with endpoint difference $\Im L$.

Moving endpoints. For the step integrals, fix the order of distinct jump positions by real part and split every convolution arc at those positions. Expanding the convolutions gives iterated analytic-kernel integrals; all integration-point differences remain in the same analytic neighborhoods. Short

tails added to the arcs show local holomorphy in their endpoints, including the smoothing variables. The arcs and their endpoints can consequently move together in small product neighborhoods. This continues the straight-contour identities needed at distinct jumps. These three checks justify the joint deformation without separating non-single-valued logarithmic terms.

Denote the smoothed step on the real comparison period by w_R , and use $b(r)$ here for a local density bound of the form $\sum_{i,k} e^{-c_6|r-r_i+kR|}$, with $c_6 > 0$ sufficiently small. The elementary smoothing estimates are

$$|w_R - w| \leq Cb(r), \quad |w_*(\gamma(r)) - w_R(r)| \leq C\sigma b(r).$$

Indeed both kernels integrate to one, so one convolves differences from the value at r , bounded along a lift by the jumps passed. The kernels including differentials differ by at most a periodized exponential times $C\sigma$. Thus $\int b \leq Cm$, $\int b^2 \leq C \sum n_d^2$. Also, writing w_R^0 for the real average, one has

$$|\bar{w}_* - w_R^0| \leq C\sigma R^{-1/2}(\|w_R - w_R^0\|_2 + \|b\|_2)$$

by averaging the unsmoothed steps.

Sharp contour and absolute assembly. The external expression in brackets in the drive is a primitive (zero at 0) with derivative $(i/\alpha) \sum_k \hat{\rho}((x+kL)/\alpha)$, evaluated at $x = x_i - i\alpha a_i$. Here $\hat{\rho}(t)$ is a positive constant times $\sinh(2\pi t/3)/\sinh(\pi t)$, by shifting the integration line by $2\pi i$ and taking the two poles. It decays as $e^{-\pi|\Re t|/3}$, uniformly at large real ends, and for $|\Im t| \leq 3/8$ has real part bounded below by a positive constant times the same decay (multiply by the conjugate denominator). Only D,O,I have nonzero drive. In general its real log on the contours is bounded above by $C_n m$. For the sharp bound take

$$R = (3/2) \log n.$$

This choice makes the weakest external drive, at the period edges, have order one: $ne^{-2R/3} = 1$. It also matches the pressure calculation in Section 6.8: after removing its analytic correction $c_* n^2 e^{-L}$, the error is $O(n^2 e^{-4R/3}) = O(1)$. The removed term itself has order $\sqrt{n}$, so it cannot be included in the bounded error.

Bend so that $\chi = 0$ except within fixed width of the edge $\{-R/2, R/2\}$, using sufficiently large constant width to achieve the small slope. At $x_i = r_i$ the main line term integrated to r_i is imaginary, and on going vertically to the shifted position it gives signed real part at least $c_7 e^{-4|r_i|/3}$, since $\xi_i a_i > 0$ for active objects. Other images contribute $O(e^{-4(R-|r_i|)/3})$; the bend changes the result also by at most $O(1/n)$ per primitive, by the fixed width and the decay. Hence the real log drive bound here is

$$Cm - c_8 \sum_{i:\xi_i \neq 0} \exp(4 \text{dist}(r_i, \text{edge})/3).$$

Before resumming M , these estimates, omitting common pressure and determinant factors, give (besides the $-2\Re \varepsilon(b_1 - b_2)^2 \log$ and factorial measures) bounds of the form

$$\exp[C_n m + CR - c_9 \sum n_d^2 - c_{10}(M - M_0)^2/R],$$

where C_n can be taken constant in the sharp contour setting. Indeed the real part of the complex pole variance integral is bounded below (before its negative multiplier) by a constant times $\|w_R - w_R^0\|_2^2 - O(\sigma^2 \|b\|_2^2)$, by the smoothing bounds. Also $|\Im \mathcal{S}|$ is controlled by the complex average bound. Its cost multiplied by $|M - M_0|$ is absorbed by a fraction of the count Gaussian and variance and box costs. The real tilt is bounded by completing a square, since $M_0 = n + O(m)$.

These bounds prove absolute summability after the stated transformations, even over all values of the original count M (the sum $b_1 + b_2$ is already determined by hole and lower counts). They upgrade the termwise reduction to an assembled fixed-sector identity. More explicitly, on the real periods one can first perform the circle expansion and shifts as above separately per fixed M, b_1, b_2 and initial lower data, moving contours term by term at fixed hole counts; disjoint bindings just convert factorials by the matching choices. The bounds apply absolutely including these terms. On straight contours in a complex strip of large real periods all the expressions continue analytically with locally uniform convergence by the non-sharp bounds. Thus the identities for each original count M continue there, including vanishing after summing a sector $M > 2n$. We can bend terms as stated; their sum over M with only the lower constraint $M \geq \max_c g_c$ is exactly $\mathcal{N}_n(e^{-L/2}, s)$. Extending this count sum to all integers is a subsequent approximation, with the same count-independent prefactors and negligible error in the sharp setting (using the unresummed bounds). If m is at most a small fraction of n , the omitted Gaussian tail costs order n^2/R ; otherwise keep part of the box cost, at least a constant times m^2/R . Unlocalized factorial measures even with $\exp(Cm)$ sum to $\exp(O(R))$, so either error is negligible.

Indeed binding p indistinguishable pairs converts the factorials for the two counts to those for the residual counts and $p!$; given the final counts they and the winding differences impose no new M constraint. Both error bounds here (relative to the common unsummed bulk factors) are negligible also against any $\exp(-CR)$ on compact tilt ranges. Straight-contour holomorphy can first be used at fixed angular parametrizations, factoring the pure contact zeros and summing exponentially decreasing Fourier remainders locally uniformly for each fixed term, before continuing the torus as above.

Count summation and localization. Poisson summation in M now gives the prefactor $(\pi/(a\varepsilon))^{1/2}e^{-s(M_0-n)}$ and a sum over integers l with phases of modulus one (M_0 real). Its exponential combined with the pole log is

$$\exp \left[-c_0 \int_{\gamma} (w_* - l - is/(2\pi))^2 dz \right].$$

Extract $\exp(s^2/(4a\varepsilon))$. What is left costs in real log at least

$$c_{11} \|w_R - l\|_2^2 - Cm - C\sigma^2 \|b\|_2^2$$

(as suppression). Indeed the real linear coefficient relative to the square for real arguments, including the differential $1 + i\chi'$, has absolute value bounded strictly below 1, since $|\Im s| < \pi$. For any such coefficient t , $x^2 + tx \geq c_{12}x^2 - C \text{dist}(x, \mathbb{Z})$ on the real line. The smoothed real step is within Cb of integer values, and replacing it by the complex smoothing costs only the controlled small errors above and a fraction of the square. Box costs again absorb quadratic errors.

Indeed (choosing the sign of l) the count transform's square exponent is $(s - i\mathcal{S} - 2\pi il)^2/(4a\varepsilon) = -c_0 L(\bar{w}_* - l - is/(2\pi))^2$, completing the variance term. After the s^2 extraction the real linear coefficient on replacing w_* by w_R is $t = (\Im s + \chi' \Re s)/\pi$; slopes can be taken small depending on the prescribed compact tilt set. The additional replacement errors in the square and linear terms integrate against $1 + i\chi'$, and are bounded by a small fraction of $\|w_R - l\|_2^2$ plus $O(\sigma^2 \|b\|_2^2 + m)$.

Finally use $g_0(r) = 2 \log(2 + \text{dist}(r, \text{edge}))$. Integration by parts against the jumps gives

$$\sum j_i g_0(r_i) = - \int (w - l) g'_0 dr \leq \frac{1}{2} c_{11} \|w_R - l\|_2^2 + C + Cm.$$

Only O has negative j , and the external cost for O absorbs the wrong sign and leaves localization for it as well. Thus one keeps factors $\exp(-g_0(r_i))$ on all objects, of bounded integral each. The

unused variance penalty sums boundedly over l . All defect sums with their bounded activities and factorials are now $O(1)$. The Gaussian sum over $b_1 - b_2$ costs $O(\sqrt{R})$, offset together with the Poisson prefactor by the original $2\pi/L$. This completes the bound, leaving only the bulk factors and the extracted harmonic twist.

6.8 Bulk calibration and puncture exponent

The defect sum is bounded after extracting its explicit pressure, determinant and harmonic twist. We now compute these factors and compare them with Proposition 6.1; this completes Theorem 1.4.

For the overall determinant, Euler summation gives

$$-\sum_{m>0} \log \det A(m\varepsilon) = L/4 + O(1).$$

Indeed the eigenvalues are $(2 \cosh v - 1)(\cosh 3v \mp 1)/\sinh 4v$; the negative log determinant extends smoothly at zero, decays exponentially, and its integral on the positive line is $\pi^2/8$, by integrating $2[\log(1 - e^{-8v}) - \log(1 - e^{-6v}) - \log(1 - e^{-v} + e^{-2v})]$. Bounded Euler error follows also on the slightly rotated rays by the derivative bound there.

Put $I = \int_0^\infty H_*(v) dv$. Poisson summation for the even extension shows

$$\varepsilon \left(\frac{1}{2} H_*(0) + \sum_{m>0} H_*(m\varepsilon) \right) = I + c_* e^{-L} + O(e^{-4\Re L/3})$$

for an absolute constant c_* . The transform at positive frequencies $2\pi k/\varepsilon$ can use the same tilted line shift as above, here passing the first two simple poles at $i\pi/4, i\pi/3$ (removable poles, if any, simply give zero), which gives this estimate including bounded imaginary parts of the period. For the bulk integral use $y = e^{-v}$:

$$v H_*(v) = \frac{2y^2(1-y^3)^2}{(1-y+y^2)(1-y^8)} = \frac{P(y)}{1-y^{24}}$$

with symmetric coefficients $P_j = P_{24-j}$, sum zero; at $j < 12$ the nonzero ones are $P_2 = P_3 = 2$, $P_5 = P_6 = -6$, $P_8 = P_9 = 8$, $P_{10} = 2$, $P_{11} = -6$. Integrate groupwise against dv/v (absolute convergence by the double zero of P). Pairing the resulting log products symmetrically about 12 gives

$$I = -\sum_{j<12} P_j \log \sin(\pi j/24), \quad \exp I = 8D_0^3/81.$$

Here one uses $\prod_{l \geq 0} (l+x)(l+1-x)/(l+1/2)^2 = \sin(\pi x)$; e.g. log differentiation gives $\pi \cot(\pi x)$ by its elementary partial fractions (residue integration on expanding half-integer squares), and the normalization is at $1/2$. To simplify, the sines at 2 and 10 have product $1/4$, at 5 and 11 product $(1+\sqrt{2})/4$, at 3 and 9 product $\sqrt{2}/4$.

Multiply the analytic numerator $\mathcal{N}_n(\mathbf{q}, s)$ by $\exp(-c_* n^2 \mathbf{q}^2)$. On $|\mathbf{q}| = n^{-3/4}$ the bounds proved above apply around the entire circle, with the pressure error after multiplication bounded in log. The correcting factor equals one at zero, so Cauchy's formula gives

$$\mathcal{N}_n(0, s) = \frac{1}{2\pi i} \int_{|\mathbf{q}|=n^{-3/4}} e^{-c_* n^2 \mathbf{q}^2} \mathcal{N}_n(\mathbf{q}, s) \frac{d\mathbf{q}}{\mathbf{q}}.$$

This bounds the original numerator $S_{2n}(1)^2 X_{n,n}(\kappa = e^s)$. The κ -free part of B_n times the bulk growth is exactly $D_0^{-2n} (4D_0^2/9)^{2n^2}$, the squared exponential factor in the denominator bound.

The remaining factors besides constants are bounded by $|\exp[L/4 + s^2/(4a\varepsilon)]|$. Since $a = 9/8$, $\varepsilon = 2\pi\alpha/L$, and $\alpha = \pi/4$, their powers on this circle satisfy, uniformly on each compact tilt set,

$$\left| \exp[L/4 + s^2/(4a\varepsilon)] \right| \leq Cn^{3/8 + 2\Re(s^2)/(3\pi^2)}, \quad \frac{3}{8} - 2 \cdot \frac{5}{48} = \frac{1}{6}.$$

Thus division by the squared homogeneous Pfaffian gives, locally uniformly in $|\Im s| < \pi$,

$$|E_n(s)| \leq C, \quad E_n(s) = X_{n,n}(\kappa = e^s) \exp[-(1/6 + 2s^2/(3\pi^2)) \log n].$$

At $s = i\pi/2$ this analytic function is exactly 1. Hence all these puncture polynomials as functions of $\eta = 2 \cosh s$ are zero-free in a fixed disk about $\eta = 0$, with Taylor coefficients of each fixed order in $\log X_{n,n}$ bounded by $O_k(1 + \log n)$ (Cauchy estimates for $\log E_n$ near the normalization point). Also

$$X_{n,n}(1) = n^{1/6 + o(1)}$$

where 1 denotes $\kappa = 1$, thus fugacity 2. For the lower bound take any subsequence and a further locally uniform limit of E_n in the strip. It is nonzero at the normalization point and hence nonzero also at points $s = it$ with $t > 0$ arbitrarily small. There $2 \cos t < 2$, so coefficient positivity gives the lower bound with exponent $1/6 - 2t^2/(3\pi^2)$. The upper bound uses $s = 0$ directly.

Here compact subsequential convergence follows e.g. from derivative bounds by Cauchy's estimates followed by a compact exhaustion and diagonal selection, with holomorphy of the limit. The same bounds at $i\pi/2$ give a neighborhood where $|E_n - 1| < 1/2$ uniformly, and η has a local analytic inverse parameter there since its derivative is nonzero.

7 Localization of surrounding polygons

We pass from the cylinder observable to polygons surrounding one face in the plane. Write $Z(R)$ for the planar gas at fugacity 2 of mutually disjoint honeycomb polygons enclosing the face center 0 and contained within distance R of it. Changing the disk cutoff by a fixed factor will not affect the exponent.

Proposition 7.1. *The disk partition function satisfies*

$$Z(R) = R^{1/12 + o(1)}.$$

There are two comparisons with the balanced cylinder of circumference $p = 2n$. Two small disk systems, one near each marked face, give the upper bound directly from $F_n(2) = n^{1/6 + o(1)}$. For the lower bound we must remove the cylinder polygons that cannot be put into two disk systems of radius $p^{1+\delta}$, for fixed $\delta > 0$. These are the essential polygons and the contractible polygons with oversized planar lifts. We will show that their total single-polygon weight is bounded, so removing them costs only a bounded multiplicative factor. The difficulty is the second family: a contractible polygon may travel far before meeting a horizontal translate.

7.1 Weighted incompatibility from the zero-free disk

Let $r_0 > 0$ be a common radius of a zero-free disk for $F_n(\eta)$ about $\eta = 0$. Available polygons on a fixed balanced cylinder have weights $w(P) = x_*^{|P|}$. Write $I(P, Q)$ for incompatibility, including $I(P, P) = 1$.

Lemma 7.2. *For every nonempty finite family $\mathcal{H}$ of available polygons,*

$$M_{\mathcal{H}} = \sum_{P \in \mathcal{H}} w(P), \quad d_{\mathcal{H}} = \frac{\sum_{P, Q \in \mathcal{H}} w(P) I(P, Q) w(Q)}{M_{\mathcal{H}}} \leq r_0^{-1}.$$

In particular, the bound does not depend on the circumference or the family.

Proof. The formal logarithm of a hard-core gas is the connected-graph expansion; see Fernández and Procacci [8, Eqs. (2.2)–(2.4)]. For a tuple $(P_1, \dots, P_k)$, its incompatibility graph has the distinct tuple positions as vertices, even when polygons repeat. Its contribution to the coefficient of η^k in $\log F_n(\eta)$ is $\prod_i w(P_i)/k!$ times the alternating sum over its connected spanning edge sets. This follows by expanding the products of $1 - I$ and grouping components. Finite total polygon weight makes the sums absolutely finite coefficientwise.

The alternating sum, multiplied by $(-1)^{k-1}$, counts acyclic orientations with unique source at vertex 1, by the fixed-root chromatic-polynomial identity of Greene and Zaslavsky [14, Theorem 7.3]. In particular, all tuples contribute with the same sign. One can check the identity by deletion–contraction at edges incident to the root: the alternating sum is the linear chromatic coefficient, and the orientation count has the same addition rule. Delete a root edge when its other endpoint has another incoming edge; contract it otherwise. These give the corresponding rooted orientations, with parallel edges simplified, and the recursion ends at the isolated-root base cases.

Every Hamilton path starting at vertex 1 supplies one such orientation, by orienting all edges in its path order. Different paths give different orientations, since the directed Hamilton path forces the total order. Summing the $(k-1)!$ orders of the remaining tuple positions therefore gives

$$|[\eta^k] \log F_n(\eta)| \geq \frac{1}{k} \sum_{P_1, \dots, P_k \in \mathcal{H}} \prod_{i=1}^k w(P_i) \prod_{i=1}^{k-1} I(P_i, P_{i+1}). \quad (7.1)$$

Thus a log coefficient bounds a weighted walk count in the incompatibility graph, not merely the mass of individual incompatible pairs.

For completeness, use the weighted inner product $\langle f, g \rangle_w = \sum_{P \in \mathcal{H}} w(P) f(P) g(P)$ and the symmetric operator

$$(Af)(P) = \sum_{Q \in \mathcal{H}} I(P, Q) w(Q) f(Q).$$

The sum in (7.1) is $\langle \mathbf{1}, A^{k-1} \mathbf{1} \rangle_w$. For odd k , the spectral moment inequality for the even power $k-1$ gives

$$\langle \mathbf{1}, A^{k-1} \mathbf{1} \rangle_w \geq M_{\mathcal{H}} d_{\mathcal{H}}^{k-1}.$$

Now freeze both $\mathcal{H}$ and its containing cylinder, and let odd k tend to infinity. Since $\log F_n$ is analytic in $|\eta| < r_0$, its coefficient root growth is at most r_0^{-1} , proving the lemma. Only the radius is uniform here; no uniformity in k is required of the fixed-order $O_k(1 + \log n)$ coefficient bounds. $\square$

To use this lemma for oversized polygons, we will force incompatibilities between two such polygons after rotating one of them. If all six rotations of the smaller polygon were compatible with the larger one, their interiors would all lie inside its period-free interior. The next geometric lemma rules this out at radius greater than $p^{1+\delta}$.

7.2 The six-rotation radius bound

Write $\omega = e^{i\pi/3}$ and $\Lambda = \mathbb{Z}[\omega]$ for the triangular face-center lattice.

Lemma 7.3. *Let V be the open Jordan interior of a honeycomb polygon Q enclosing 0 . Suppose that an open set U contains $\omega^r V$ for $0 \leq r \leq 5$, and that*

$$U \cap (U + kp) = \emptyset \quad (k \in \mathbb{Z} \setminus \{0\})$$

for $p \geq 1$. If $D = \max_{z \in Q} |z|$, then

$$D \leq Cp \log(D + 2)$$

with an absolute constant C .

Proof. Choose a piecewise linear path γ in V from 0 to radius $D - O(1)$, with $O((D + 1)^2)$ bounded-length segments. Such a path follows the connected set of interior faces and can then be perturbed while keeping clearance from Q . Except at its start, require it to avoid

$$\bigcup_{r=1}^5 (1 - \omega^r)^{-1} p\Lambda.$$

Take its endpoint and intersections generic. The endpoint will then project to a regular point in the interior of a quotient triangle below. Bounded D can be absorbed in the constant.

From horizontal separation to lattice separation. Let $S(t)$ be the union of the six rotated prefixes of γ up to time t . We claim that

$$S(t) \cap (S(t) + \lambda) = \emptyset \quad (\lambda \in p\Lambda \setminus \{0\}). \quad (7.2)$$

Suppose instead that there is a first contact. Only finitely many relative translates can meet these bounded networks. A contact between tips at different centers would put the current point of γ in one of the excluded fixed-point sets. Every contact is therefore between a tip and an older point. A given tip can meet older points from only one other center: otherwise those older points already gave a contact at an earlier time.

Rotations and translations act transitively on all tips. Consequently the contact graph on the centers $p\Lambda$ has the complete orbit of edges with displacement l , for some $l \in p\Lambda \setminus \{0\}$, and no other displacement orbit. Its components are the cosets of $l\Lambda$ in $p\Lambda$. The networks carrying different components are disjoint, while each component is connected by its contact edges. If there were two components, one would contain a path continued periodically in direction l , inside a band of finite width, and another such a path in direction ωl . Left-to-right and bottom-to-top crossings of a sufficiently large parallelogram in these two directions must meet, a contradiction. Thus $l\Lambda = p\Lambda$, so l/p is a unit of Λ . Since the edge set contains the full rotation orbit, it contains a direct contact with horizontal displacement p . Both networks lie in the corresponding translates of U , contrary to the hypothesis. This proves (7.2).

The physical quotient and the auxiliary slope. The physical quotient

$$O = \mathbb{C}/(p\Lambda \rtimes \langle \omega \rangle)$$

is a sphere with cone marks A, B, C of orders $6, 3, 2$. It is the doubled flag triangle of the period-lattice equilateral tiling: the three vertices are a tiling vertex, a triangle center and an edge midpoint. The image of 0 is A . Mark also the regular endpoint projection q .

If two points of γ have the same image in O , their lifts differ by a rotation about 0 followed by a period translation. The translation is zero by (7.2). Radius is therefore single-valued on the

projected path. Split its generic segments at their intersections and erase loops in this finite graph to obtain a simple arc α from A to q , avoiding B, C . Lifting the retained pieces amounts to rotating tails about 0, so its terminal lift still has radius $D - O(1)$. The number of pieces is polynomial in $D + 1$.

We now classify this arc in an auxiliary, topological model. Set

$$\mathcal{P} = \mathbb{R}^2 / (2\mathbb{Z}^2 \rtimes \{\pm 1\}), \quad \mathcal{T} = \mathbb{R}^2 / (2\mathbb{Z}^2).$$

The pillowcase $\mathcal{P}$ is a sphere, and $\mathcal{T}$ is its double cover branched at the four corners. Identify the four marked points of O with these corners, with A at 0. This is the familiar four-punctured-sphere model for primitive torus slopes; see Farb and Margalit [7, Sections 1.2 and 2.2.5]. We give the arc comparison needed here, since it must preserve radius in the physical cover, not length in this auxiliary one.

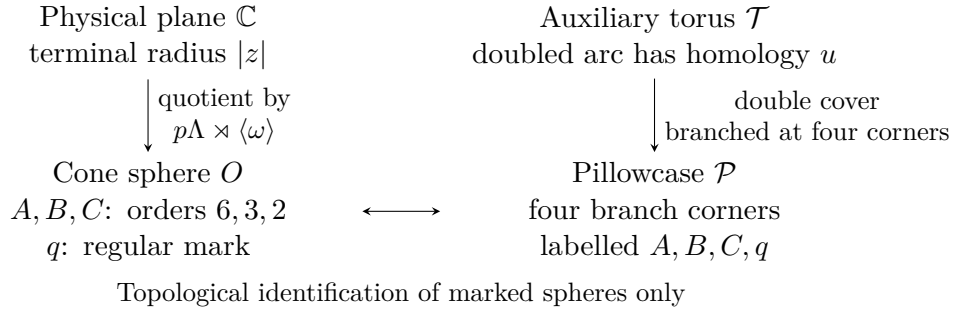


Figure 3: The two covering constructions in the radius bound. The auxiliary torus classifies the endpoint arc by a primitive vector u ; its physical terminal radius is measured in the plane on the left. The identification of the marked spheres does not identify their metrics or their covers.

Use the tetrahedral triangulation of O obtained by joining q to the three vertices of its flag-triangle face. On $\mathcal{P}$, split the two squares by opposite diagonals, and identify corresponding triangles affinely. Each bounded physical segment meets boundedly many period tiles because $p \geq 1$, and boundedly many of their subdivisions. Its image in each auxiliary triangle is a segment of bounded length. Hence the image of α has length polynomial in $D + 1$, uniformly in q . This uses bounded piece counts and triangle diameters, not a uniform derivative bound for the affine identifications when a triangle becomes thin.

Lift α in the auxiliary plane from 0 to an integer point u , whose parity is the terminal corner. The two lifts in $\mathcal{T}$ form a simple closed curve; a plane lift of this curve runs from 0 to $2u$. Its homology vector in the period-2 basis is thus u , and $|u| \leq (D + 2)^{C_0}$ for some absolute C_0 . It is nonzero by parity and primitive: a simple curve of nonzero homology on a torus is nonseparating, and cutting along it supplies a dual curve of intersection number one. Notice that u is not the displacement of the original physical lift.

The straight segment from 0 to this primitive u projects to a simple arc with the same endpoint corners. It has no internal corner, and $(s \pm t)u \in 2\mathbb{Z}^2$ forces $s \pm t \in 2\mathbb{Z}$, which rules out any other identification of its interior points. To compare it with α , take the boundary of a narrow disk neighborhood of each arc, containing just its two endpoint marks. Each boundary lifts to a pair of parallel torus curves of slope u .

Put these two sphere boundaries in minimal transverse position relative to the four marks. If they intersect, their number of crossings is even and at least two. Their union has no cut vertex: after deleting any crossing, the two remaining paths still meet at another crossing. The complementary

faces are therefore disks bounded by simple even-sided polygons, and each bigon has distinct vertices. Euler's formula gives

$$\sum_{\text{faces } F} (4 - \# \text{sides of } F) = 8.$$

An unmarked bigon could be removed by an isotopy. Since there are only four marks, the formula forces exactly four bigons, each containing one mark, and all other faces to be quadrilaterals. In the branched double cover every marked bigon becomes a quadrilateral, while each unmarked quadrilateral lifts to two. The resulting torus has a quadrangulation of valence four. Gluing unit squares gives a flat torus on which the two curve families run straight and cross perpendicularly. Its complete simply connected flat cover develops onto the Euclidean plane; its deck maps are translations, since they are orientation-preserving and fixed-point-free. The crossing families consequently have different homological slopes, contradicting their common slope u .

The boundaries can thus be made disjoint. As they enclose the same pair of marks, the annulus between them contains no mark, and they are isotopic relative to the marks. This disk isotopy preserves terminal lift radius in O . Indeed a disk containing only an endpoint arc and the marks A, q lifts above it to disks branched at the preimages of A . In the component about 0, the only monodromy is rotation about 0; q is regular. All arcs in this disk from A to q therefore have equal terminal radius. Such disk neighborhoods apply locally throughout the isotopy, and the same observation compares the two arcs once they lie in one disk. Thus the straight auxiliary arc has the same physical terminal radius as α .

Reducing the shear word by cone monodromy. The Euclidean algorithm sends the unit horizontal vector to u by a word in powers of the horizontal shear $H(x, y) = (x + y, y)$ and quarter turns, with $O(1 + \log |u|)$ blocks. These integer linear maps act on the pillowcase. Every shear exponent can be reduced modulo 12 without changing the terminal physical radius of the final endpoint arc.

To see this, H^2 is isotopic relative to the corners to one Dehn twist about a curve separating them into pairs. Deform $(x, y) \mapsto (x + 2y, y)$ to $(x, y) \mapsto (x + F(y), y)$, where F is odd, $F(y + 2) = F(y) + 4$, is flat at $2k$ near each integer row $y = k$, and rises by 2 between successive rows. The two twisting strips on the torus are identified to one annulus on the pillowcase. For any curve separating the four marks into pairs, its side containing q contains exactly one cone mark. Its monodromy in the physical cover has order dividing 6. Six twists consequently insert only trivial six-turn based loops at all crossings of an endpoint arc with the twisting annulus.

When a shear block is replaced inside a word, the final arcs differ by a conjugate of such a sixfold twist. Its supporting curve still separates the marks into pairs, and the conjugated twist fixes each mark. This justifies the reduction for internal blocks as well as terminal ones: the endpoint marks remain the same at every comparison. The shortened word has length $O(\log(D + 2))$ in a fixed finite collection of matrix maps.

An additive radius cost for each pure generator. Some matrices permute the corners, and these cannot all be lifted as maps of the physical plane with its differently ordered cones. Separate that permutation first. Choose fixed representatives for the matrix classes modulo 2. If $W = g_1 \cdots g_m$ is the shortened word and r_i represents the class of $g_1 \cdots g_i$, with $r_0 = 1$, then

$$h_i = r_{i-1} g_i r_i^{-1}, \quad W = h_1 \cdots h_m r_m.$$

The h_i belong to a fixed finite collection of pure, corner-fixing matrices. The representative r_m acts first on the unit arc; it has the same corner permutation as W , so the resulting arc runs from

A to q . Its bounded complexity gives an initial physical radius $O(p)$. We do not assert a global physical-plane lift for this nonpure map.

Each pure map, transported to O , does lift to a homeomorphism $\tilde{f}$ of the physical plane fixing 0. Off the four marks, the lifting subgroup is normally generated by the peripheral-loop powers of orders 6, 3, 2, together with the loop about the regular mark q : upstairs these are circuits of individual punctures. A pure map preserves these generators up to conjugacy. It therefore preserves the subgroup, lifts off the punctures, and extends across them by the local cyclic-cover picture.

Conjugation by $\tilde{f}$ preserves the translation subgroup of the deck group: its nonidentity elements are exactly the infinite-order deck transformations. Thus for a real-linear lattice automorphism L ,

$$\tilde{f}(z + \lambda) = \tilde{f}(z) + L\lambda, \quad \lambda \in p\Lambda. \quad (7.3)$$

It also intertwines rotation about 0 by ω with rotation by ω or ω^{-1} . Such a real-linear L is a similarity; as it is a lattice automorphism, $|\det L| = 1$, so it is orthogonal.

The remaining bound is on one period cell, uniformly in the regular mark q . From 0 to a generic point of that cell, choose boundedly many straight pieces in an $O(p)$ region, avoiding lifted marks after departure. They cross boundedly many flag triangles and their subdivisions, even when some subtriangles are thin. After the affine identification, the path has boundedly many bounded straight pieces in the auxiliary square model. Applying a fixed matrix and subdividing again preserves that bound. Mapping back, every piece lies in a physical triangle of diameter $O(p)$, so the path lifts to length $O(p)$. Limits handle nongeneric points. This also proves the bounded-complexity estimate for the initial representative arc. It follows that $|\tilde{f}(z_0)| \leq Cp$ on a period cell at 0, with one constant for the finite collection of pure generators.

Write $z = z_0 + \lambda$. Equation (7.3) and orthogonality now give the global *additive* bound

$$|\tilde{f}(z)| \leq |z| + C'p.$$

Starting from the representative arc and applying the $O(\log(D+2))$ pure lifts, we obtain $D - O(1) \leq Cp \log(D+2)$. Since $p \geq 1$, this proves the lemma. $\square$

7.3 Removing oversized polygons

The geometric lemma also gives a form that will be used for polygons in a strip. Here a pure lattice period may point in any direction; it need not be the horizontal period of the original balanced cylinder.

Corollary 7.4. *Fix $\delta > 0$. Suppose $\mathcal{U}$ is a family of planar honeycomb polygons enclosing 0 whose interiors are disjoint from their nontrivial translates by a common pure lattice period of length p . For all sufficiently large p ,*

$$\sum_{\substack{P \in \mathcal{U} \\ \max_{z \in P} |z| > p^{1+\delta}}} x_*^{|P|} = O(1).$$

The constant is independent of the family and the period direction.

Proof. For the geometric argument, rotate coordinates to make the given period horizontal; the six-rotation lemma is unchanged by this rotation. For the weighted-degree argument, use an ordinary balanced cylinder in the original lattice coordinates. Take a finite subset of oversized polygons, of mass B , and put it and all six lattice rotations into this cylinder with a sufficiently large circumference, so that every polygon encloses one marked face and not the other and has its

original weight. This containing cylinder need not have the given period, and its circumference may depend on the finite subset. Lemma 7.2 is uniform in that choice.

For two chosen polygons P, Q with area of P at least area of Q , suppose all rotations of Q are compatible with P . For each rotation, its Jordan boundary and that of P are disjoint and enclose 0, so its closed disk lies strictly inside the disk of P , by the area ordering. The interior of P can therefore serve as U in Lemma 7.3, giving $D_Q \leq Cp \log(D_Q + 2)$. For fixed $\delta > 0$ and large p , this contradicts $D_Q > p^{1+\delta}$.

The area-ordered pairs have total weight at least $B^2/2$. Assign to each one an incompatible rotation of Q . A pair in the union with rotations has at most six preimages, so its incompatibility numerator is at least $B^2/12$, while the union has mass at most $6B$. The weighted-degree bound gives $B \leq 72r_0^{-1}$. Exhaustion by finite subsets proves the claim. $\square$

Proof of Proposition 7.1. On the cylinder of circumference $p = 2n$, essential polygons separating the marked faces fall into two families, according to which cylinder end is on the same side as each mark. Opposite families cannot coexist, and translation by n , swapping the marks, exchanges their weights. If a finite subset of one family has mass B , its union with its translated image has mass $2B$ and incompatibility numerator at least $2B^2$. Lemma 7.2 gives $B \leq r_0^{-1}$. Thus the total single-polygon weight of essential polygons is $O(1)$.

Every remaining cylinder polygon is contractible. Lift its disk and translate the enclosed marked face to 0. Its interior is disjoint from all nonzero translates by the horizontal period p . For each of the two marked-face families, Corollary 7.4 shows that the polygons whose lifted radius exceeds $p^{1+\delta}$ have total weight $O(1)$.

For the upper comparison, put two independent planar disk gases of radius cp , for a fixed sufficiently small $c > 0$, around the two marked faces. Their disks are disjoint on the cylinder, so all their polygons qualify and

$$Z(cp)^2 \leq F_n(2) = n^{1/6+o(1)}. \quad (7.4)$$

For the lower comparison, remove essential and oversized polygons from a cylinder configuration. Ignoring every avoidance restriction involving the removed polygons bounds their possible contribution at fugacity 2 by

$$\prod_{P \text{ removed}} (1 + 2w(P)) \leq \exp \left(2 \sum_{P \text{ removed}} w(P) \right) = O(1).$$

The retained polygons lift into the two planar disk systems of radius $p^{1+\delta}$. Dropping restrictions between these systems yields

$$F_n(2) \leq C_\delta Z(p^{1+\delta})^2. \quad (7.5)$$

The upper comparison gives exponent at most $1/12$, and the lower one gives exponent at least $1/(12(1+\delta))$. Monotonicity fills the radii between the cylinder scales. First let the radius tend to infinity for fixed δ , then let $\delta \downarrow 0$, proving the proposition. $\square$

In the strip application below, a transverse lattice period of length comparable to the strip height makes each polygon interior period-free. Corollary 7.4 therefore supplies the same oversized-polygon removal before Proposition 7.1 is applied to the retained disk system.

8 Vertex visits and finite chord means

The analytic argument has produced a partition function for polygons. We now insert a visited vertex into a path. The conversion is a finite exploration identity, valid before any infinite-volume

limit. Its positive expansion will also justify the deletion comparison needed for the later transfer from arches to bridges.

A finite vertex set D is *filled* if every lattice vertex strictly inside a simple polygon in its induced honeycomb graph also belongs to D . Boundary ports are midpoints of edges with one endpoint in D and one outside. An excursion joins two distinct boundary ports and visits distinct vertices of D . We order its endpoints, so reversal gives a second excursion. For $v \in D$, define

$$T_D(v) = \sum_{\gamma \ni v} x_*^{|\gamma|} e^{i\sigma W(\gamma)}, \quad \sigma = 3/8.$$

Reversal conjugates each term, so $T_D(v)$ is real. Its individual terms need not have positive real part in a general filled set. For a face f incident to v , let $Z_D(f)$ be the sum over systems of vertex-disjoint polygons in D enclosing its center, with weight $2x_*^{|P|}$ per polygon and with the empty system included.

Proposition 8.1 (Finite visit recursion). *For every finite filled vertex set D and $v \in D$,*

$$\begin{aligned} T_D(v) = 3 + \sum_{\substack{P \subset D \\ v \in P}} 2x_*^{|P|-1} [-\cos(\sigma(2\pi - \tau_v(P)))] \\ + \sum_{\substack{P \subset D \\ v \in P^\circ}} 2x_*^{|P|} T_{P^\circ}(v). \end{aligned} \quad (8.1)$$

Here P° is the strict interior vertex set and $\tau_v(P) \in \{\pi/3, -\pi/3\}$ is the counterclockwise turn at v . Every coefficient on the right is positive. Thus $T_D(v) \geq 3$, it is monotone under inclusion of filled sets containing v , and there are absolute constants $c, C > 0$ such that

$$c \sum_{f \sim v} Z_D(f) \leq T_D(v) \leq C \sum_{f \sim v} Z_D(f).$$

If D is cut out by admissible tiling lines in a convex polygon and $A_D(v)$ is the positive ordered excursion weight through v , then

$$\cos(3\pi/8)A_D(v) \leq T_D(v) \leq A_D(v).$$

Proof. At a fresh vertex reached along one edge, take either nonbacktracking continuation with weight $x_* e^{i\sigma\pi/3}$ or $x_* e^{-i\sigma\pi/3}$. Their sum is one. Stop at an exit port or just before choosing a continuation at the first repeated vertex. The exploration tree is finite and replacing a node by its children preserves total weight. A prefix stopped on first arrival at a vertex has not yet taken its weight or its turn.

Start at a boundary port. A first repeat consists of a simple stem followed by a polygon attached at its end. The stem is outside the polygon: otherwise the outside neighbor of its starting port would be inside that polygon, contradicting filling. The two orientations of the polygon add turns $\pm 4\pi/3$ relative to the stem. Their phase sum is $2\cos(4\pi\sigma/3) = 0$. This pairing also cancels after restricting to histories that have visited v , because reversal preserves their vertices. Continuing a first-arrival prefix at v has total terminal weight one. Hence the sum of first-arrival prefixes equals the excursion weight through v . Summing starting ports gives $T_D(v)$.

Now start at v along each of its three edges, with initial weight one and with neither weight nor turn at v . Stop at an exit, a return to v , or the first repeat of another vertex. The three explorations have total weight 3. Their exit terms reverse the prefixes just counted, so their sum is the conjugate

of $T_D(v)$ and hence $T_D(v)$. A return to v traces a polygon P through it. Its turn omits exactly $\tau_v(P)$ from the total 2π , and its weight omits exactly one factor x_* . The two orientations therefore contribute

$$2x_*^{|P|-1} \cos(\sigma(2\pi - \tau_v(P))).$$

For a first repeat away from v , let P be the attached polygon. If v is outside P , exterior-stem cancellation applies. If v is inside, the unused edge at the attachment points into the polygon. Arriving from the stem replaces the counterclockwise turn $-\pi/3$ there by $+\pi/3$. The two traversals add turns $\pm 8\pi/3$, both with phase -1 , and contribute $-2x_*^{|P|}$ times the stem weight. For fixed P , these stems are exactly paths from v , with no weight or turn there, to boundary ports of P° . Filling gives $P^\circ \subset D$; planarity shows that an edge leaving its interior ends on P . Completing the last half-edge therefore introduces no extra weighted vertex. Prefix reversal in this smaller filled set gives stem sum $T_{P^\circ}(v)$. The terminal balance is

$$3 = T_D(v) + \sum_{P \ni v} 2x_*^{|P|-1} \cos(\sigma(2\pi - \tau_v(P))) - \sum_{v \in P^\circ} 2x_*^{|P|} T_{P^\circ}(v),$$

which proves the recursion.

If P encloses one of the three faces incident to v , its turn there is $+\pi/3$; if it encloses two, the turn is $-\pi/3$. The omitted-turn angles are $5\pi/3$ and $7\pi/3$. Set

$$c_1 = -\cos(5\pi/8) > 0, \quad c_2 = -\cos(7\pi/8) > 0.$$

Iterate the recursion. It terminates since a strict interior has fewer vertices than its container. The result sums outer-to-inner chains of disjoint polygons strictly enclosing v , with factor $2x_*^{|P|}$ for each. In the final interior D' the remaining multiplier is

$$3 + \frac{2}{x_*} \sum_{\substack{Q \subset D' \\ v \in Q}} c_{m_Q} x_*^{|Q|}, \quad m_Q \in \{1, 2\}.$$

This positive expansion proves both positivity and filled-set monotonicity: inclusion only adds eligible chains and through-polygons.

The same strict chains enumerate $\sum_{f \sim v} Z_D(f)$. A polygon strictly enclosing v encloses all three incident face centers. A polygon through v encloses exactly m_Q of them. Conversely, a polygon enclosing an incident face has v in its interior or on its boundary, because the open face does not meet the polygon and approaches v . A vertex-disjoint system has at most one polygon through v , and it must be innermost. Thus its final multiplier is

$$3 + 2 \sum_{\substack{Q \subset D' \\ v \in Q}} m_Q x_*^{|Q|}.$$

The two ratios $c_m/(x_* m)$ are fixed positive numbers. Termwise comparison proves the incident-face bounds.

Finally close a convex-domain excursion by the complementary counterclockwise boundary arc. Its turn is the lifted change of outward boundary normal from the initial port to the terminal one, minus π . That change lies in $[0, 2\pi]$, so $|W| \leq \pi$. Each phased real part is therefore between $\cos(3\pi/8)$ and one times its positive weight. Summation proves the last assertion. $\square$

The proposition estimates visits without choosing the wall on which the excursion ends. We first apply it to hexagon chords and mixed strip exits; only afterward do we isolate bridges.

8.1 Exponential lateral tails in a strip

Before normalizing the finite chord laws, we control excursions that travel sideways while remaining in a thin strip. This estimate also justifies the subsequent lateral exhaustions with a length insertion.

Lemma 8.2 (Lateral tail). *Consider critical exits from a fixed bottom port in a strip of h layers, or in a convex tiling-line truncation of that strip. Let $X(\gamma)$ be the largest horizontal distance of a visited vertex from the source. There are absolute constants $C, c > 0$ such that*

$$\sum_{X(\gamma) \geq t} x_*^{|\gamma|} \leq C \exp\{-ct/(h+1)\} \quad (t \geq 0).$$

The same bound, with a polynomial prefactor, holds with a first-length insertion. In particular the contribution with $X \geq h \log^2 h$ to any first-length mass is smaller than every fixed inverse power of h as $h \rightarrow \infty$.

Proof. For the right tail, take two further tiling cuts with outward normals $(\sqrt{3}/2, 1/2)$ and $(\sqrt{3}/2, -1/2)$, in coordinates along and above the bottom side. Let E_+, E_- be the sets of old exits lost when the corresponding half-plane is imposed. Flux subtraction puts their phased sums on the distinct rays $e^{-i\sigma\pi/3}$ and $e^{-2i\sigma\pi/3}$: the turn from the source to a new cut is fixed. Differently directed cuts do not share ports. Since vertical coordinates range over $O(h+1)$, their intercepts can be staggered by $O(h+1)$ and rounded to the tiling grid so that

$$\{X_+ > r + C(h+1)\} \subset E_- \subset E_+ \subset \{X_+ > r\},$$

where X_+ is maximum rightward displacement. All old exit weights have phased real parts comparable to their positive weights by convexity. Separation of the two rays implies that the absolute weight of $E_+ \setminus E_-$ is at least a fixed positive fraction of the weight of E_- . This gives fixed-factor decay in each increment $C(h+1)$, starting from the uniformly bounded total exit mass. Reflection gives the left tail.

In this subtraction, an exit is lost exactly when it visits a removed vertex. A newly clipped exit attempts such a vertex along the new cut; an old exit still valid contributes identically to the two flux sums, even if its port belongs to both boundaries. Thus no boundary term was omitted by the nesting of the cuts. The constants are uniform in a convex lateral truncation, and monotone exhaustion gives the full-strip statement.

Finally a simple path in h layers and horizontal range X visits at most $C(h+1)(X+1)$ vertices. The tail integral of the preceding exponential bound consequently gives, after adjusting C, c ,

$$\sum_{X(\gamma) \geq t} |\gamma| x_*^{|\gamma|} \leq C(h+1)(h+t+1)e^{-ct/(h+1)}.$$

At $t = h \log^2 h$ this is smaller than any fixed power of h^{-1} . □

8.2 Macroscopic chords in a hexagon

For specificity take regular lattice hexagons of integer side $R \rightarrow \infty$. Say an arc is macroscopic when its diameter is at least $R/10$. Their absolute critical mass over all boundary sources is

$$R^{3/4+o(1)}.$$

For the lower bound use sources in the central third of one side. Exits not to the initial bottom side all have macroscopic reach. Flux and comparison of bottom returns with arches in a containing flat strip of $O(R)$ layers give non-bottom mass at least $b_{O(R)}$.

For the upper bound per source, impose a cut parallel to its starting side at height $h = \lfloor R/\log^2 R \rfloor$. Flux subtraction bounds the mass of exits removed by this cut by a constant times b_h , since its new exit prefixes are strip bridges. A retained macroscopic path must reach horizontal distance comparable to R , so Lemma 8.2 bounds its weight by $Ce^{-c\log^2 R}$. Summing over the $O(R)$ boundary sources proves the upper normalizing exponent.

Under the normalized macroscopic arc weights, bulk vertices (say in the central third) now have visit probability

$$R^{-2/3+o(1)}.$$

Indeed all visiting paths there qualify, and the surrounding-polygon sums in Proposition 8.1 lie between disk sums at comparable radii. Proposition 7.1 gives their exponent. The same upper bound holds throughout the hexagon. Summing the probabilities yields the mean visited mass, or length,

$$\mathbb{E}|\gamma| = R^{4/3+o(1)}.$$

8.3 A fixed-source strip version

There is also an annealed dimension statement with one actual boundary point fixed. In the full planar strip of h layers take all the critical exits from a fixed bottom port (thus bridges to the top or arches back to bottom), conditioned to get at least one quarter of the way up. Then

$$\mathbb{E}|\gamma| = h^{4/3+o(1)}.$$

Changing this to any fixed positive fraction strictly below one half gives the same result. The exit mass of this ensemble before normalization is $h^{-1/4+o(1)}$: lower by the bridges, upper for the arches by the height-tail bound. This conditioning could also confine horizontal range to $h\log^2 h$ about the start without changing the exponent. For bridges alone the argument gives the mean upper bound $h^{4/3+o(1)}$ (and hence the usual in-probability upper-exponent bound), not by itself the matching bridge lower bound.

The pointwise estimate behind this statement will also be needed for the bridge lower bound. It sums over all boundary positions, rather than fixing a source.

Lemma 8.3 (Strip visit weights). *Let $A_{S_h}(v)$ be the positive weight of ordered strip excursions through v , with both endpoints free on the two horizontal boundaries. Uniformly over strip vertices,*

$$A_{S_h}(v) \leq h^{1/12+o(1)}.$$

For every fixed $\delta > 0$, uniformly over vertices at least δh layers from both horizontal boundaries,

$$A_{S_h}(v) = h^{1/12+o(1)}.$$

The same upper bound holds for the ordered excursion weight $A_D(v)$ in every convex lateral truncation D , including excursions with lateral endpoints. Its lower bound holds uniformly when, in addition to the horizontal clearance above, v is at distance at least δh from every lateral side.

Proof. Convexly truncate the strip at lateral distance W on both sides using tiling lines of the two slanted side orientations. For a face within bounded distance of the strip, fugacity-2 systems using strip vertices and surrounding that face have sum at most $h^{1/12+o(1)}$, uniformly in the truncation. Indeed every enclosing polygon interior lies in the slab and is disjoint from its translates by nonzero multiples of $p\omega$, for a sufficiently large integral $p \asymp h$. Corollary 7.4 applies to this transverse period: its geometric argument rotates the period direction to horizontal, while its degree bound uses a

sufficiently large balanced cylinder containing any finite test subset and all its rotations. Loops of radius greater than $p^{1+\alpha}$ therefore cost at most a bounded exponential factor. The remaining loops are controlled by Proposition 7.1. First let $h \rightarrow \infty$ for fixed $\alpha > 0$, then let $\alpha \downarrow 0$. For faces at least a fixed multiple of h from the walls and the lateral ends, an inscribed disk gives the matching lower exponent.

Proposition 8.1 now gives both estimates in every finite convex truncation. We also need a tail bound for its side sources. Use the two slanted tiling cuts facing toward the far part of the strip, with both outward normals pointing laterally that way. Their distinct angles from the initial inward direction lie strictly between $-\pi$ and π , so the separated-ray argument of Lemma 8.2 bounds the weight reaching lateral distance t by $Ce^{-ct/(h+1)}$. Displacement toward the near end is limited by $O(h+1)$, and length is bounded by $C(h+1)(1 + \text{lateral span})$. Thus each side source has bounded length mass at fixed h , uniformly in the truncation width.

Let $W \rightarrow \infty$ at fixed h . Excursions between horizontal boundary ports exhaust the strip excursions. For a fixed marked vertex, paths from the $O(h)$ distant side ports have vanishing weight by the same tail bound. This proves the stated full-strip estimates. $\square$

We finish Proposition 1.3 by averaging over horizontal sources. In a finite truncation D , counting every visited vertex gives the exact identity

$$\sum_{v \in D} A_D(v) = \sum_{\gamma \text{ an ordered excursion in } D} |\gamma| x_*^{|\gamma|}.$$

Divide by horizontal width and let that width tend to infinity at fixed h . Lemma 8.3 bounds the result above by $h^{13/12+o(1)}$. A central band of $\asymp h$ layers supplies the same lower exponent. For the ensemble conditioned to reach ϑh layers above its starting wall, where $0 < \vartheta < 1/2$ is fixed, choose this band symmetric about the midline and more than ϑh layers from either wall. Every path contributing a visit to that band then meets the conditioning event, from either horizontal starting side.

The slanted side sources disappear in this average: there are only $O(h)$ of them, and the tail estimate in the preceding proof bounds each one's length mass at fixed h . Their contribution divided by width therefore vanishes.

Each horizontal wall has one source port per unit length. Translation invariance and top-bottom symmetry therefore identify the limiting average with twice the fixed-source length weight. One can first restrict to bounded path range, pass to the large-width limit, and then remove that restriction using the exponential tails. Thus the conditioned ensemble has unnormalized length weight $h^{13/12+o(1)}$. Its mass is $h^{-1/4+o(1)}$ by Proposition 5.1, giving the asserted mean. The same tails make the discarded first moment at horizontal range $h \log^2 h$ smaller than every fixed power of h , proving the optional confinement statement.

9 A marked tripod comparison for bridge mass

We have proved the mixed exit mean, but its length mass might still be carried mainly by arches returning to the same wall. This section proves that crossing bridges retain the bulk visit exponent and completes Theorem 1.1. The first step is an exact comparison of intersecting pairs, obtained by marking a tripod. The second is a geometric lower bound on how many translates of bridges hit a deep arch. Fix a vertex v in the central third of the strip of h layers. Sum over all boundary positions here. Count each bridge once (say bottom to top); let m be their weight through v . Count each same-side arch on either boundary also once, with total weight through v denoted by u .

Absolute weights are always products of x_* per visited vertex. Write

$$\theta = \pi/8, \quad k = \cos \theta, \quad c = \cos(3\theta) = \sin \theta, \quad p = e^{i\theta}, \quad g = \cos(2\theta), \quad \phi = 2g - 1 = c/k.$$

9.1 Deletion and first contact

An active outer path is phased by $\exp(i\sigma W)$ as before. It may be stopped just before first attempting entry to v , without using a weight there; call such prefixes η . The phased full exits through v sum to the sum on these prefixes by deletion flux. Likewise for first hit (meaning first attempted vertex, not taking its weight) of another obstacle. For disjoint obstacles v, X , the phased sum of full exits hitting both equals the sum of prefixes to first X that have visited v , plus prefixes to first v that have visited X : just use deletion for $v, X, v \cup X$. All actual path intersections below mean shared visited vertices.

At fixed h these operations in the unbounded strip can be done by absolute convergence. Indeed embed the strip by projection into a tilted cylinder, with a period along another tiling direction and large enough vertical shift for injectivity on the strip. On that cylinder strip-bridge weights in the longitudinal direction decay exponentially in the number of layers, as above. All SAWs from a fixed vertex there have absolutely summable critical weights with exponential tails in longitudinal span D : lengths are $O_h(D + 1)$, and the weak-bridge unfolding count (decomposition at extremal heights on the two halves, as above) uses subexponentially many lists with sublinear number of spans, whose sum is at least D . Positive weak bridges again extend by boundedly many vertices to cut lines without repetitions and each costs at most a constant times exponential decay in its span. Thus the loss besides that exponential is subexponential. This gives absolute convergence also with polynomial factors and for connected finite products of paths with one site anchored. Flux calculations can equivalently truncate laterally first (also when a fixed finite crosscut is removed).

Let J be the absolute sum for a bridge and an arch (arch counted once, on either side) intersecting each other with at least one through v , and J_{CC} the analogous ordered sum for two bridges. Each pair receives the product of its two path weights, so a vertex shared by the two independently chosen paths occurs in both factors. These sums are finite by the anchored convergence just proved. Our first target is

$$J \leq C(J_{CC} + m). \tag{9.1}$$

Once this comparison is established, lateral tails will bound J_{CC} in terms of m , while translations will bound J below in terms of u . The following complex sums are introduced solely to prove (9.1).

9.2 The marked tripod identity

Use triples of boundary ports a, b, c' , where $b < c'$ are ordered on one horizontal side and a is on the opposite. Positive order is east along the bottom, reversed along the top. We sum over both choices of side. A tripod consists of simple arms from these ports to a center vertex, mutually disjoint until the center, all weights included there once, the arm turn W_j from j measured up to arrival (before turning at the center). By planarity their cyclic order is forced. Since joining b to a or c' to a forms a bridge (turn zero in total), and the join turns are $+\pi/3, -\pi/3$ respectively,

$$e^{i\sigma W_a} = p e^{i\sigma W_b} = \bar{p} e^{i\sigma W_{c'}}.$$

Figure 4 shows the first-contact construction. Only the cyclic order, rather than the pictured shape, enters the phase relations.

Let L_* sum all tripods containing v , using the first (common) phase $e^{i\sigma W_a}$.

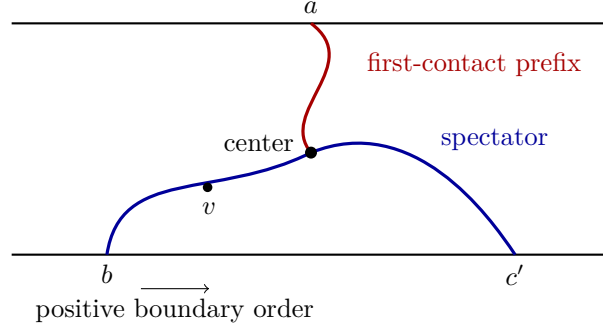


Figure 4: A contact tripod, shown schematically. The arch from b to c' is the spectator and a prefix from the opposite port a reaches it for the first time at the center. The center is weighted once. Reassigning the active arm gives the two bridge-spectator terms. The vertex v may lie on any arm or at the center; the restrictions in the proof keep these cases separate.

The following sums use independent spectator and active paths, with only the active path phased. A full exit includes all its visited vertex weights; a prefix η stops just before v and omits its weight. The port order and the sum over both choices of side are as above.

Sum	Spectator	Active path and contact condition
A	Arch $b \rightarrow c'$ containing v	Full exit from a , hitting the spectator
E_A	Arch $b \rightarrow c'$ avoiding v	Full exit from a , hitting both the spectator and v
R_A	Arch $b \rightarrow c'$ avoiding v	Prefix η from a , hitting the spectator before its stop
Z_b	Bridge $c' \rightarrow a$ containing v	Full exit from b , hitting the spectator
$Z_{c'}$	Bridge $b \rightarrow a$ containing v	Full exit from c' , hitting the spectator
E_C	Bridge avoiding v , counted once	Full exit from any boundary source, hitting the spectator and v
R_C	Bridge avoiding v , counted once	Prefix η from any boundary source, hitting the spectator

Set $Z = Z_b + Z_{c'}$. In E_C and R_C , the active source may lie on either boundary and may coincide with a spectator endpoint.

Hitting a marked spectator is represented by first contact by deletion. This gives exactly the contact tripod weights; in particular $A, pZ_b, \bar{p}Z_{c'}$ in common phase restrict L_* to v not exclusively on the open arm a, b, c' , respectively. The contact representation uses the free third edge for the incoming active prefix, since its source differs from the spectator endpoints. When the spectator avoids v , the term in which v is visited before first contact is instead the complementary restriction in L_* , with phase depending on the active arm. Thus

$$R_A = E_A + A - L_*, \quad R_C = E_C + Z - 2kL_*.$$

This uses the both-obstacles formula above; if the active source coincides with a bridge spectator endpoint, all exits and prefixes to v hit that spectator with no possible v before first contact, consistent with the second equation.

For another balance, let the spectator now be a prefix η from the top weighted with *conjugate* phase, and grow the opposite active path in $S \setminus \{v\}$. Its valid ends are an original outer boundary port or an attempt at v , stopped before taking its weight. Revisit pairs from an outer source still

cancel, with or without deleting the spectator vertices. Use also the bottom spectator and top active balance, add, and take real parts. The real sum of products for valid leaves hitting those spectators is

$$\Re(2cR_A + R_C) + 2P, \quad P = \Re \sum_{\eta_B \cap \eta_T \neq \emptyset} w(\eta_B)w(\eta_T)e^{i\sigma(W_B - W_T)}$$

where subscripts denote the lower/upper source and w denotes absolute path weight. Outer leaves are arches avoiding v (both orientations, averaging to phase c) or bridges avoiding v (phase 1); the other leaves are the intersecting prefixes.

Write u_B, u_T for the once-counted arch weights through v on the bottom and top, so $u = u_B + u_T$. By deletion, the two full prefix sums are the nonnegative real numbers

$$M_B = m + 2cu_B, \quad M_T = m + 2cu_T.$$

Disjoint opposite-side prefixes join at v to give each bridge through v once. Their product omits exactly the weight x_* at v , and their arrival-turn difference is $\pm\pi/3$, with real phase k . Subtracting these disjoint pairs from the full product gives

$$P = M_B M_T - \frac{k}{x_*} m \geq -\frac{k}{x_*} m. \quad (9.2)$$

At first contact the two shoulders from opposite sides form a bridge λ avoiding v . The incoming active edge is free: neither the edge toward the spectator's source nor the edge toward its stopped-at v could have brought the opposite active path to its first contact. The spectator suffix starts along the third edge from the center and goes to v , avoiding λ . The bridge takes the center weight once; the suffix takes neither that weight nor the weight at v . Conversely these objects give all contacts once per choice of active side.

Write $\delta = \pm\pi/3$ for the bottom-to-top join turn at the center, and W' for the suffix turn starting along its outgoing edge. The shoulder-turn difference is $-\delta$; the turn into the suffix from the top is δ , and from the bottom $-\delta$. The two phase factors therefore add to $2ge^{-i\sigma W'}$. For fixed λ , the real suffix sum without $2g$ is exactly the real sum of prefixes to v in $S - \lambda$ from the new boundary ports on the unused outward edges of λ . Length zero is included when appropriate.

Write $H(S; v)$ and $H(S - \lambda; v)$ for the real prefix sums to v from the respective whole boundaries. Original prefixes avoiding λ are unchanged, so the new-boundary suffix sum equals the original-boundary prefix sum hitting λ , minus

$$\Delta_\lambda = H(S; v) - H(S - \lambda; v) \geq 0.$$

For the inequality, reverse to exits from v and use the positive nesting expansion of Proposition 8.1, whose final multipliers are $3 + (2/x_*) \sum_Q c_{m_Q} w(Q)$. The domain $S - \lambda$ is filled: a removed bridge connected to the outer boundary cannot enter the strict interior of a polygon avoiding it. For fixed λ , take the nesting comparison first in convex lateral truncations with and without its vertices, and then pass to the strip; side contributions anchored at v vanish by the convergence already proved. When summing $w(\lambda)\Delta_\lambda$, use its representation as hit prefixes minus new-boundary prefixes. Both are connected terms; no sum of disconnected boundary terms is needed. The first-contact calculation consequently gives

$$\Re(2cR_A + R_C) + 2P = 2g \left[\Re R_C - \sum_{\lambda \not\equiv v} w(\lambda) \Delta_\lambda \right].$$

Using the earlier equations (the L_* terms cancel since $\phi k = c$),

$$2c\Re(A + E_A) + 2P \leq \phi\Re(E_C + Z).$$

9.3 Comparison of intersecting path pairs

We now compare the phased expressions with the positive pair sums. Including only bridge exits as active paths gives $\Re(A + E_A) \geq J$; the remaining exits have positive real weights. Also $\Re(E_C + Z) \leq 2cJ + 2J_{CC}$: allow active sources coinciding with a marked spectator endpoint, adding nonnegative real weight, and combine both orientations of every active path. Together with (9.2), the last phased inequality becomes

$$2c(1 - \phi)J \leq 2\phi J_{CC} + \frac{2k}{x_*}m.$$

Since $\phi < 1$, this proves (9.1).

Let μ_h be the unnormalized measure on bottom-to-top bridge shapes from one fixed bottom port, of total mass b_h . Write $X(\beta)$ for the largest horizontal distance of a visited vertex from that source, as in Lemma 8.2, and set $T = h \log^2 h$ for large h . If both bridge shapes have $X \leq T$, then for any fixed marked bridge the other shape has at most CT translations hitting it. Choosing which member contains v only changes the constant. Thus

$$J_{CC} \leq CTb_h m + R_h,$$

where R_h is the contribution of pairs with at least one $X > T$.

We bound this remainder absolutely, without an estimate on m . A fixed-source shape β_1 has at most $|\beta_1|$ translations placing a visited vertex at v , and $|\beta_1| \leq C(h+1)(X_1+1)$, where $X_i = X(\beta_i)$. For each such translate, a second shape has at most $C(X_1 + X_2 + 1)$ translations hitting it. Hence, uniformly in v ,

$$\begin{aligned} R_h &\leq C(h+1) \iint_{\max(X_1, X_2) > T} (1 + X_1 + X_2)^2 d\mu_h(\beta_1) d\mu_h(\beta_2) \\ &\leq C(h+1)(T+h+1)^2 e^{-cT/(h+1)} = O(h^{-100}). \end{aligned}$$

The second line follows by integrating the exponential tail in Lemma 8.2 for the first two moments of X , using $b_h \leq 1$ and splitting according to which $X_i > T$. Since $b_h = h^{-1/4+o(1)}$, we obtain

$$J_{CC} \leq h^{3/4+o(1)}m + O(h^{-100}). \quad (9.3)$$

9.4 Translation hits on a deep arch

The upper pair bound pays one bridge crossing mass for each allowed horizontal translation. We need a matching lower bound for bridges hitting each arch through the marked vertex. Here the error must be uniform in the arch, whose length and horizontal span are unrestricted.

Lemma 9.1 (Translated bridge hits). *Let α be any same-side arch meeting the central third of the strip. Let $\mathcal{H}_h(\alpha)$ be the total weight of strip bridges, summed over all bottom sources, that intersect α . For every fixed $\epsilon \in (0, 1)$,*

$$\mathcal{H}_h(\alpha) \geq h^{3/4-\epsilon-o(1)},$$

uniformly over all such arches. In particular $\mathcal{H}_h(\alpha) \geq h^{3/4-o(1)}$ uniformly.

Proof. By reflection in the strip midline, it suffices to treat a bottom arch. Translate its start to the fixed bottom source, put $W = h^{1-\epsilon}$, and consider its prefix up to first crossing layer $r = \lfloor h/4 \rfloor$. Such a crossing exists because the arch meets the central third.

Equal-height contact intervals. We first compare two prefixes α_0, β_0 from the bottom to their first crossings of the same cut. Parametrize them on $[0, 1]$; their heights lie strictly between the two cut heights internally. There is a compact connected set of equal-height parameter pairs containing $(0, 0)$ and $(1, 1)$, whose projections onto both parameter intervals are all of $[0, 1]$. Consequently its horizontal-difference image is an interval of translations giving contact. It contains the start and final offsets, and also an offset matching any prescribed point on either prefix with a point on the other.

This is the piecewise-linear mountain-climbing construction [21, Theorem 5.5]; we give the contact argument needed here. On the parameter square, triangulate so that the height difference is piecewise linear. For sufficiently small positive regular values its level set has exactly two boundary endpoints, near the diagonal corners, joined by an arc. Indeed the two prefix heights are strictly monotone near their extremal endpoints and stay away from these extremal heights on every compact interior parameter interval. A subsequential Hausdorff limit of the compact arcs is connected, lies in the zero set, and contains both corners. Such a subsequence can also be obtained by diagonal selection of the dyadic boxes met by the arcs. Each coordinate projection is connected and contains 0 and 1, hence equals $[0, 1]$. Taking horizontal differences proves the interval assertion. At integer translations, geometric contact of the lattice walks gives a shared vertex. A contact at the intermediate cut is also a shared vertex of the full walks, since that port lies inside their lattice edges.

Two direct alternatives. Suppose first that at some intermediate cut the arch prefix has two crossings at horizontal distance at least W . Every bridge shape then has at least $W - O(1)$ hitting translates. To see this, use the bridge crosscut's left/right side indicator at the two points, with any fixed convention for points on it. Their separation is an integer. Summing the signed indicator difference over integer translates telescopes to that integer in absolute value. Any discrepancy forces the intervening arch segment to meet the bridge. Summing bridge shapes therefore gives hitting mass at least $(W - O(1))b_h$.

Suppose next that the arch's displacement a at first crossing of some layer $1, \dots, r$ satisfies $|a| \geq W$. A bridge and its reflection about its start have first-crossing displacements d and $-d$ there, and

$$\max\{|a - d|, |a + d|\} \geq |a| \geq W.$$

For at least one member of this reflection pair, the contact interval between the start and final offsets yields $W - O(1)$ hitting translates. Reflection preserves weights, so again the hitting mass is at least a constant times Wb_h .

In the remaining case every first-crossing displacement is less than W and every same-cut crossing spread is less than W . Thus all cut crossings of the arch prefix lie within $2W$ of its start. The full prefix stays within $2W + O(1)$: inside a cut-to-cut layer a walk between ports follows the internal zigzag monotonically.

Wide bridge prefixes against a confined arch. We will show that fixed-source bridges of weight at least $h^{-1/4-o(1)}$ make lateral distance greater than $4W$ before their first crossing of layer r . We first bound the exceptional confined pieces that enter the marking argument.

For $h/10 \leq j \leq h/6$, let e_j be the weight of fixed-source j -bridges confined to lateral distance $4W$ from their start. Then

$$e_j \leq C \exp(-ch/W). \tag{9.4}$$

Surround the width- $8W$ tube by a tube of width $O(W)$ whose side walls alternate the two slanted upward tiling steps between successive horizontal layers, outside the desired confinement with a safety margin. Join these walls horizontally on top and bottom. Starting east along the positive bottom boundary, its unwrapped tangent angles stay in $[0, 2\pi]$: on the right wall they are $\pi/3, 2\pi/3$, and on the left $4\pi/3, 5\pi/3$. For a bottom source, closing an exit by the boundary gives turn $B - \pi$,

where $B \in [0, 2\pi]$ is the cumulative tangent turn from source to exit. Thus all original exits have turns in $[-\pi, \pi]$ and positive comparable real weights. This supplies the needed positivity even though the tube is not convex.

Now cut by lines with the two slanted upward normals, of angles $\pm\pi/3$ from vertical. Boundary-source leaf cancellation still holds after either cut, since a polygon cannot surround the bottom source. The genuinely new exits lie on the cut and stay in the convex wedge between its half-plane and the half-plane above the bottom, so their turns are fixed. The lost exit sums consequently lie on two separated rays. Place the intercepts at the source's lateral coordinate near $s + C'(W + 1)$, stagger them by $O(W + 1)$, and round to tiling lines. Since all lateral coordinates are restricted to $O(W + 1)$, the two lost events are nested between reaching heights $s + C(W + 1)$ and s . The separated-ray argument of Lemma 8.2, with the positive original exit weights just established, gives fixed-factor decay per such height increment, starting at $O(W + 1)$. The confined j -bridges are among the top exits of this tube. This proves (9.4).

Under the fixed-source bridge measure μ_h defined above, let $N(\gamma)$ count layers $j \in [[h/10], \lfloor h/6 \rfloor]$ that it crosses exactly once and whose initial j -bridge reaches lateral distance greater than $4W$. At a single-crossing cut, the two bridge pieces occupy disjoint open slabs. Their weights multiply exactly, since ports carry no weight. Proposition 5.1 and (9.4) give

$$\begin{aligned} \int N d\mu_h &= \sum_j b_{h-j}(b_j - e_j) = h^{1/2+o(1)}, \\ \int N^2 d\mu_h &\leq 2 \sum_{j_1 \leq j_2} b_{j_1} b_{j_2-j_1} b_{h-j_2} \leq h^{5/4+o(1)}. \end{aligned}$$

For the second line, dropping the lateral excursion restrictions leaves three bridge pieces. Use $b_0 = 1$ on the diagonal and $\sum_{d \leq h} b_d = h^{3/4+o(1)}$; the two outside factors each have order $h^{-1/4+o(1)}$, uniformly over the chosen layers. Hence unnormalized Cauchy-Schwarz yields

$$\mu_h(N > 0) \geq \frac{(\int N d\mu_h)^2}{\int N^2 d\mu_h} \geq h^{-1/4-o(1)}.$$

Every bridge counted here has the required lateral excursion before first crossing r . Apply the contact-interval construction to the two prefixes ending at r , matching that prescribed excursion point of the bridge to the confined arch prefix. The resulting offset differs from the initial offset by at least $2W - O(1)$, and hence gives at least order W integer hitting translates. Their total weight is at least $Wh^{-1/4-o(1)} = h^{3/4-\epsilon-o(1)}$.

All three alternatives give this bound with constants and errors independent of the arch. For each fixed ϵ , first let $h \rightarrow \infty$; only then let $\epsilon \downarrow 0$. This proves the uniform exponent assertion. $\square$

9.5 The bridge first moment

Lemma 9.1 gives $J \geq h^{3/4-o(1)}u$. Combining this with (9.1) and (9.3) yields

$$u \leq h^{o(1)}m + O(h^{-100-3/4+o(1)}).$$

On the other hand, $2(u + m) = A_{S_h}(v)$: the ordered excursion convention counts each path twice. Lemma 8.3 therefore gives $u + m = h^{1/12+o(1)}$ in the central third, uniformly in v . The negligible additive error above cannot carry this mass, so

$$\begin{aligned} m(v) &= h^{1/12+o(1)} && \text{uniformly in the central third,} \\ m(v) &\leq h^{1/12+o(1)} && \text{throughout the strip.} \end{aligned}$$

To pass from a marked vertex to a fixed source, let $\mathcal{V}_h$ contain one representative of each horizontal-translation orbit of strip vertices. Translating a bridge's bottom source to the fixed source identifies $m(v)$ with the weighted number of visits to the orbit of v . Thus the unnormalized fixed-source length mass is exactly

$$\ell_h = \sum_{v \in \mathcal{V}_h} m(v).$$

There are $O(h)$ such orbits, with order h represented in the central third. The pointwise upper and lower estimates give $\ell_h = h^{13/12+o(1)}$. Finally, dividing by $b_h = h^{-1/4+o(1)}$ gives

$$\mathbb{E}|\gamma| = \frac{\ell_h}{b_h} = h^{4/3+o(1)}.$$

Lemma 8.2 makes both the discarded mass and discarded first moment at horizontal range $h \log^2 h$ negligible, so the same conclusion holds with that confinement. This completes Theorem 1.1.

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